Payment Infrastructure and Policy Effectiveness: Evidence from Public Programs in Togo[*]

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Abstract

Public programs in lower-income countries are increasingly delivered through digital payments, but rural economic life still runs on cash. Households must convert cash into digital money to pay, or a digital transfer into cash to spend it, and either conversion requires a trip to an agent. We show that this last-mile problem is costly enough to weaken policy effectiveness. In rural Togo, the transport cost alone of reaching an agent amounts to 19% of what the median customer pays each month for pay-as-you-go solar electricity. Exploiting the staggered rollout of a nationwide solar subsidy, we find its effect on adoption is nearly three times larger where an agent is within walking distance. A subsequent government-led agent expansion raises adoption by as much as this differential, consistent with complementarity and isolating the role of payment access from potential confounders. The same friction, now on the cash-out side, attenuates the effects of Togo's emergency cash transfers, delivered digitally through the same agents. Because operators place agents by transaction volume and public programs often target places where volume is low, this wedge between private and social returns suggests a role for public investment in payment infrastructure.

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Digital payment systems are becoming the default infrastructure for delivering public programs in lower-income countries: payments for services are increasingly collected digitally, and cash transfers are deposited directly into mobile wallets (Lindert et al., 2020). On the supply side, digital delivery reduces leakage and lowers administrative costs (Muralidharan, Niehaus and Sukhtankar, 2016). However, daily economic life in much of rural Africa still functions on cash, so digital delivery leaves a last-mile problem: households must convert cash into digital balances to access subsidized services, and convert digital transfers back into cash to consume. Both conversions typically happen at physical cash-in and cash-out points, where each trip can be expensive, especially where these access points are sparse. While the supply-side gains have been well documented, whether these household-side costs are large enough to dampen program effectiveness is ultimately an empirical question. The same program can succeed in one place and disappoint in another (Duflo, 2017; Hanna and Olken, 2018; Meager, 2019; Vivalt, 2020), but how much of this variation reflects the cost of reaching the payment network remains largely unknown.

We study this question in rural Togo, where two nationwide programs depended on the same cash–digital interface (mobile money). The CIZO program subsidized electricity access through pay-as-you-go solar home systems, which households paid for digitally (cash-in); Novissi, an emergency cash transfer program, deposited benefits directly into recipients' mobile money accounts (cash-out). Both programs depended on the network of agents who convert cash into digital balances and back. The cost of reaching an agent thus creates a wedge between each program's nominal value and its value to households. We show that the CIZO subsidy's effect on solar adoption is nearly three times larger where an agent is within walking distance, and that the subsequent government-led placement of agents in unserved villages raises adoption there. Consistent with the same mechanism, Novissi's treatment effects are larger in cantons with better agent coverage.

Rural electrification is central to our analysis: the stakes are large, and the payment structure of off-grid solar makes the wedge potentially important. Nearly 800 million people lack electricity, overwhelmingly in rural South Asia and Sub-Saharan Africa (International Energy Agency, 2022), and off-grid solar has become the central margin of expansion (Burgess et al., 2023; Lang, 2025*a*): extending the grid to sparsely populated areas is prohibitively costly, and by 2040 an estimated 70% of rural Africans gaining electricity will do so off-grid (Africa Progress Panel, 2017). Existing work documents low take-up even at heavily subsidized prices and emphasizes limited willingness to pay, so that only large subsidies move adoption (Lee, Miguel and Wolfram, 2020; Burgess et al., 2023; Burlig and Preonas, 2022). We offer a complementary explanation, which is that the posted price understates what households actually pay. Under the pay-as-you-go (PAYGO) contracts, which now finance most solar home system sales (The World Bank, 2024), households buy electricity in small and frequent increments, and each payment typically requires a trip to an agent. The transport cost therefore recurs with every purchase, so the effective price of electricity is the posted price plus a transaction cost that rises with distance to the network and with payment frequency. In our sample, the median village lies 5.7 kilometers from the nearest agent, and for the median customer these repeated trips amount to 19% of the monthly price of electricity access, rising to 39% at the 75th percentile. These costs are of the same order as the subsidy's price cut. Part of what looks like low willingness to pay is a high and unmeasured component of price.

Our main solar analysis exploits two sequential interventions that lowered the effective cost of electricity through different channels. The CIZO subsidy, among the first demand-side solar subsidies in Sub-Saharan Africa, rolled out across districts in a staggered way between March and July 2019, and cut the price of solar electricity by 18–44% depending on the solar kit. The government-led *Réseau Granulaire* expansion, launched in September 2019, placed mobile money agents in remote villages that previously lacked nearby access; at baseline, only 27% of villages in our sample had an agent within 3 kilometers, the walking threshold we use throughout. While the subsidy lowered the posted price of electricity, the agent expansion lowered the cost of paying. Because the two policies were introduced at different times, we can separate their effects. Each rollout was itself staggered across locations, so both event studies use the interaction-weighted estimator of Sun and Abraham (2021).

We combine four data sources. First, proprietary administrative records from Togo's largest solar provider cover more than 50,000 customers in over 2,000 rural villages and provide uniquely rich customer-level data, tracking each customer's application and installation, every subsequent payment, and daily electricity use.[1] Second, government data on the universe of mobile money agents, with GPS coordinates and opening dates, let us measure each village's distance to the nearest agent and pin down the timing of agent arrival. Third, an original survey of rural moto-taxi fares allows us to put a monetary value on the cost of reaching an agent. Fourth, anonymized data from the randomized evaluation of the Novissi emergency cash transfer program (Aiken et al., 2022), generously shared with us and merged with the agent data, let us test whether the same friction operates beyond subsidy programs.

A simple framework organizes the analysis around two barriers to adoption: a price barrier and an access barrier. On the extensive margin, a household adopts only if it is willing to pay the posted price and able to complete the repeated payments the PAYGO contract requires, at a transport cost that rises with distance to the nearest agent. Completing a payment is itself a chain of steps: a working phone, arranging transport if the agent is beyond walking distance, an agent who is present and holds float, and a phone signal at the moment of the transaction. As in an O-ring production function (Kremer, 1993), the payment fails if any step fails, and remoteness both adds steps and makes each less reliable, so access falls with distance by more than the transport fare alone implies. The subsidy relaxes the price barrier; agent expansion relaxes the access barrier. Because adoption requires clearing both, relaxing one raises the share of households for whom the other binds, so the marginal effect of each policy is larger where the other is already in place: the two instruments are complements. On the intensive margin, PAYGO lets households choose how often to pay, so a price cut allows them to consolidate payments and make fewer trips. The

[1] In 2016, these villages had very low electrification rates, with only about 7% of rural households accessing any source of electricity (Government of Togo, 2016).

framework yields testable predictions: the subsidy's effect on adoption should rise with proximity to an agent and drop sharply once an agent is no longer within walking distance; agent arrival should raise adoption by at least that differential where the subsidy is already in place; and existing customers should pay less often when the price falls, especially near an agent. Because the same friction is incurred in reverse when a program disburses rather than collects, it also predicts that digital cash transfers are worth less where agents are far, and most visibly so for outcomes that require converting the balance into cash.

Two results show that the payment infrastructure substantially affects the reach of the subsidy. First, the subsidy more than doubled solar adoption, consistent with contemporaneous evidence from the same setting (Lang, 2025*a*). However, the average conceals sharp heterogeneity: the effect is nearly three times larger in villages with an agent within 3 kilometers at baseline (0.96 versus 0.35 additional applications per village-month), a gap significant at the 5% level and unchanged by controls for village wealth, population density, and market access. Because pre-existing agent access is not randomly assigned, this comparison could still reflect unobserved village characteristics. The subsequent agent expansion then provides complementary event-study evidence. The *Réseau Granulaire* expansion began after the subsidy was already nationwide, so identification comes from the timing of each agent's arrival across the 100 villages that gained a first nearby agent. Agent arrival raises adoption by 0.64 applications per village-month after the subsidy was nationwide. In these same villages the subsidy had raised adoption by 0.33, so the arrival of an agent did roughly twice what an 18–44% price cut had done, and more than the subsidy's average effect across all villages (0.47). That agent effect is also similar to the subsidy's differential effect across villages with and without an agent (0.61 applications per village-month), the lower bound our second prediction requires: an agent's arrival under the subsidy carries both that differential effect and whatever effect the agent would have had in the subsidy's absence. Variation in the timing of payment access alone thus reproduces the entire cross-sectional gap. By exploiting different margins of the effective price at different times, the two designs show that payment access is an important determinant of adoption, and ultimately of the effectiveness of the subsidy program.

The results are consistent with the subsidy and payment infrastructure acting as complements. Because the expansion occurred only under a nationwide subsidy, we never observe an agent's effect in the subsidy's absence, so the interaction between the two policies is not directly identified. The arrival effect nonetheless bounds it. For the two instruments to be additive rather than complements, an agent alone would have to raise adoption by the full 0.64, more than three times the pre-subsidy adoption gap between villages with and without one (0.20), a descriptive gap that itself likely overstates an agent's causal effect.

Three further findings connect the estimated effects to the cost of reaching agents and rule out the leading alternatives. First, the subsidy–agent interaction turns on the walking threshold: subsidy effectiveness drops sharply once the nearest agent lies beyond 3 kilometers and changes little with additional distance, matching the step structure of rural transport costs. Second, the pattern is not an artifact of who lives near agents: the subsidy effect is larger near an agent within groups defined by livelihood, age, and gender. Third, households save on the number of trips. The subsidy reduced payment frequency by 19% without changing electricity use, i.e., the same consumption financed with fewer, larger payments. The response was larger near agents, where higher baseline payment frequency left room to consolidate while remote customers were often already at the one-payment-per-month floor. Overall, these responses are consistent with households timing their payments as if each trip were costly. We further rule out supply-side responses by the provider, differential demand for electricity, coordination through third-party payers, and easier subsidy receipt where an agent is present.

The same friction reappears on the cash-out side. We combine the randomized evaluation of Novissi, Togo's nationwide emergency cash transfer (Aiken et al., 2022, 2025), with our agent data. Because transfers were paid into mobile money accounts, beneficiaries had to reach an agent to convert them into cash.[2] Each standard deviation of agent access raises Novissi's effect on financial health by 0.05 standard deviations ($p < 0.05$), from an insignificant effect one standard deviation below mean access to 0.08 one standard deviation above it. The pattern is consistent with households protecting essential food consumption and absorbing cash-out costs on the residual margin: Novissi substantially improves food security regardless of agent access, whereas cheaper cash-out allows more of the transfer to remain available for savings and other financial buffers. The interaction for health care access is positive but imprecise, and it is zero for the index of outcomes that do not require cashing out. Although agent access is not randomly assigned, the heterogeneity survives a horse race against proximity to financial institutions, proximity to markets, and population density: only the agent-access interaction remains, and it is even larger. The specificity of the pattern to mobile money agents argues against generic remoteness as the explanation and points instead to payment access as a broader bottleneck in the effectiveness of digitally delivered policy.

These findings point to a potential market failure in the provision of payment infrastructure. A mobile money agent is typically a local shopkeeper who holds both cash and digital money and earns a small commission on each deposit and withdrawal. Becoming an agent requires a substantial upfront commitment of working capital (Suri, 2017), which pays off only where transaction volumes are sufficiently high (Cull et al., 2017). Private operators therefore have the strongest incentives to expand in high-volume markets. Public programs, however, often target poor and remote areas where transaction volumes are low. Operators do not internalize the value that agent coverage creates by improving the reach and effectiveness of publicly financed programs. This wedge between private returns and social value suggests a rationale for public investment in underserved areas.[3] The *Réseau Granulaire* expansion we study can be understood as one such intervention. An illustrative extrapolation of the agent-arrival estimate to all underserved villages would move adoption within the provider we study from about half to roughly 70% of the pace

[2] A single cash-out trip costs roughly 12% of the monthly transfer at the median canton distance and about 19% where no village in the canton lies within walking distance of an agent.

[3] Under-provision of this kind is characteristic of payment networks, where adoption on one side raises returns on the other (Higgins, 2024).

required by Togo's 2030 electrification target. The results therefore motivate viewing payment infrastructure as complementary public capital: investment in last-mile access raises the social return on a broad portfolio of programs, beyond the direct gains from financial inclusion.

Our first contribution is to show that payment infrastructure can substantially increase the effectiveness of public programs by alleviating the transaction cost on the household side. Work on financial inclusion and mobile money documents direct welfare effects through risk sharing, consumption smoothing, labor supply, and investment (Burgess and Pande, 2005; Suri, 2011; Jack and Suri, 2014; Suri, 2017; Fink, Jack and Masiye, 2020; Breza and Kinnan, 2021; Lee et al., 2021; Fonseca and Matray, 2022; Berkouwer et al., 2023; Riley, 2024; Batista and Vicente, 2025). The closest precedent is Muralidharan, Niehaus and Sukhtankar (2016), who show that digitizing payments in India reduced supply-side leakage as the state transferred funds to households. Our channel operates on the demand side, in the cost households pay at the cash–digital interface. Aker et al. (2016) compare delivery technologies and find that recipients paid through mobile money in Niger traveled half as far as those collecting cash at a distribution point. In Mexico, Bachas et al. (2018) show that expanding access to a broader ATM network sharply reduced travel and opportunity costs, with larger reductions in distance associated with greater account use. Here, we hold the delivery technology fixed and show that digital delivery does not eliminate the friction: it remains binding for households farthest from an agent. These are the households public programs typically target, e.g., Novissi was aimed at Togo's poorest cantons, and a fifth of the cantons in the evaluation sample have no village within walking distance of an agent.

Second, we introduce a structurally different friction into the take-up and ordeal literature (Bertrand, Mullainathan and Shafir, 2006; Currie, 2006; Kleven and Kopczuk, 2011; Alatas et al., 2016; Deshpande and Li, 2019; Homonoff and Somerville, 2021). In most of this literature, the friction is a hassle cost—paperwork, stigma, or travel to an enrollment office—paid once at the point of claim, and the evidence comes overwhelmingly from higher-income settings. The cost we study is paid at every transaction with the program, so its cumulative weight grows with payment frequency and can rival the posted price of the service itself. It is also governed by the density of the payment network rather than by the design of the application process, placing its reduction in the domain of infrastructure policy. Its incidence also inverts the screening logic for ordeals: an ordeal can improve targeting when its cost is lowest for those with the greatest need (Nichols and Zeckhauser, 1982), whereas here, cost rises with remoteness, which weighs most heavily on precisely the areas these programs prioritize. The friction has precedent in social policy: in the paper-voucher era of US food assistance, obtaining and redeeming benefits carried real time and travel costs, which the shift to electronic delivery later reduced (Currie, 2006; Zhou et al., 2026). That reduction was made possible because benefits delivered electronically could also be spent electronically; where households transact in cash, electronic delivery instead moves the cost to the conversion between cash and digital balances. Finkelstein and Notowidigdo (2019) note that the shape of the cost distribution, and not only its level, matters for the welfare consequences of take-up interventions. The ordeal we study carries a market price, so we observe that distribution directly rather than infer it. The cost is near zero within walking distance of an agent, jumps discretely beyond it, and subsidy effectiveness falls with distance in the same pattern.

Third, we contribute to work on electricity demand and PAYGO electrification in lower-income settings (Lee, Miguel and Wolfram, 2020; Grimm et al., 2020; Burgess et al., 2023; Gertler et al., 2023; Gertler, Green and Wolfram, 2024; Lang, 2025*a*,*b*). Prior work documents low take-up, even at heavily subsidized connection prices across both grid and off-grid markets. PAYGO contracts, which finance most solar home system sales in Sub-Saharan Africa, were designed to relax upfront liquidity constraints by replacing a large one-time payment with a stream of small recurring ones. We show that this design magnifies exposure to a different friction: each payment is made through a network that can be costly to reach, creating a wedge between posted and effective prices that suppresses adoption and dampens measured price elasticities. Lang (2025*a*), studying the same subsidy, establishes that adoption is highly sensitive to the posted price. We take that sensitivity as our starting point and show that the payment-access cost is a substantial component of the effective price.

The remainder of the paper is organized as follows. Section 1 describes the solar setting and the subsidy and agent-expansion interventions. Section 2 presents the conceptual framework. Section 3 describes the data and empirical strategy. Section 4 details the empirical specifications. Sections 5 and 6 present the main results on adoption and mechanisms driving the complementarity. Section 7 presents results from the Novissi cash transfers policy. Section 8 discusses policy implications and Section 9 concludes.

1 Context and Policies

1.1 Payment Infrastructure: Mobile Money and Digital Payments

Togo, a country of 8.5 million with a GDP per capita of $2,200 (PPP, 2017), entered the period we study with low financial inclusion. In 2017, two years before the subsidy rolled out, only 45% of adults held an account at a formal financial institution or with a mobile money provider, and ownership was concentrated in urban areas (World Bank Group, 2017). More than half the population lived below the international poverty line of $1.90 a day, disproportionately in rural areas.The households that programs like CIZO and Novissi target were therefore also the ones with the least access to the formal financial system.

Access has been improving, mostly through mobile money and under an explicit government strategy of digital transformation. Between 2011 and 2021, mobile penetration rose from 40% to 78% and internet penetration from under 5% to 75%.Mobile money accounts grew from 1% of adults in 2014, the first year Findex measured them, to 21% in 2017 and 36% in 2021, and accounted for most of the rise in financial inclusion over the period (Klapper, Singer and Starita, 2025).[4] The government also began paying social protection through mobile money at scale; Novissi, the emergency cash-transfer program we study, paid informal workers digitally during the COVID-19 pandemic, further familiarizing households with mobile money.

These gains did not remove the physical layer connecting cash and digital balances. Two operators, Togocel and Moov, run the country's mobile money services (TMoney and Flooz), but most rural households still earn and spend in cash, so funding a wallet requires visiting an agent to deposit cash and have the account credited, and converting a digital transfer into cash requires the reverse trip. Agents concentrate where transaction volumes are high, typically in larger villages and small towns, so rural households often face long trips to reach one; at the start of our study period, the median village in our sample lies 5.7 km from the nearest agent (Appendix Figure A3). For households that live far from an agent and pay frequently because of liquidity constraints, this distance is a sizable recurring transaction cost.

1.2 Rural Electrification in Togo

At the start of our study period, electricity access in Togo was low and starkly unequal across space. In 2016, roughly 45% of urban households were connected to the grid but just 7% of rural households had electricity, leaving almost one million households, the vast majority rural, without access (Government of Togo, 2016).

In response, the government launched *CIZO* ("light up" in Mina) in 2017, an initiative to electrify rural households with solar home systems, and in 2018 adopted a national electrification strategy targeting universal access by 2030.

Extending the grid to sparsely populated areas is costly, and low average consumption limits its financial viability (Blimpo and Cosgrove-Davies, 2019), so the strategy anticipates at least half of newly electrified households being served off-grid, primarily by solar home systems and mini-grids; the solar home system component targets roughly 550,000 mostly rural households.

These systems are sold on pay-as-you-go contracts and paid for through mobile money, so electrification proceeds not through a one-time connection but through a stream of small recurring digital payments, hence the strategy's reach depends on the payment infrastructure.

1.3 Solar Home Systems

The recent rise of cost-effective off-grid solutions, particularly solar panels, has changed the landscape of rural electrification. The cost of photovoltaic modules has fallen from well over $70 per watt in the late 1970s to around $2 per watt in 2010 and roughly $0.19 per watt in 2020, a reduction of more than 99%. These dramatic cost declines have created new opportunities to electrify

[4]Growth continued past our study period: by 2024, 48% of adults held a mobile money account and 57% held an account of any kind, near the Sub-Saharan African average of 58%. Accounts at financial institutions did not grow between 2017 and 2024 (34% to 32%), so mobile money accounts for all of the net gain. The urban–rural gap in account ownership that remains (66% versus 53% in 2024) is a gap in bank accounts; mobile money ownership is nearly uniform across geography (51% urban, 46% rural) (Klapper, Singer and Starita, 2025).

remote areas where grid extension is prohibitively expensive and have made stand-alone solar home systems (SHS) a viable technology for rural households in low-income countries. Within Togo's electrification strategy, SHS have thus emerged as a central off-grid technology for reaching dispersed rural households.

Off-grid solar solutions differ from grid connections along several dimensions. *(1) Flexibility.* Off-grid systems can be deployed quickly at the household or village level, without large investments in transmission and distribution infrastructure. *(2) Usage.* SHS typically support low- to medium-power appliances—such as general lighting, phone charging, radios, fans, and small televisions—whereas grid connections or larger solar installations are needed for high-power appliances and productive or industrial use. *(3) Reliability.* Urban areas in Togo suffer frequent grid outages (Kpemoua, 2016), so well-sized solar systems can offer more reliable power, although shortages may still occur in periods of consecutive heavy rainfall, especially for medium-power appliances. While off-grid solar presents a flexible and potentially more reliable alternative.

These constraints have spurred the development of *pay-as-you-go* (PAYGO) SHS offered by private companies across Sub-Saharan Africa, which allow households to acquire solar systems through a rent-to-own contract rather than a large one-off purchase. Poor rural households in Togo typically rely on subsistence agriculture, face substantial income risk, and have limited access to formal credit. Under the PAYGO model, customers first select a bundle and make a modest upfront downpayment, which secures the installation of the solar panel, battery, and chosen appliances in their home as well as the first month of electricity. After installation, customers make recurring payments over time, renting access to the system while gradually building ownership. The total stream of payments is structured so that, after roughly three years of regular payments and continuous usage, the customer owns the system outright. This study uses data from the largest SHS provider in Togo. The "basic" kit bundle consists of a small solar panel, a battery, three light bulbs, and a phone charger. The downpayment, which covers installation and the first 30 days of electricity, is 8,960 CFA (approximately $15), followed by a daily rate of 160 CFA (about $0.27), equivalent to roughly $8 per month (using an exchange rate of 600 CFA per US dollar). The SHS provider also offers "intermediary" and "premium" kits with additional appliances: the intermediary kit includes a radio, while the premium kit includes both a radio and a TV.[5]

Unlike most grid electricity payment systems, the PAYGO rent-to-own model allows customers flexibility in when and how much they pay over time. Customers purchase electricity in daily units, gaining unlimited access to their SHS for the number of days purchased. If access time expires, the solar company can remotely lock the SHS, preventing further use until additional days are bought—a form of "digital collateral" (Gertler, Green and Wolfram, 2024). Customers who do not make payments for over sixty consecutive days are considered in default, and the company may repossess the SHS after 120 days of inactivity (though this is not common in practice). The combination of remote lockout and the credible threat of repossession makes PAYGO contracts

[5]The downpayment for the intermediary kit is 12,320 CFA (about $20.50) and the implied monthly payment is about $11. The downpayment for the premium kit is 21,000 CFA ($35), and the implied monthly payment is about $19.

enforceable even in settings with weak institutions.

PAYGO SHS contracts in Togo are built around the country's expanding digital payments infrastructure. Customers pre-pay for electricity access using their mobile phones via mobile money: they initiate a payment from their mobile money wallet, which immediately credits their SHS account and adds days of electricity access.[6] The mobile operators, and where their agents are located, are therefore crucial for both SHS providers and customers.

1.4 Policy 1: Nationwide Solar Home System Subsidy

In 2019, the Government of Togo launched a nationwide demand-side subsidy for solar home systems called *CIZO Cheque* as part of the CIZO program to achieve universal access to electricity by 2030. As described in Section 1.2, the national electrification strategy foresees roughly half of new connections coming from off-grid technologies, with SHS playing a central role in sparsely populated rural areas. This was among the first government subsidy in Africa explicitly designed to support solar energy payments. The first off-grid SHS company only began operating in Togo in 2017, so the government introduced a large public subsidy to accelerate SHS adoption and make monthly payments more affordable for low-income households.

The Togolese program is distinctive in that it offers an untargeted end-user subsidy, rather than a pure supply-side results-based financing scheme. The subsidy takes the form of a uniform price reduction of \$0.12 (CFA 70) per day of electricity purchased (or \$3.50 per month), regardless of the SHS bundle chosen. For the basic kit described in Section 1.3, this corresponds to a reduction in the effective monthly cost from about \$8 to \$4.50, a price cut of roughly 44%. For the intermediary kit, the subsidy reduces the monthly cost from around \$11 to about \$8 (a 30% reduction), and for the premium kit from roughly \$20 to about \$16 (an 18% reduction). Given that monthly GDP per capita in Togo was around \$57 in 2019, the subsidy represents a sizeable transfer for rural households.

Operationally, the subsidy leaves the downpayment and installation unchanged. Customers still pay the full upfront cost, including the first month of electricity. For subsequent months, the government matches customers' electricity purchases up to a ceiling of roughly \$3.50 per month: customers pay their usual PAYGO tariff, and the government tops up the remaining amount so that a full month of access is unlocked. Customers who do not make payments, or who default on their contracts, do not receive any subsidy. The subsidy is implemented digitally, via a national platform linking SHS providers, mobile operators, and La Poste, consistent with Togo's broader digitalisation strategy.

A key feature of this policy for our purposes is its staggered rollout. Between March and July 2019, the subsidy was introduced in three successive cohorts of districts: Phase 1 in March, Phase 2 in May, and Phase 3 in July (see Panel A of Figure 1). Each cohort became eligible for the subsidy

[6] In Rwanda, Lang (2025*b*) finds that most PAYGO customers visit a mobile money agent for every solar payment, rather than saving money in their mobile wallets. We find a similar pattern from qualitative interviews and field visits in rural Togo, a comparable context.

on its phase-specific start date, while the rest of the country continued to face pre-subsidy prices. This scattered rollout across space and time provides quasi-experimental variation in exposure to the subsidy, which we exploit to identify its causal effects using an event-study design described in Section 4.1.

All rural households using PAYGO SHS were eligible to benefit from the subsidy, conditional on making payments.[7]

1.5 Policy 2: Mobile Money Agent Network Expansion Policy

The second policy we study is a large-scale, supply-side expansion of the mobile money agent network, designed primarily to expand access to digital financial services in rural areas and, in doing so, to lower the costs of using mobile money. In late 2019, shortly after the SHS subsidy rollout, the Togolese government launched the financial inclusion program *Réseau Granulaire* (Granular Network). Under this program, the national postal operator, La Poste, recruited, trained, and hired its own agents to provide mobile money services, deploying them across more than 200 villages nationwide. The program explicitly targeted rural localities that previously lacked nearby access to mobile money agents. Importantly, the subsidy was already in place nationwide when the agent expansion began, so we identify only the effects of mobile money agent arrival, conditional on the presence of the subsidy.

The expansion was implemented in a decentralised and staggered manner. Regional supervisors at La Poste were given discretion to identify villages with poor access to agents, subject to broad program guidelines, and new agents were then rolled out progressively across villages from September 2019 onwards. This process created substantial variation in the timing of the first local agent arrival across villages. In total, 100 villages in our data received their first mobile money agent through this program between September 2019 and May 2021 (see Panel B of Figure 1). We define a village as gaining access when an agent first opens within 3 km of the village centre, a distance within which customers can typically walk without incurring monetary travel costs.

The timing of the agent expansion exhibits two distinct phases due to the COVID-19 pandemic (see Appendix Figure A2). From September 2019 to March 2020, there was a continuous expansion of the mobile money agent network, with new agents opening every couple of months in different villages. With the onset of COVID in April 2020, the program was abruptly paused, and no new agents were added in rural areas for the remainder of 2020. The expansion then resumed in early 2021, with a second wave of deployments. We exploit this staggered rollout in our empirical design, by using the discrete timing of the first agent arrival in each village to study how improved access to mobile money agents affects SHS adoption.

[7] In practice, not all customers actually received the subsidy on every payment. To operationalize the program and promote digital financial inclusion, the government required that SHS payments be made through a *personal* mobile money account that was linked to the customer's solar account. This design facilitated automated top-ups and digital monitoring, but it also created frictions in the early months of the program. Technical issues in reconciling beneficiary lists meant that some customers were not correctly flagged as eligible when they made payments, and therefore did not receive the top-up. As a result, the majority, but not all payments were effectively subsidised, especially at the start of the rollout (see Appendix Figure A1).

1.6 Policy 3: Emergency Cash Transfers

In April 2020, the Togolese government launched *Novissi* ("solidarity" in Ewe), a digital cash-transfer program for informal workers affected by the COVID-19 pandemic, initially covering urban areas (Aiken et al., 2022). In late 2020, the program was extended to the 100 poorest rural cantons, with eligibility determined by machine-learning predictions of poverty from mobile-phone data. Aiken et al. (2025) evaluate this phase through an embedded randomized controlled trial in which approximately 49,000 registered individuals were randomly assigned to receive or not receive the transfer. Beneficiaries received monthly transfers of 8,620 FCFA (USD \$15.50 = \$38 PPP) for women, and 7,450 FCFA (USD \$13 = \$33 PPP) for men, for five months, paid into mobile money accounts. Transfers were delivered exclusively through mobile money, so receiving a payment required a registered account and converting it into cash required a trip to an agent.

2 Conceptual Framework

This section develops a simple framework that organises the empirical analysis. Sections 2.1 and 2.2 introduce our general framework for the role of payment infrastructure. In its simplest form, payment infrastructure governs the *accessibility* of a program, and thus is essential for its effectiveness. Sections 2.3 and 2.4 discuss the effectiveness of the subsidy and cash transfers, highlighting additional channels through which these policies can interact with the payment infrastructure in complementary ways. Section 2.5 spells out the predictions.

2.1 Payment Infrastructure, Transaction Costs and Accessibility

Digital payments in Togo run through the mobile money payment infrastructure, while transactions in rural areas remain overwhelmingly in cash. Every digital interaction therefore passes through physical infrastructure: to pay a bill, cash must be converted into digital balance at a mobile money agent; to spend a digital transfer, it must be converted back.

Let $\tau(d)$ denote the *unit transaction cost* of making a payment at distance d from the nearest agent. We think of this primarily in terms of transportation costs: reaching an agent is free within walking distance and requires paid transport beyond it, so τ jumps at the walking threshold and rises far more slowly thereafter.[8]

Transaction costs also govern the *accessibility* of the payment system, $\alpha(\tau) \in [0, 1]$, with $\alpha_\tau \leq 0$. At its simplest, the accessibility term could denote the share of households with an operational mobile money account, an effective measure of financial inclusion. Appendix Figure A4 draws on a nationally representative household survey and shows that among rural households, mobile money account ownership declines only modestly with distance to the nearest agent. Moreover, among rural households below the poverty line, a group well represented among SHS customers,

[8] The transport-cost survey we conducted documents one-way moto-taxi fares of roughly 305 CFA plus 40 CFA per kilometer, implying a round trip of roughly 610 CFA plus 85 CFA per kilometer, in 2019 CFA (Section 3.5).

the gradient is even flatter. Ownership alone therefore does not entirely capture accessibility.

This is because $\alpha(\tau)$ admits richer interpretations, and account ownership is only the first step to effective access. We conceptualise accessibility analogously to an O-ring production function (Kremer, 1993). Let $K(\tau)$ denote the number of steps that the household must complete to make a transaction, which depends on the transaction cost, capturing its level of remoteness and distance from an agent. These steps include knowing, or obtaining help with, the appropriate mobile-money procedure; arranging transportation when necessary; reaching an agent who is present and holds the required cash or digital money; and obtaining a mobile signal when the transaction is attempted. Existing evidence suggests that these constraints frequently prevent attempted transactions from being completed.[9]

The probability that the full transaction can be completed is then $\alpha(\tau) = \prod_{k=1}^{K(\tau)} q_k(\tau)$, where $q_k(\tau)$ denotes the probability that step k succeeds. Transaction costs enter twice. First, as transaction costs increase, the number of steps also increase. Second, higher transaction costs may also lower the probability that each step succeeds (e.g. when agents are distant, there may also be few nearby alternatives when an attempt fails). Modest reductions in the reliability of several steps therefore compound. A failed attempt is also more costly when retrying requires another paid trip, and recurring payments expose households to these risks month after month. This distinction between monetary transaction costs and process accessibility is consistent with Alatas et al. (2016). In their cash-transfer experiment, the farther registration site imposed only a modest monetary transport cost, yet moving registration closer increased applications by approximately 15%.

2.2 A General Framework for Payment Infrastructure and Policy

The interaction between payment infrastructure and policy effectiveness includes two components. Consider a digital policy instrument x that targets an outcome $T(x)$ in a population facing transaction cost τ. The actual realised outcome, determined directly by the accessibility of the target population, is thus:

$$A(x, \tau) = \alpha(\tau) \times T(x).$$

The policy's effectiveness—how much the realised outcome shifts with the policy instrument—is therefore increasing in $\alpha(\tau)$, and hence decreasing in τ. The intuition is that as the payment infrastructure expands, lowering transaction costs and improving accessibility, a policy's reach widens, allowing it to have a far greater impact. Throughout, we assume that $T_x \geq 0$. The policy and the payment infrastructure are therefore complements. Formally, we have that:

$$A_x = \alpha(\tau) \times T_x(x).$$

[9] Using mystery-shopping exercises in Bangladesh, Tanzania, and Uganda, Innovations for Poverty Action (2023) report that 29 percent of attempted transactions failed across the three countries: the agent was absent in 17 percent of visits and, conditional on the agent being present, 14 percent of transactions still failed. The estimated opportunity cost of the time lost to failed transactions was nearly ten times the direct monetary transaction cost in Bangladesh, nearly three times as large in Uganda, and nearly twice as large in Tanzania.

This relation is at the core of our conceptual framework: it is our first-order accessibility channel. However, to capture richer and more complex interactions, we can also extend our framework to allow for an additional second-order channel through which the payment infrastructure can interact with a policy's effectiveness. Let τ also enter directly into the target function: $T(x, \tau)$. Then, there is an additional term for policy effectiveness:

$$A_{x\tau} = \underbrace{\alpha_\tau(\tau) \times T_x(x, \tau)}_{\text{direct (access)}} + \underbrace{\alpha(\tau) \times T_{x\tau}(x, \tau)}_{\text{indirect (additional)}}$$

The first term is the direct accessibility channel, while the second is an additional indirect channel that can amplify or dampen a policy's effectiveness. Complementarity requires that this cross-partial be negative.[10]

2.3 The Effectiveness of Subsidies for Solar Home Systems

We now apply this framework to subsidies for solar home systems. The government's objective in subsidising solar home systems is to increase rural electrification, so we measure the subsidy's effectiveness by the number of households it induces to adopt. When deciding to adopt, households must factor in the posted monthly price of electricity $\overline{p}$, the unit transaction cost τ, and the number of monthly payments $\overline{n}$, which we assume for now is fixed (we relax this below).[11] The policy instrument is a subsidy s on the price of electricity, which reduces the price to $\overline{p} - s$ without affecting the unit transaction cost τ. The full monthly cost of electricity, P, is thus:

$$P = \overline{p} - s + \overline{n} \times \tau. \tag{1}$$

Let $T(P)$ denote the demand of households for electricity at cost P, where T behaves as a standard twice-differentiable convex demand function with $T_P < 0$ and $T_{PP} \geq 0$. This is the government's target outcome: the level of rural electrification. We can express this as a function of the subsidy and the transaction cost: $T(P(s, \tau))$, noting that the subsidy lowers the total price, $P_s = -1 < 0$, and transaction costs increase it, $P_\tau = \overline{n} > 0$. Because the system requires digital payments, village-level adoption is scaled by the accessibility term:

$$A(s, \tau) = \alpha(\tau) \times T(P(s, \tau)) \tag{2}$$

It is straightforward to show that both the subsidy and a policy of lowering transaction costs (by introducing an agent to the village) will increase adoption: $A_s \geq 0$ and $A_\tau < 0$.[12] However, our primary object of interest is how the effectiveness of the subsidy, A_s, is affected by the pay-

[10] This is because increasing τ has a negative effect on accessibility with $\alpha_\tau \leq 0$, while we assume that the policy instrument x has a positive effect on the target outcome with $T_x \geq 0$.

[11] Households are able to pay less than the full monthly cost of electricity in exchange for less electricity usage. This does not impact our conceptual framework as these preferences *after* adoption would be factored into the demand curve.

[12] The inequality is weak for the subsidy in case there is no accessibility at all with $\alpha(\tau) = 0$. Transaction cost reductions have two effects: increasing accessibility and lowering the total price.

ment infrastructure τ. Under what conditions is the subsidy's effectiveness enhanced by a more inclusive payment infrastructure with lower transaction costs? Equivalently, when are reductions in the price through the subsidy s and in transaction costs τ complements for adoption, so that $A_{s\tau} < 0$? This cross-partial derivative consists of two terms, the direct access term and an indirect curvature term:

$$A_{s\tau} \;=\; \underbrace{\alpha_\tau(\tau) \times (-T_P)}_{\text{access (direct)}} \;+\; \underbrace{\alpha(\tau) \times (-\overline{n}\, T_{PP})}_{\text{curvature (indirect)}} \;<\; 0. \tag{3}$$

The first term, the direct access effect, is negative, as higher transaction costs reduce accessibility and thus lower the effective reach of the direct subsidy effect $(-T_P)$. The second, indirect term is weakly negative because demand is convex ($T_{PP} \geq 0$). Intuitively, convex demand is flatter at higher costs, so where transaction costs push up the monthly cost of electricity (P), a one-unit price cut moves fewer households to adopt; the term drops out entirely if demand is linear and so is not necessary for complementarity.

Therefore, our **first prediction** (P1) is that the subsidy's effect on adoption is larger in villages with a mobile money agent, where transaction costs are lower, than in villages without one. As the cross-partial is symmetric ($A_{s\tau} = A_{\tau s}$), introducing an agent equivalently raises adoption by more once the subsidy is in place than it would in the subsidy's absence. The differential effect of the subsidy in villages with and without an agent should equal the differential effect of an agent's arrival with and without the subsidy.

We cannot measure the second margin directly, since we are unable to identify the effect of an agent arriving in the absence of the subsidy in our data. We therefore turn to a weaker implication. Our **second prediction** (P2) is that introducing an agent where the subsidy is in place raises adoption by at least the subsidy's differential effect. The reason is that the effect of an agent's arrival under the subsidy decomposes into two parts: the subsidy's differential effect, and the effect an agent would have had without the subsidy. The latter is weakly positive, since any reduction in transaction costs raises adoption. Equivalently, the agent's arrival effect under the subsidy is an upper bound on the complementarity term.

We can further exploit the shape of transaction costs in distance. As set out in Section 2.1, the fixed moto-taxi fare makes τ jump once an agent lies beyond walking distance, while the per-kilometer component raises it only gently thereafter. Moreover, going beyond walking distance also results in a discrete increase in the number of steps needed for payment, worsening accessibility. Our **third prediction** (P3) is that the subsidy's effect is correspondingly non-linear in distance to the nearest agent: it should drop sharply once the nearest agent lies beyond walking distance, and then flatten.

2.3.1 An Additional Channel: Endogenous Payment Frequency

So far we have taken the payment plan as fixed. We relax that restriction here because the PAYGO system gives customers flexibility over when they pay. The number of payments is an endogenous choice depending on the subsidised price and the unit transaction cost: $n(\overline{p}-s,\tau) \geq 1$.[13] Households can choose to pay often in small amounts, or accumulate cash and pay infrequently in larger instalments. This is the familiar trade-off in inventory models of transaction demand, between the fixed cost of each trip and the cost of assembling a larger sum for each payment (Baumol, 1952; Tobin, 1956). Complementarity through this channel requires $P_{s\tau} > 0$, so that the subsidy lowers the total cost by more where transaction costs are low. With frequency fixed at $\overline{n}$ the subsidy and transaction costs are additively separable and $P_{s\tau} = 0$; allowing $n(\overline{p}-s,\tau)$ to vary gives

$$P_{s\tau} \;=\; -\big(n_p \;+\; n_{p\tau} \times \tau\big). \tag{4}$$

The first term is a *level* effect. A subsidy lowers the bill, which households can then cover in fewer and larger payments, economising on trips ($n_p \geq 0$). This points towards substitutability: the subsidy itself reduces exposure to transaction costs, and does so most where those costs are highest. The second term is a *flexibility* effect: how strongly frequency responds to the subsidy itself depends on transaction costs. Complementarity through this second term requires frequency to respond more strongly to the subsidy when an agent is near than when one is far ($n_{p\tau} \leq 0$): near an agent, payment schedules are flexible and can adjust, whereas far from one there is little room to adjust downwards. A lower unit transaction cost also raises payment frequency directly, since frequent, smaller payments become cheaper ($n_\tau \leq 0$).

Our **fourth prediction** (P4) is that the subsidy reduces payment frequency by more in villages with a mobile money agent than in villages without one. This would establish the flexibility effect; whether it outweighs the level effect, and so whether endogenous payment frequency is on net a further source of complementarity, is a quantitative question we return to in Section 6.3.[14]

2.4 The Effectiveness of Cash Transfers

We now apply the same framework to monthly emergency cash transfers, with the transfer c as the policy instrument. Households withdraw once a month, so a single unit transaction cost is netted directly out of what they receive and the effective transfer is $c - \tau$. Let $W(m)$ denote welfare, increasing in the resources m that a household receives ($W_m > 0$), and normalise $W(0) = 0$. The effect of receiving the transfer is: $\alpha(\tau) \times W(c-\tau)$. Higher transaction costs reduce the program's accessibility, and thus its impact (direct access effect), and also reduce the effective size of the transfer (indirect additional effect). Both channels operate through the cash-out transaction, so they scale outcomes that require converting the balance into cash, but less so for outcomes that

[13] Households could in principle prepay for more than a single month, but this is very rare in our data.

[14] For completeness, endogenous payment frequency rescales the two effects in Equation (3) by $-P_s$ rather than one, and with P_τ in place of $\overline{n}$. This leaves their signs unchanged, since $P_s < 0$ and, provided payment frequency is inelastic with respect to τ, $P_\tau > 0$. It also adds a third term:

$$A_{s\tau} \;=\; (-P_s)\Big[\underbrace{\alpha_\tau(\tau)\times(-T_P)}_{\text{access (direct)}} \;+\; \underbrace{\alpha(\tau)\times(-P_\tau\, T_{PP})}_{\text{curvature (indirect)}}\Big] \;+\; \underbrace{\alpha(\tau)\times T_P\, P_{s\tau}}_{\text{payment plan (indirect)}}\,.$$

respond to receipt of the transfer itself. Our **fifth prediction** (P5) is therefore that the effects of the cash transfer are larger where transaction costs are low, and more so for outcomes that require converting the transfer into cash (food security, financial health, and health care access).

2.5 Predictions

The framework yields five predictions, each tested below.

P1. The subsidy's effect on adoption is larger in villages with a mobile money agent than in villages without one (Section 5.2).

P2. Introducing an agent where the subsidy is in place raises adoption by at least the subsidy's differential effect by mobile money access (Section 5.3).

P3. The subsidy's effect is non-linear in distance to the nearest agent, dropping sharply once the nearest agent lies beyond walking distance and then flattening (Section 6.2).

P4. The subsidy reduces payment frequency by more in villages with a mobile money agent than in villages without one (Section 6.3).

P5. The effects of the cash transfer are larger where transaction costs are low, and more so for outcomes that directly require the cash, such as food security, financial health, and health care access (Section 7).

3 Data and Summary Statistics

We combine four data sources: transaction-level records from a solar provider covering 52,878 households in 2,212 villages from November 2017 to April 2021; La Poste's subsidy records, which show whether each payment was topped up; the national registry of roughly 13,000 mobile money agents, with locations and yearly presence, together with the arrival dates of agents posted under the *Réseau Granulaire* expansion; and a transport-cost survey we conducted of 123 moto-taxi routes out of 34 villages. Together, these data link every payment to its village, to whether the subsidy reached it, and to the distance to the nearest agent at the time. Section 7 describes the Novissi evaluation data, and Table 1 reports summary statistics.

3.1 Solar Home System Data

We obtain household-level administrative data from one of the two largest providers of off-grid solar home systems (SHS) in Togo at that time. The data include information on customer applications and installation dates, as well as household-level registration details such as basic characteristics and previous energy sources. In addition, we observe transaction-level records on customer payments and SHS usage, which allow us to track repayment behaviour and patterns of electricity consumption over time.

The company data covers 52,878 households across 2,212 villages from November 2017 up to April 2021. During the study period, there were two major off-grid solar companies in Togo, which eventually expanded to five in 2020. Given the similar products and pricing structures, we believe the data are reasonably representative of the SHS population in Togo.

Table 1 reports summary statistics. Panel A describes the full sample of 52,878 households. Adopters are overwhelmingly rural smallholders: 71% report agriculture as their main income source, and 92% previously relied on traditional lighting—battery lanterns, flashlights, or kerosene lamps and candles—while fewer than 2% previously used a modern energy source (grid, generator, or solar), so the system is the first modern source of electricity for virtually all customers. The average customer is 39 years old, and only 3% of registered applicants are women.

Panel A also details the bundle choices customers made at registration. They chose between three options: (i) the basic kit (a small solar panel with three light bulbs and a phone charger; 49%), (ii) the intermediate kit (basic kit plus a radio; 20%), and (iii) the premium kit (intermediary kit plus a TV; 31%). Conceptually, we consider that households first decide whether or not to adopt the solar panel at the proposed basic kit's price. After making this decision, they add additional items at their convenience like a radio or TV, increasing the overall price.

From the transaction-level data, we construct several variables to track electricity usage over time. We measure the number of payments made each month, the average payment size, and how these payments translated into electricity consumption, as indicated by the usage share. A 100% usage share means the customer purchased electricity for every day of the month. Panel B shows that in a typical month customers make at least one payment 61% of the time and have a usage rate of 71%. Notably, only 23.6% of households make multiple payments in a month; the rest make just a single payment, suggesting that many buy electricity in bulk.

Panel C aggregates the adoption data to the village level: 18% of villages record at least one new monthly customer application, with an average of 0.52 monthly applications.

3.2 Subsidy Data

We obtain administrative data on the SHS subsidy from La Poste, which operates the national CIZO platform on behalf of the Government of Togo. The dataset contains transaction-level records for all subsidy disbursements (top-ups) processed during our study period, allowing us to observe, for each SHS payment, whether a subsidy was actually credited. On average, around 80% of SHS payments made by customers received the intended top-up, indicating that the CIZO system successfully channelled subsidies for the vast majority of transactions. At the same time, this implies that roughly 20% of payments were not subsidised in practice; as a result, effective subsidy outreach was somewhat more limited (see Appendix Figure A1).

3.3 Mobile Money Agent Network Data

We obtain administrative data from the Ministry of Public Service Efficiency and Digital Transformation, which cover the full universe of approximately 13,000 mobile money agents in Togo that are part of the Togocel and MOOV networks. The dataset reports the longitude and latitude of every registered agent, together with information on their presence over time, up to 2022, at a yearly frequency. These data allow us to construct time-varying village-level measures of access to mobile money agents.

Throughout the paper, we define a village as having a mobile money agent—or easy access to one—if at least one agent is located within a 3 km radius of the village centre. We use 3 km because it is a plausibly walkable distance and is conservative, given that we lack village boundary data and households can live quite far from the village centre. Moreover, villages are sometimes located close to one another, so this radius-based measure performs well in practice.[15] In January 2019—prior to the subsidy rollout—only 602 villages (27% of our sample) had a mobile money agent, while the remaining 1,610 villages had limited access with an average distance of 8.5km (Panel D of Appendix Table A5). Panel A of Figure 1 shows the spatial distribution of villages by agent access and reveals substantial variation across the country.

3.4 Mobile Money Agent Policy Expansion

We obtain administrative data on the large-scale mobile money agent expansion program implemented by La Poste and the Togolese government (Section 1.5). These data cover all La Poste mobile money agents recruited under the *Réseau Granulaire* initiative and report the geographic coordinates and an approximate date on which the agents first became operational. We use this information to identify the villages that gained access to a La Poste agent through the program and to construct village-level timelines of agent arrival for our empirical analysis. Consistent with our definition of a village having a mobile money agent, we say that the policy introduces an agent to a village when a new agent is posted within 3 km of the village centre.

3.5 Transport Costs Data

Rural residents, including SHS customers, frequently rely on motorbike taxis to reach neighbouring villages when walking is too time-consuming. This is often the primary mode of reaching a mobile money agent located in another village. To quantify the costs that beneficiaries face when visiting a mobile money agent to pay for electricity, we hand-collected route-level transport data from moto-taxi drivers in 34 rural villages across all five regions of Togo. For each origin village, we recorded the one-way fare, distance, and travel time to nearby destination villages, yielding 123 origin–destination routes.[16] The data are consistent with a substantial fixed cost in motorized travel: the linear fit between fare cost and distance in Appendix Figure A7 has an intercept of CFA 306 and a comparatively modest slope of CFA 42 per kilometer. This implies a discrete increase in travel costs once a moto-taxi becomes necessary. Table A1 reports summary statistics for these routes. The median one-way moto-taxi fare is 580 CFA for a median distance of 7.4 kilometers, with a median cost of approximately 83 CFA per kilometer. Travel times are also substantial: the median one-way trip takes 20 minutes by moto-taxi and 80 minutes on foot.

[15] In our analysis, we also show that the results are robust to alternative distance thresholds.

[16] Since the survey was conducted in 2023, we deflate reported prices to 2019, the period of analysis.

3.6 Control Variables

We include three main controls. First, we measure village-level wealth across Togo by using data from Meta's Relative Wealth Index, a global geospatial database developed by Chi et al. (2022). This index combines household survey data with non-traditional sources like satellite imagery, cellular network data, topographic maps, and privacy-protected Facebook connectivity data. Machine learning algorithms predict relative wealth at a fine resolution of 2.4 km². Importantly, this data captures *relative* wealth within each country, enabling us to compare wealth across villages in Togo. Since not every village directly corresponds to a wealth cell, we calculate the wealth score for each village using the nearest cell. Second, we measure village-level population density—defined using a 3km radius around the village coordinates—using gridded data from WorldPop.[17]Third, we measure the village-level distance to the nearest market using data on the universe of markets for Togo in order to capture market proximity and access.

4 Empirical Specifications

In this section, we lay out the core empirical specifications and identification strategies for the subsidy and mobile money agent policies. Our primary analysis is conducted at the village level, which is the smallest geographic unit available. For both policies, we use an event-study design that exploits the staggered rollout of the interventions over time and across locations. We estimate dynamic effects using the interaction-weighted (IW) estimator of Sun and Abraham (2021) as our main specification, which is designed for settings with staggered treatment timing and is robust to treatment effect heterogeneity across cohorts and cross-lag contamination. We adopt this approach in place of a standard two-way fixed effects (TWFE) model, which can produce misleading estimates under staggered rollouts (Borusyak, Jaravel and Spiess, 2021; De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021). Nevertheless, we also report results from alternative estimation strategies for our main outcomes as a robustness check. Throughout we use the "last treated" group as the control group, as recommended by Sun and Abraham (2021).

Our identification strategy for both policies relies on two key assumptions. First, the parallel trends assumption, which requires that, in the absence of the policy, the difference in outcomes between treated and control villages would have evolved similarly over time (Angrist and Pischke, 2008). Second, we assume that there was no anticipatory behaviour. We provide a partial test of these assumptions by evaluating pre-trends, using the same number of pre-treatment periods as post-treatment periods.

[17] We use WorldPop population density estimates for Togo, which provide the number of people per square kilometer at 30 arc-second resolution (approximately 1 km at the equator). The data are produced in WGS84 using a Random Forest-based dasymetric redistribution approach and are adjusted to match official United Nations population estimates from the 2019 Revision of the World Population Prospects.

4.1 Event-Study Analysis of the Subsidy Policy

The subsidy policy was announced in February 2019 and rolled out over six months, reaching different groups of districts at three different times (see Figure 1). This staggered rollout provides natural variation to evaluate the initial impact of the policy. The timeline for the phased rollout is as follows:

- *Phase 1*: March 2019: Subsidy begins in 11 districts (826 villages).
- *Phase 2*: May 2019: Subsidy begins in an additional 13 districts (677 villages).
- *Phase 3*: July 2019: Subsidy begins in the final 12 districts (709 villages).

The Government of Togo first launched the policy in districts with the lowest electrification rates (0–10%), then extended it to districts with 11–20%, and finally to those with 21–40%, based on 2016 data. Appendix Table A2 shows increasing baseline electrification rates across the three district groups according to the 2016 government figures.[18] By 2019, however, the 2016 data were outdated, and more arbitrary considerations influenced the timing of rollout. In particular, the national coordinator sometimes delayed the subsidy in districts that were thought to have better grid access by 2019.

Although the timing of the district-level rollout was not random—since districts were grouped by baseline electrification rates—these differences in overall access need not translate into systematic differences among off-grid rural households, which are relatively homogeneous across Togo, a small country. Appendix Table A3 reports baseline characteristics (up to February 2019) by subsidy rollout phase for pre-existing customers and villages. Most household characteristics (Panel A) – including income sources and previous energy use – and payment behaviour (Panel B) are not statistically different across waves. The waves differ only along a few, quantitatively small dimensions: customers in later-treated districts are about 2-3 years older, slightly more likely to be female applicants, more likely to hold premium systems, and have marginally higher usage shares (by two percentage points or half a day per month).

At the village-level, adoption rates at both the extensive and intensive margins are similar across waves (Panel C). However, consistent with the district targetting, later-treated districts are in general denser, wealthier, and closer to mobile-money agents (Panel D).[19] Taken together, this suggests that, despite differences in village characteristics, the rural households in which SHS are adopted are relatively similar across rollout phases.

We estimate the following village-level regression:

$$Y_{vdt} = \lambda_t + \theta_v + \sum_{e \in \{1,2\}} \sum_{l=-3, \neq -1}^{3} \beta_e^l \mathbb{1}\{Sub_{vd} = e\} \cdot D_{vdt}^l + \epsilon_{vdt}, \tag{5}$$

where Y_{vdt} denotes an outcome for village v in district d during month t. In our main specification,

[18] Since the data were provided in ranges, we use the midpoint of each range to calculate district averages. All districts in Group 1 are in the 0–10% range, hence there is a 5% mean with no standard deviation.

[19] Importantly, our identification strategy does not require that the waves be comparable in levels.

Y_{vdt} is the number of applications to purchase a SHS in village v in month t. The terms λ_t and θ_v are month and village fixed effects, respectively. The indicators $\mathbb{1}\{Sub_{vd} = e\}$ are dummies for whether village v in district d belongs to subsidy rollout cohort $e \in \{1, 2\}$, corresponding to districts that became eligible in March and May, respectively. Following Sun and Abraham (2021), villages in the last set of districts to become eligible (Group 3, July rollout) serve as the "last treated" control group. The variables $D^l_{vdt} = \mathbb{1}\{t - T^{\text{sub}}_{vd} = l\}$ are relative-time indicators that equal one if month t is l periods away from the month T^{sub}_{vd} in which village v first became eligible for the subsidy. We exclude the pre-period $l = -1$ as is standard practice. We include three post-treatment periods, since all districts were eligible by July 2019 as well as three pre-treatment periods. We cluster standard errors at the district level.

The parameters of interest are the interaction-weighted dynamic effects $\hat{\beta}^l = \sum_{e\in\{1,2\}} \omega^l_e \hat{\beta}^l_e$, where the weights ω^l_e reflect the sample share of cohort e in relative period l. For $l \geq 0$, $\hat{\beta}^l$ captures the impact of the subsidy l months after implementation as a weighted average of cohort-specific treatment effects. To obtain a "static" average treatment effect and its associated standard error, we follow Sun and Abraham (2021) and average the post-treatment dynamic effects:

$$\hat{\beta}^{avg} = \frac{\sum_{l=0}^{3} \hat{\beta}^l}{4}. \tag{6}$$

4.2 Event Study Analysis of the Mobile Money Agent Policy

We leverage the government-led expansion of mobile money agents, described in Section 1.5, to estimate the causal effect of improved agent access using an event-study design. As shown in Appendix Figure A2, 100 villages with no prior mobile money agent gained access to their first agent at different times between September 2019 and May 2021 under this policy. In total, 358 villages lie within 3 km of one of the newly deployed agents, but 258 of these already had a mobile money agent nearby before the program and therefore do not experience a first arrival.[20] The expansion proceeded continuously from September 2019 until it was abruptly halted in March 2020 with the onset of COVID, and then resumed in January 2021.

Our event-study design exploits both the staggered timing of agent arrivals during the first phase (September 2019–March 2020) and the COVID-induced pause, with villages that eventually received an agent in the second phase (January 2021–May 2021) serving as the eventually-treated control group. The estimating panel runs from July 2018 to December 2020, which provides seven bimonthly pre-periods for the earliest treatment cohort and follows treated villages through the end of 2020. As additional analysis, we also estimate a specification that uses only villages receiving an agent in January 2021 as the control group. Since the recorded agent arrival dates are approximate rather than exact, we aggregate the data into 2-month periods and estimate the event-study at this bimonthly frequency. The staggered expansion thus generates six treatment co-

[20]We use annual data on the universe of mobile money agents that were not part of the government-led expansion and require that no agent is observed within 3 km of the village in any year up to 2021 in order to classify it as having no prior nearby agent.

horts defined by the two-month period in which the agent arrives, with the number of villages in parentheses: September 2019 (44), November 2019 (5), January 2020 (13), March 2020 (26), January 2021 (6) and May 2021 (6).

Appendix Table A4 reports baseline characteristics for the first and second phases in the expansion sample. Household characteristics (Panel A) are similar across the two groups: age, applicant gender, income sources, previous energy use, and kit choices do not differ meaningfully.[21] There are some differences in payment behaviour, with second-phase customers more likely to make multiple payments, but the usage shares are similar (Panel B). At the village level, the adoption rate (Panel C) and most village characteristics (Panel D) are quite similar across the two groups, and by construction no sample village had an agent within 3 km at baseline.

We estimate the following village-level regression:

$$Y_{vt} = \delta_t + \phi_v + \sum_{e \in \{1,2,3,4\}} \sum_{l=-7, \neq -1}^{7} \tau_e^l \mathbb{1}\{Agent_v = e\} \cdot A_{vt}^l + \varepsilon_{vt}, \tag{7}$$

where Y_{vt} denotes an outcome for village v in bimonth t. In our main specification, Y_{vt} is the number of monthly SHS applications in village v in bimonth t, where we take an average across both months so that our outcomes are comparable to the monthly analysis. The terms δ_t and ϕ_v are time and village fixed effects, respectively. The indicators $\mathbb{1}\{Agent_v = e\}$ are dummies for whether village v belongs to treatment cohort $e \in \{1, 2, 3, 4\}$, where a cohort is defined by the bimonth in which the village first received a mobile money agent. The variables $A_{vt}^l = \mathbb{1}\{t - T_v^{\text{agent}} = l\}$ are relative-time indicators that equal one if bimonth t is l periods away from the bimonth T_v^{agent} in which village v first became treated. As before, we omit the pre-period $l = -1$ and include seven pre-treatment and eight post-treatment periods in the event window. We cluster standard errors at the village level.

The parameters of interest are the interaction-weighted dynamic effects $\hat{\tau}^l = \sum_{e=1}^{4} \omega_e^l \hat{\tau}_e^l$, where the weights ω_e^l denote the sample share of cohort e observed in relative period l. For $l \geq 0$, $\hat{\tau}^l$ captures the impact of gaining access to a mobile money agent l bimonths after arrival as a weighted average of cohort-specific treatment effects. To obtain a "static" average treatment effect, we average the post-treatment dynamic effects:

$$\hat{\tau}^{avg} = \frac{\sum_{l=0}^{7} \hat{\tau}^l}{8}. \tag{8}$$

4.3 Customer-Level Event Study Analysis

We also evaluate the effects of these policies on customer-level outcomes for pre-existing customers. We implement the same regression specifications, Equations (5) and (7), but rather than using village fixed effects, we use customer fixed effects. We restrict the sample in two ways. First,

[21] While the equality tests for previous energy source yield p-values below 0.05, the magnitudes are small and not economically meaningful.

we include only customers who had already adopted a solar home system in the period immediately before the policy change (February 2019 for the subsidy policy and July 2019 for the mobile money agent expansion). Therefore, while the sample is unbalanced for some of the pre-policy periods, it is balanced in the post-policy periods. Second, we focus on relatively active customers, defined as those who used their system for at least half of the days in that pre-policy period. This restriction ensures that our results are not driven by customers who had effectively disengaged from the product prior to the policy change, and who are thus unlikely to be affected by the policy.

5 Adoption Effects of the Subsidy and of Agent Arrival

We proceed in four steps. First, we document the large and positive impact of the subsidy on SHS adoption. Second, we examine heterogeneity by pre-subsidy access to mobile money agents and show that the subsidy's effect is nearly three times larger in villages with an agent (Prediction P1). This heterogeneity is suggestive, as it may reflect other unobserved differences across villages. Third, we therefore turn to the mobile money agent expansion policy and estimate the causal effect of the arrival of an agent, which raises adoption by at least the subsidy's differential effect by agent access (Prediction P2). Fourth, we draw out what these estimates jointly imply for the complementarity between the two policies.

5.1 Subsidy Effects on Adoption

We exploit the phased rollout of the subsidy across districts to estimate its causal impact on solar home system adoption, measured using the number of new monthly applications. We begin by plotting the evolution of solar home system adoption separately for each of the three district groups in Appendix Figure A5, covering the two-year period from September 2018 to August 2020. Four features stand out. First, adoption increases sharply and immediately when the subsidy is introduced in a given group. Second, there is no clear evidence that households in any of the groups anticipated the subsidy, as already shown in Lang (2025*a*). Third, outside the rollout windows, the three groups move closely together, displaying similar peaks and troughs over time. Fourth, there is a clear level shift in adoption when comparing the pre-subsidy period, when adoption is relatively flat, to the post-subsidy period, when all districts are treated and average adoption is considerably higher. Taken together, these patterns suggest that the identifying assumptions underlying our event-study analysis are plausible.

We then plot the event-study coefficients and show a sharp and immediate increase in applications following the introduction of the subsidy, with the effect continuing to rise over the next two months before stabilising (Appendix Figure A6 and Equation (5)). There is no evidence of differential pre-trends prior to the policy change. We report the average effects in Panel A of Table 2. The estimates show that the subsidy increases adoption by 0.47 new applications per village-month on average, more than doubling adoption relative to the pre-policy control mean of 0.40.[22]

5.2 Subsidy Effects by Access to Mobile Money Agents

To provide evidence that transaction costs shape both adoption and policy effectiveness, we examine whether the effect of the subsidy on adoption differs by pre-subsidy access to mobile money agents. As described in Section 3, households must pay for electricity through a mobile money account, while rural customers still transact primarily in cash. Mobile money agents therefore play a key role in converting cash into digital balances and facilitating electricity payments. We classify a village as having a mobile money agent if there is an agent located within 3 km of the village centre in December 2018, three months before the subsidy rollout.

Figure 2 plots the event-study coefficients separately for villages with and without a mobile money agent. The subsidy effect is substantially larger in villages with an agent. Two to three months after the subsidy is introduced, adoption rises by roughly 1.3 to 1.5 additional applications per village-month in villages with a mobile money agent, compared with only around 0.3 to 0.6 additional applications in villages without one.[23] Reassuringly, for both groups, there is no evidence of differential pre-trends prior to the subsidy rollout.

Table 2 reports the corresponding average effects separately by baseline access to mobile money agents. In Panel B, for villages with a mobile money agent, the subsidy increases adoption by 0.96 applications per village-month on average. In Panel C, for villages without a mobile money agent, the corresponding effect is only 0.35. The average treatment effect is therefore around 2.7 times larger in villages with an agent, indicating that the subsidy is substantially more effective where households have easier access to mobile money services. The difference across the two groups of 0.61 is also statistically significant: we reject equality of the effects at the 5% level. This provides evidence for Prediction P1.

5.2.1 Robustness

Wealth, Population Density, and Market Access. Access to a mobile money agent could reflect a range of other potential confounders. In particular, agents may be more likely to operate in villages that are wealthier, more densely populated, or closer to markets, each of which could independently affect SHS adoption. Wealthier villages may contain more households able to afford a system, denser villages may facilitate information diffusion, and villages closer to markets may face lower search and transaction costs more generally. Panel D of Appendix Table A5 shows that these differences are indeed present: villages with access to an agent are wealthier, more densely populated, and closer to markets. In columns (2) to (4) of Table 2, we therefore augment our main empirical specification by interacting time-invariant measures of village wealth, population

[22] These findings are in line with the contemporaneous estimates of Lang (2025*a*), whose identifying variation is the subsidy's rollout across districts. We study the village-level variation in agent access.

[23] The standard errors are larger for villages with a mobile money agent because only around a quarter of villages in our sample had an agent.

density, and distance to markets with time fixed effects. The estimated average effect of the subsidy is virtually unchanged, and, importantly, the heterogeneous effects in Panels B and C are not attenuated. These controls, however, can only account for observable differences: villages with and without an agent may still differ along unobserved dimensions, so this heterogeneity remains descriptive rather than causal.

Duration of Agent Presence. One way to partially address this concern exploits the history of the agent network. Villages that gained an agent only shortly before the subsidy were, until recently, themselves villages without one. If unobserved village characteristics rather than agent presence drove the larger subsidy effect, their subsidy effect should resemble that in unconnected villages. Because our administrative data record agent locations annually from before the 2013 launch of mobile money in Togo, Appendix Table A6 orders villages by when their agent arrived relative to the subsidy.[24] Among the 62 villages connected only during 2018, the year preceding the subsidy, the effect is 0.68 (column 2): smaller than in villages served since before 2018 (column 1), but still roughly double the 0.35 in villages without an agent (Panel C of Table 2). A second comparison holds selection fixed. Villages with no agent during the rollout that gained one during 2020 had a subsidy effect of 0.32 (column 3), essentially identical to the 0.32 in villages connected only later or not at all (column 4), so being the kind of village that operators eventually select does not itself generate a larger subsidy effect. The doubling between columns (2) and (3) therefore reflects the agent having arrived in time, though agent placement is not randomly assigned and these comparisons remain suggestive rather than conclusive.

Cutoff Distance. Our results are also robust to the distance threshold used to define mobile money access. The main specification uses 3 km, which we consider conservative. Appendix Table A7 shows that varying the cutoff—down to 2 km, or up to 4 km—yields quantitatively similar heterogeneous subsidy effects, with differences that are statistically significant at the 5% level at the 2 and 3 km thresholds, but only at the 10% level at 4 km, consistent with this distance starting to become relatively far to walk. Point estimates are in fact larger under narrower definitions of proximity (see also Appendix Figure A8).

Alternative Estimation Strategies. Appendix Table A11 re-estimates the subsidy effect using the estimators of Callaway and Sant'Anna (2021), Borusyak, Jaravel and Spiess (2021) and De Chaisemartin and d'Haultfoeuille (2020) alongside our main interaction-weighted specification (Sun and Abraham, 2021), holding the sample, fixed effects, clustering and last-treated comparison group fixed so that only the estimator changes. The effect ranges from 0.81 to 0.96 applications per village-month in villages with an agent within 3 km at baseline (Panel A) and from 0.33 to 0.41 in villages without such access (Panel B), all significant at the 1% level, so the subsidy remains two to three times as effective where an agent is nearby under every estimator.

[24]We exclude the 136 villages whose agent arrived during 2019, since the annual agent data cannot establish whether the agents had arrived before or after the subsidy rollout.

5.3 Mobile Money Agent Effects on Adoption

To move beyond descriptive heterogeneity, we next exploit the mobile money agent expansion policy to estimate the causal effect of improved access to a mobile money agent on SHS adoption. By the time this policy was launched, the SHS subsidy had already been rolled out nationwide, so the estimates isolate how the arrival of a mobile money agent affects adoption conditional on the subsidy being in place. If the heterogeneous subsidy effects documented above reflect a causal role for mobile money access, then the arrival of an agent should raise adoption in previously underserved villages and reduce the adoption gap relative to villages that already had an agent.

We exploit the staggered rollout of the policy across villages to estimate these effects using the number of new monthly SHS applications. Our event-study design draws on both the staggered arrival of agents during the first phase of the rollout (September 2019–March 2020) and the COVID-induced pause, with villages that were only treated later, between January and May 2021, serving as the eventually treated control group.

We plot the event-study coefficients in Figure 3. Although the estimates are relatively imprecise, given the much smaller sample of treated villages in this analysis (100, compared with 2,212 in the subsidy analysis), the figure shows a positive effect following the arrival of a mobile money agent.[25] There is also no evidence of differential pre-trends prior to agent arrival.

In Panel A of Table 3, we find that the arrival of a mobile money agent increases SHS adoption by 0.64 applications per village-month on average, with the estimate statistically significant at the 1% level. Because our last treated group combines villages that received an agent in January 2021 with those treated only in May 2021, this specification pools cohorts that are separated by four months. In Panel B, we therefore use only the January 2021 cohort as the eventually treated control group and exclude the May 2021 cohort. Under this alternative specification, the estimated effect rises to 0.67 applications per village-month and remains significant at the 5% level. Both estimates indicate that households respond to the arrival of a mobile money agent by adopting a solar home system. Moreover, these magnitudes slightly exceed the gap in subsidy effects between villages with and without a mobile money agent (0.61 applications per village-month), our Prediction P2. These results thus lend further support to a causal interpretation of the subsidy's heterogeneity.

5.3.1 Robustness

Wealth, Population Density, and Market Access. We perform a similar set of robustness checks as in Section 5.2 by flexibly controlling for baseline village wealth, population density, and distance to markets. In columns (2) to (4) of Table 3, we augment our main specification by interacting these time-invariant village characteristics with time fixed effects. The estimated effects remain virtually unchanged, with no meaningful attenuation in the point estimates. The estimates remain statistically significant at least at the 5% level across all columns.

[25] Our analysis is conducted at the two-month level, but we divide the adoption outcome by two so that it is expressed in village-month applications and is therefore directly comparable to the subsidy effects.

Cutoff Distance. Our results are largely robust to the distance threshold used to define access to a mobile money agent. Appendix Table A8 shows that tightening the cutoff from 3km to 2km leaves the effect positive, statistically significant, and of similar magnitude (0.57); widening it to 4km, however, yields a smaller estimate, a similar pattern to that of the subsidy and consistent with 4 km lying beyond typical walking distance.[26]

Representativeness of the Mobile Money Agent Villages. Villages selected for the mobile money agent expansion may differ systematically from other villages without an agent, which would complicate our comparison with the subsidy effects. The event-study sample comprises only 100 villages—around 6% of villages without a mobile money agent at the time of the subsidy launch—raising the question of whether the effects of agent arrival in this selected subsample generalise to the broader population of unconnected villages. To assess this, we exploit the fact that the subsidy rollout provides independent variation even within this subsample, and estimate the effect of the subsidy on adoption using these 100 villages alone. Appendix Table A9 shows an average effect of 0.33 applications per village-month. The estimate is imprecise, as expected with only 100 villages (s.e. 0.33), but its magnitude is very close to the estimate of 0.35 for all villages without mobile money access (Panel C of Table 2) and stable across specifications that flexibly control for wealth, population density, and market proximity. That expansion villages exhibit subsidy responses that are virtually identical to those of the broader population of villages without an agent suggests that they are not systematically different along dimensions relevant to subsidy adoption. This, in turn, strengthens the interpretation that the heterogeneous subsidy effects documented in Section 5.2 reflect the causal role of mobile money infrastructure, rather than pre-existing differences across village types.

COVID Effects. The COVID-19 pandemic is unlikely to confound our subsidy estimates, since the rollout was completed in July 2019, well before its onset. It could, however, confound the agent expansion: as households avoided travel and mobile money became a channel for government cash transfers, proximity to an agent may have raised adoption through a COVID-specific mechanism unrelated to the payment friction we study. The GiveDirectly–Novissi transfers (Section 7) are one example, although they were delivered only in November 2020, late in our sample.

Appendix Figure A5 shows that aggregate adoption dipped only modestly in mid-2020, remained well above pre-subsidy levels, and recovered quickly. More directly, Appendix Table A10 re-estimates the agent event study on shortened samples. Restricting the data to May 2020 (Panel A) leaves the adoption effect virtually unchanged—0.63 applications per village-month against 0.64 in the main specification—and significant at the 5% level. Ending the sample in March 2020 (Panel B), at the onset of the pandemic, leaves few post-treatment periods; the point estimate falls modestly to 0.52 but remains significant at the 5% level in the main specification, and at the 10% level with controls. Both panels are robust to controls for wealth, population density, and

[26]Changing the threshold alters both the sample of ever-treated villages and the phase to which a village is assigned: because a village's nearest agent within a wider radius may have arrived at a different date, moving the cutoff can shift a village between the first-phase cohorts (September 2019–March 2020), which serve as our treated group, and the second-phase cohort (January–May 2021), which serves as our control group.

market proximity. The post-2020 data thus add precision, but we find no evidence that the pandemic drives the estimated effects of agent expansion.

Alternative Estimation Strategies. As with the subsidy, we re-estimate the agent arrival effect under each of the alternative estimators. Panel C of Appendix Table A11 reports point estimates of a similar magnitude throughout, ranging from 0.48 to 0.64 applications per village-month.

5.4 Complementarity between the Subsidy Policy and Payment Infrastructure

The agent-arrival results provide additional evidence consistent with the subsidy and access to payment infrastructure acting as complements: introducing both policies raises adoption by more than the sum of their individual effects. On its own, the subsidy heterogeneity is suggestive but not decisive—the subsidy raises adoption around three times as much in villages with access to a mobile money agent, yet agent placement is not random. The key observation is that once the subsidy is nationwide, the arrival of an agent lifts adoption by an amount that slightly exceeds this differential subsidy effect by mobile money agent access. We now explain precisely why this pattern supports complementarity, and where the argument reaches its limits.

An ideal test would cross-randomise the two policies, generating four cells—villages with neither policy, each policy alone, and both—and four margins: each policy introduced with and without the other. Complementarity is the difference between the joint effect and the sum of the two individual effects, and mechanically this difference equals the heterogeneous effect of the subsidy by agent access or, equivalently, the heterogeneous effect of an agent by subsidy status: the heterogeneity is the complementarity. Our natural experiments identify three of the four margins. The subsidy adds 0.96 applications per village-month in villages with agent access and 0.35 in villages without (Panels B and C of Table 2), and agent arrival under the subsidy adds 0.64 (Table 3); the effect of an agent in the absence of the subsidy is unobserved.[27]

Each policy therefore offers a different estimate for the complementarity term. The subsidy-side heterogeneity is fully observed: the subsidy effect is 0.61 larger in villages with agent access than without. The agent-side heterogeneity cannot be formed, but the observed margin still carries information. An agent's arrival is unlikely to reduce adoption, so the effect of an agent alone is plausibly non-negative, and the agent-side complementarity can therefore be at most the arrival effect itself when there is a subsidy: 0.64. However, a standalone effect above 0.64 would eliminate the complementarity altogether. Can we bound this effect further beyond zero? The pre-subsidy adoption gap between villages with and without agent access is 0.20 (control-group means in Panels B and C of Table 2), and this descriptive gap likely overstates the causal standalone effect insofar as it partly reflects selection. Even treating all of it as causal leaves a complementarity term

[27] Combining margins across the two designs treats them as comparable and additive in levels. The two rollouts began only months apart, and the subsidy effect within the eventual expansion villages (0.33; Appendix Table A9) is nearly identical to that in all villages without an agent (0.35).

of 0.45.[28]

The key takeaway is what the arrival effect rules out. Had it been close to zero, any complementarity would have required the effect of an agent alone to be negative. This is difficult to rationalize and the reasonable conclusion would have been that the 0.61 subsidy gap reflected unobserved village characteristics correlated with agent presence. Instead, the arrival effect of 0.64 slightly exceeds the 0.61 gap, consistent with complementarity with a small standalone effect.

6 The Payment-Infrastructure Mechanism

The previous section established that the subsidy's effect on adoption is significantly larger where an agent is present and that the arrival of an agent raises adoption by about as much as this differential. In this section, we present additional evidence that payment infrastructure is the core mechanism behind these results. We first quantify the transaction costs of reaching an agent and benchmark them against the price of electricity (Section 6.1). We then show that the subsidy's effect follows the non-linear shape of these costs in distance (Section 6.2, Prediction P3), and that customers respond to the subsidy by consolidating payments, as they would if each trip were costly (Section 6.3, Prediction P4). Finally, we consider alternative explanations for the complementarity and show that the evidence weighs against each of them (Section 6.4).

6.1 The Magnitude of Transaction Costs

Rural households in Togo transact primarily in cash, yet PAYGO electricity payments must be made via mobile money, requiring a trip to a mobile money agent to convert cash into a digital balance. The median household in our sample is located approximately 5.7 km from the nearest agent, making each payment round trip costly. Using the transport cost data we collected across 123 origin-destination village pairs in rural Togo (see Section 3.5), we investigate the role of transaction costs as a mechanism for the complementarity documented above.

We approximate the transaction costs customers faced when paying for electricity in January and February 2019, before the subsidy was introduced. Consistent with our definition throughout the paper, which treats 3 kilometers as the walking threshold, we set monetary transaction costs to zero when the nearest mobile money agent is within 3 kilometers and apply the moto-taxi cost when the distance exceeds this threshold. For each customer i and month t, we compute the monthly transaction cost as the number of payments made in that month multiplied by the customer's distance to the nearest mobile money agent, the median moto-taxi fare per kilometer in the transport-cost survey, and a round-trip factor of two.[29] We then average the January and February values to obtain a customer-level measure of baseline monthly transaction costs.

[28] Alternatively, on the subsidy side, we could instead use a more conservative subsidy effect of 0.68 based on villages that gained an agent only during 2018, just before the subsidy (column 2 of Appendix Table A6), set against the 0.32 in villages that gained an agent only during 2020 and so had none during the rollout (column 3), which implies a smaller gap of 0.36.

[29] We restrict this calculation to customers making payments, as transaction costs are otherwise undefined.

These costs are substantial. The median customer incurred an estimated CFA 927 per month in travel costs to reach a mobile money agent, rising to CFA 1,872 at the 75th percentile. Relative to the monthly price of electricity access, this amounts to 19% for the median customer and 39% at the 75th percentile. These magnitudes underscore the severity of the payment-infrastructure friction and help explain why subsidies and mobile money access are complements. A subsidy that reduces the electricity bill by up to 44% may generate limited additional adoption if households must still incur large recurring travel costs to reach an agent and make a payment. By the same logic, reducing the distance to mobile money agents represents a large effective price reduction: given the baseline transaction costs we estimate, the agent expansion likely lowered the full cost of electricity access substantially, consistent with its large effects on SHS adoption.

These estimates are necessarily approximate, and there are reasons to think they could overstate or understate the true burden. On the one hand, our calculation may overestimate transaction costs if customers combine payment trips with other errands, so the marginal cost of reaching an agent is lower than the full fare (see Section 6.4.4). Some households may also have bicycles or walk longer distances, reducing or eliminating the moto-taxi cost. These calculations also assume that customers must visit an agent every time they make a payment. While this is likely true for a majority of customers, those who already have money loaded into their mobile money account could in theory pay for electricity without making this trip.

On the other hand, several factors suggest the estimates are conservative. First, we measure distance from the village centre, while rural households often live on dispersed farms further out. Second, the fare captures only the monetary cost and excludes the opportunity cost of travel time, which could otherwise be devoted to farming or household production. Third, it does not account for repeat trips required when agents are absent or mobile signal fails—both common in rural settings. Fourth, it abstracts from the cost of arranging transport itself: in remote villages, moto-taxis are not always available, and locating a driver takes time.

Whichever direction dominates, the burden remains large. Even at half our measure, whether because trips are combined with errands, some households walk, or one loaded balance covers two payments, and with every omitted cost set to zero, the median customer would still spend about CFA 460 a month reaching an agent, roughly 10% of the monthly price, and the 75th-percentile customer about CFA 940, roughly 20%. Against a bill cut by the largest subsidy rate of 44%, these shares rise to roughly 17% and 35%.

6.2 Non-Linear Subsidy Effects by Distance

Transportation costs τ are non-linear with a step-function cost structure: a discrete jump at the walking threshold, followed by a relatively flatter cost schedule beyond it based on moto-taxi rates per kilometer. Moreover, at this walking threshold, we might also expect accessibility $\alpha(\tau)$ to worsen, precisely because beyond this distance, there are additional steps needed to make each payment. This makes payments less accessible beyond the monetary costs alone. The subsidy's effects should thus follow this same non-linearity, our Prediction P3.

To test this prediction, Figure 4 reports heterogeneous subsidy effects separately for four distance bands: villages with a mobile money agent within 0–3 km, and, among villages without an agent, bands of 3–6 km, 6–9 km, and 9+ km. The pattern is a step rather than a gradient. In the 602 villages with an agent within 3 km, the subsidy raises adoption by 0.96 applications per village-month. Beyond that threshold the effect is two to four times smaller and does not decline further with distance: 0.33 in the 3–6 km band (559 villages), 0.22 in the 6–9 km band (450 villages), and 0.44 beyond 9 km (601 villages). Each outer band is individually significant, but none is distinguishable from the others, so the estimates trace no systematic gradient in distance once an agent is out of walking range.[30] This is what the step-function cost structure predicts: crossing the walking threshold imposes a discrete fare and adds steps to every payment, whereas each further kilometer adds only a small per-kilometer charge. Appendix Figure A8 shows broadly consistent, though noisier, patterns using even more granular 1 km distance bands.

6.3 Subsidy Effects on Payment Frequency

Section 2.3.1 derived an intensive-margin prediction: a lower electricity price should lead liquidity-constrained households to make fewer, larger payments—and this bulk-purchasing response should be larger where a nearby agent makes flexible payment schedules feasible. Table 4 tests both parts. Across all villages (Panel A), the subsidy leaves the probability of making any payment unchanged—the relevant margin is not whether customers pay, but how often—while the number of payments per month falls by 0.26 from a pre-policy mean of 1.32 ($p < 0.01$) and the probability of multiple payments falls by 3.0 percentage points. Splitting by baseline agent access (Panels B and C), the reduction is concentrated where an agent is within 3 km: 0.31 versus 0.19 payments (difference significant at 5%), and 4.6 versus 1.5 percentage points for the probability of multiple payments. In the language of Section 2.3.1, the level effect is present everywhere, while the flexibility effect appears in the gap between the two panels—concentrated at the walking-distance threshold, as in Section 6.2, consistent with our Prediction P4.

These estimates pin down the condition under which the intensive-margin cross-partial $P_{s\tau}$ from Section 2.3.1 is positive. discretizing, let $s = 2{,}000$ CFA denote the monthly subsidy, Δn the subsidy-induced change in monthly payments (-0.31 with an agent within 3 km, -0.19 without), τ the per-trip transaction cost in a village without an agent, and $\Delta\tau$ the reduction in that cost when an agent is present:

$$P_{s\tau} \;\approx\; \underbrace{\frac{\Delta n}{s}\bigg|_{\text{agent}}}_{\text{level}} \;+\; \underbrace{\frac{\tau}{\Delta\tau} \times \frac{\Delta n\big|_{\text{no agent}} - \Delta n\big|_{\text{agent}}}{s}}_{\text{flexibility}} \;=\; \frac{-0.31}{2{,}000} \;+\; \frac{\tau}{\Delta\tau} \times \frac{-0.19-(-0.31)}{2{,}000}. \tag{9}$$

The cross-partial is therefore positive if and only if $\Delta\tau/\tau < 0.39$. The two terms pull in opposite

[30] Treating the band estimates as independent, pairwise differences among the three outer bands have t-statistics below 1.2. Each outer band differs from the 0–3 km band by more, although the comparison with the 9+ km band is only marginally significant ($t = 1.86$), reflecting its wider standard error.

directions: the level term works against complementarity, because each avoided trip saves less where trips are already cheap, while the flexibility term works for it, because the consolidation response is larger where an agent is near. On net, the subsidy lowers the total price of electricity (price + transaction costs) by more in villages with an agent whenever agent access reduces the full per-trip cost by less than 39%. This is plausible: travel is only one link in the payment chain, and the costs of queueing, agent liquidity, and network availability remain even when the agent is in the village. We stress that this is a back-of-the-envelope exercise as it treats the two panels as a discrete change in τ holding all else fixed. However, it suggests that endogenous payment frequency reinforces the complementarity documented on the adoption margin.

Lastly, Panel D examines how agent arrival changes the payment behaviour of existing customers. The response is limited; point estimates for the number of payments are small, positive, and not statistically significant.

6.4 Alternative Explanations

We consider five checks to help rule out alternative explanations for the observed complementarity between the subsidy and mobile money agents. First, we examine whether supply-side responses by the solar company could explain our results. Second, we ask whether the complementarity reflects the composition of customers in agent villages rather than payment infrastructure itself. Third, we consider demand-side explanations, whereby a nearby agent raises the value of owning a phone and thereby the demand for home electricity. Fourth, we assess whether households can circumvent transaction costs by coordinating payments through a single traveller. Fifth, we test whether greater awareness of the product or easier subsidy receipt in agent villages accounts for the results. In each case, the evidence weighs against the alternative explanation and in favour of the payment-infrastructure channel.[31]

6.4.1 Supply-Side Constraints

We address the possibility that supply-side factors may have differentially influenced adoption during the subsidy period; e.g., that increased adoption could reflect greater attention from the solar company toward villages eligible for the subsidy with greater mobile money access. We present two arguments against this hypothesis. First, our adoption measure is based on the number of applicants, not connected customers. While supply issues might affect the number of connected customers, leading to installation delays, they are less likely to influence the number of applicants. Our results are consistent across both measures, and the lack of significant discrepan-

[31] Lang and Ly (2026) document higher electricity uptake in sub-Saharan African districts with mobile money access and attribute the effect primarily to income from remittances and transfers, with savings also responding. Those channels operate through the account rather than the agent, and the arrival design holds the account margin fixed. Mobile money was available in expansion villages before any agent arrived, SHS payments are made through mobile money by construction, and account ownership among rural households varies little with distance to the nearest agent (Appendix Figure A4). A household financing its payments through digital transfers would need no trip to an agent and no cash-in. Agent arrival changes the cost of converting cash into an electronic balance.

cies between them suggests that supply-side effects are minimal (Appendix Table A12). Second, the solar company relies on local shops (approximately one shop per district), which are responsible for supplying their respective areas and operate independently. We observe no reduction in supply in control districts during the subsidy rollout, suggesting no reallocation of resources.

6.4.2 Customer Composition

A further concern is that the complementarity reflects differences in customer composition rather than payment infrastructure itself: villages with mobile money agents may simply contain households whose demand is more responsive to a price reduction. We first compare the baseline characteristics of pre-subsidy customers across village types in Appendix Table A5. Age, gender, previous energy sources, and pre-policy payment behaviour exhibit no statistically significant differences between villages with and without access to a mobile money agent. Customers in agent villages are, however, somewhat less likely to earn their income from agriculture and more likely to hold premium kits, consistent with the differences in wealth and population density across village types documented in the same table.

These differences have a testable implication. If composition drove the complementarity, the subsidy effect within any given segment would not differ by agent access, and the village-level gap would arise only from mixing. A village-level payment friction instead dampens the response of every customer in the village, so the amplification should appear within each segment. We therefore disaggregate adoption by kit type (Appendix Table A13) and by income source, age, and gender (Appendix Table A14). The amplification is strikingly uniform. The subsidy effect in agent villages is roughly two-and-a-half to three times the effect in villages without an agent for basic, intermediary, and premium kits alike—though with the smaller per-kit samples, equality is rejected only for premium kits—and is larger in agent villages in all six demographic subgroups, with equality rejected in three of six columns at the 5% level. No single segment drives the complementarity, with differential effects by mobile money agent access even among agricultural households and those with basic kits, the segments over-represented in villages without an agent. This is difficult to reconcile with any composition of observable customer types. Panel C of both tables makes the same point with the arrival of an agent itself: these event studies include village fixed effects. Thus, the village and its pool of potential adopters are held fixed, and adoption nonetheless rises across segments, most strongly for basic kits, whose small payments are most exposed to per-trip costs. The amplification thus appears within every customer segment and within villages as their payment infrastructure changes, consistent with a village-level friction rather than the customers those villages happen to contain.

6.4.3 Demand and Usage

A phone and a solar home system are complementary goods: households need a phone to make payments for their SHS, and electricity is in turn needed to keep the phone charged. While phones could be charged elsewhere, doing so at home is most convenient. By extension, a mobile money agent in the village could increase the value of owning a phone by expanding access to digital services, thereby shifting demand for a solar home system upward. This would provide an alternative explanation for the complementarity between the subsidy and mobile money agents.

We explore this potential mechanism by examining electricity usage of the solar home system among existing customers in Appendix Table A15. The flexible payment structure allows households to miss payments and thus be locked out of their system, preventing the solar panel from charging the battery and, in turn, from powering lamps or charging devices. Among our sample of active customers, pre-policy usage is high at around 95%, which is only partly mechanical: while our inclusion criterion requires that customers have electricity for at least half the month in the pre-policy period, the pre-policy mean covers all pre-periods. We find no meaningful effect of the subsidy on usage in villages with or without access to a mobile money agent across all three outcomes capturing the extensive and intensive margins of use. Similarly, the arrival of a mobile money agent has no effect on usage. If the presence of an agent increased the value of owning a phone, for which electricity is a complementary good, we would expect usage to rise. Instead, the point estimates are again close to zero and not statistically significant, although the joint pre-trend tests reject in this specification, so we treat the arrival estimates on usage as suggestive.

Furthermore, we observe hourly wattage use, which allows us to confirm these patterns more directly. On average, households consume about 76 Wh per day, roughly consistent with a small solar panel powering a few LED lamps and a phone. Usage is fairly stable across days (median coefficient of variation ≈ 0.30), and consumption is heavily concentrated in the evening and night hours, which together account for about 62% of total use, consistent with lighting as the primary application. Patterns are broadly similar across households, with only a small minority exhibiting irregular or daytime-intensive use suggestive of business activity. Although customers in areas with a mobile money agent consume slightly more watt-hours on average, this difference is quantitatively modest. These patterns help to rule out the possibility that the complementarity between the subsidy and the mobile money agent is driven by differential demand for electricity, which would be reflected in usage of the solar home system.

6.4.4 Coordination Frictions

In the presence of transaction costs, customers could, in theory, coordinate to mitigate these costs by pooling resources and delegating payments to a subset of individuals. For example, customers might ask one person in the village—e.g., someone who already travels to a market with a mobile money agent—to take their cash and complete the cash-in on their behalf. Such arrangements would generate substantial savings for other customers but require coordination. If widespread, this type of coordination would effectively neutralise transaction costs as a barrier to adoption.

We leverage the fact that all payments are made by phone, so we can observe both the phone number used for the payment and the solar home system account into which the payment was made. We then match the payment phone number to the set of phone numbers recorded for each customer (each customer has at least one phone number on record, and up to three). This allows us to identify cases in which a customer paid for their SHS using a third party's phone, and thus to measure the extent of delegated payments and coordination.[32]

Payment coordination is, if anything, a force that should attenuate our main mechanism rather than explain it. If households in remote villages can routinely delegate trips to a subset of travellers, then the absence of a nearby agent should matter less for both adoption and payment behaviour. We do find baseline evidence of delegated payments. In villages with access to a mobile money agent, roughly two-fifths of customers regularly rely on third-party payments, and a similar share of payments are made through third parties, indicating that payment frictions are reduced but not eliminated. The corresponding averages are slightly higher—45% and 42%, respectively—in villages without access to a mobile money agent.

The subsidy appears to lower measured coordination because it reduces payment frequency, and therefore the number of occasions on which delegation is useful (see Appendix Table A16, columns 1 and 2). The reduction occurs in villages with and without an agent. However, the reduction in coordination is not meaningfully larger in villages without agent access. In addition, coordination does not respond when a local agent arrives: Appendix Table A16 shows near-zero effects of agent arrival on the share of paying customers using third parties and on the share of payments made by third parties.

We therefore interpret coordination as an imperfect coping device rather than as the mechanism behind our main results. Households do try to economize on trips, but doing so does not eliminate the payment-infrastructure barrier that drives the larger subsidy response in high-access villages. Taken together, these findings indicate that coordination does not neutralize the transaction costs behind the complementarity.

6.4.5 Information and Subsidy Receipt

We address concerns that the increase in adoption might be driven by information channels. First, greater awareness of the SHS product—rather than the price reduction itself—could raise adoption if marketing intensified around the subsidy launch. Information barriers are particularly relevant when assessing policy impacts, especially in rural areas of low-income countries where knowledge of new technologies is often limited. To investigate this, we collected detailed marketing campaign data from the solar company over time (bi-monthly) at the shop level. We find no significant increase in information campaigns during the subsidy launch; if anything, the number of campaigns decreased everywhere. We further discuss this issue in Appendix Section B where we explore the role of marketing campaigns and show that they do not appear to drive our results. However, this is one type of information campaign, and we cannot fully rule out the information channel potentially coming from public officials, which might have targeted treated areas first.

[32] There are two situations in which this measure will fail to capture payment coordination. First, payments may be made regularly by someone outside the village (for example, a migrant household member who routinely pays for electricity). In such cases, the payment pattern reflects household arrangements rather than within-village coordination to overcome transaction costs. Second, a customer may use a phone number that is not registered with the company; these payments will not be flagged as third-party transactions in our data, even if they involve coordination.

Second, a mobile money agent in the village could, in principle, complement the subsidy by helping customers actually receive it, especially given the low levels of financial literacy in these settings. We assess this mechanism by examining the impact of the mobile money agent policy on subsidy disbursement in Appendix Table A17. We examine two outcomes. The first is the share of payments that receive the subsidy, i.e., payments matched by La Poste and disbursed to the SHS company. The second is the share of new customers whose first payment receives the subsidy, which isolates newly adopting customers from the existing customer base. Neither responds to agent arrival: the effect of agent arrival on the share of payments receiving the subsidy is negative and statistically insignificant, and the effect on the share of new customers receiving the subsidy is small and statistically insignificant. If anything, the signs run against the concern that agents facilitate subsidy receipt. Taken together, these results provide no evidence that the presence of a mobile money agent raises subsidy disbursement rates for either existing or new customers, making this an unlikely driver of the observed complementarity.

7 External Validity: Digital Cash Transfers

The solar analysis studies one direction of the conversion between cash and digital money: to pay for electricity, households convert cash into digital balances at an agent. Togo's Novissi emergency cash transfers, disbursed during COVID-19, run the same conversion in reverse: benefits arrive as digital balances that beneficiaries convert into cash through the same agent network. The two programs thus span both directions of the friction in Section 2—the cost of reaching an agent is incurred whether the state collects a payment or disburses a subsidy. If payment infrastructure, rather than anything specific to solar electricity, drives the results above, the same access measure should shape the transfer's effectiveness. Appendix C provides additional details.

Novissi targeted poor cantons and disbursed monthly transfers directly into beneficiaries' mobile money accounts (8,620 CFA for women and 7,450 CFA for men). Because beneficiaries needed to reach a mobile money agent to cash out, the effective value of the transfer should be lower in places with worse agent access. This yields a sharp prediction: treatment effects should be larger where mobile money access is better for outcomes that depend on purchasing power, but not necessarily for outcomes such as mental health that may respond to the reassurance of support itself.

We combine the randomized evaluation data from Aiken et al. (2025) with our administrative data on the location of mobile money agents. After matching canton names, we link 9,284 individuals in 236 cantons to a canton-level measure of mobile money access: the share of villages in the canton with an agent within 3 km, standardized to mean zero and unit variance.[33] Access varies substantially across the Novissi sample. On average, 47.8% of villages in a canton are within 3 km

[33] We are unable to match 227 individuals, leaving a matched sample of 97.6% of the original evaluation.

of an agent; 19.5% of cantons have no village within 3 km, while 17.8% have full coverage.[34] We estimate a simple interaction specification that regresses each standardized outcome on treatment, mobile money access, and their interaction, controlling for strata, enumerator, and survey-week fixed effects and using the original sampling weights.

We group the study's seven outcomes by whether realizing them requires converting the transfer into cash. Food security, financial health, and health care access do, since each is financed by spending the transfer, so their treatment effects should depend most on the cost of reaching an agent. The remaining four (financial inclusion, mental health, perceived status, and labor supply) can respond to the transfer without a cash-out, and we combine them into a single index following the construction in Aiken et al. (2025).

Table 5 reports the results. Panel A presents the main treatment effects on the matched sample: Novissi improved food security by 0.07 standard deviations and the index of other outcomes by 0.06, with smaller and insignificant average effects on financial health and health care access.[35] Panel B presents the heterogeneous effects by mobile money access. A one-standard-deviation increase in canton-level access raises the treatment effect on financial health by 0.05 standard deviations, significant at the 5% level, and on health care access by 0.04, which is not significant. The interaction for food security is positive but small, and the interaction for the index of outcomes that do not require cashing out is zero. One interpretation is that households prioritize food spending: the transfer is put towards food first regardless of how costly it is to reach an agent, so the effect on food security is the largest of the average effects but does not vary with access. What is left over is what differs. Where cashing out is less costly, more of the transfer survives the trip, and the residual accumulates as savings, which is where the heterogeneity emerges. The heterogeneity is thus confined to outcomes that require converting the transfer into cash, consistent with our Prediction P5.

The magnitudes are economically plausible. Using our transport-cost survey, a round trip to an agent costs roughly 650 CFA plus 90 CFA per kilometer, expressed in 2021 CFA to match the year of the transfers. At the median canton-level distance in the Novissi sample, 3.4 km, cashing out once costs roughly 950 CFA, or about 12% of the average monthly transfer of 8,035 CFA. At the 75th percentile distance, 5.5 km, that rises to about 1,140 CFA, or 14% of the transfer. In the 19.5% of cantons where no village is within 3 km of an agent, the median distance is 9.4 km, implying a cash-out cost of about 1,490 CFA—roughly 19% of the transfer—before accounting for time. These are substantial losses in effective transfer value.

Panels C and D probe the access measure. The financial-health interaction is similar when access is defined at 2 km (0.044) or 4 km (0.041), and at 4 km the health care interaction becomes

[34] The friction described here is household-level—whether a beneficiary can walk to an agent—but the Novissi data identify only each respondent's canton, so we proxy it with the canton share of villages within 3 km. The interaction in Table 5 therefore captures how treatment effects vary with *canton-average* access, not the household threshold itself. This is conservative: because treatment is randomly assigned, the gap between a beneficiary's own access and their canton average is orthogonal to treatment—it can only bias the estimate toward zero. The estimates are thus a lower bound on the underlying village-level friction.

[35] Results are very comparable to those with the full sample in Aiken et al. (2025).

significant (0.046). Because cantons with more agents are also closer to markets and financial institutions and more densely populated, with correlations between 0.33 and 0.49, Panel D lets the mobile money interaction compete in a single regression with treatment interactions for each of these. The mobile money interaction on financial health rises to 0.078 (s.e. 0.033), while none of the three alternatives is distinguishable from zero. What matters is proximity to an agent where the transfer can be converted into cash, not proximity to services in general.

We view this evidence as suggestive since mobile money access is not cross-randomized in the Novissi setting. Still, the same payment infrastructure that amplifies the solar subsidy also appears to amplify the value of digital cash transfers. In both contexts, policy benefits are delivered digitally, yet their realized value depends on the cost of reaching the last-mile cash-in or cash-out point.

8 Policy Implications

The preceding sections show that a digitally delivered subsidy doesn't affect households uniformly in a cash-based economy. The *effective* price reduction depends on the cost of reaching the payment network. This has two implications, one for how much the subsidy can plausibly contribute toward Togo's 2030 electrification target, and one for the role of public investment in payment infrastructure.

8.1 Benchmarking the Subsidy Estimates Against the 2030 Target

Togo's target of 550,000 solar home system households by 2030 provides a benchmark for the magnitude of our estimates: on the government's timetable, it requires roughly 150,000 additional households every three years. We use our estimates to conduct an illustrative accounting exercise. The steady-state extrapolation holds the per-village treatment effect fixed over 36 months, ignoring market saturation as the adoptable population is drawn down, entry by competing providers, and whether agent placements remain viable at low rural transaction volumes. All three would reduce the cumulative effect, so the projections can be considered as upper bounds, but they are still informative about orders of magnitude.

In our sample of 2,212 villages, the pre-policy control mean of 0.40 new applications per village-month implies about 32,000 applications over 36 months in the absence of the subsidy. Applying the average treatment effect of 0.47 applications per village-month over the same horizon adds roughly 38,000, bringing the total to about 69,000 within the provider we study—just under half the required pace. Even allowing for activity by other solar providers, the subsidy alone is thus unlikely to close the remaining electrification gap.

How much more could better payment access deliver? One benchmark asks what would happen if every village responded to the subsidy as today's villages with an agent do. At the high-access effect of 0.96 applications per village-month, the subsidy would induce roughly 76,000 applications over 36 months, or about 108,000 in total once baseline demand is included—still short of the roughly 150,000 needed every three years to stay on pace.

The second benchmark is more conservative and tied directly to the causal access estimate. Applying the agent-arrival effect of 0.64 applications per village-month to the villages that initially lacked an agent within 3 km adds roughly 37,000 applications over 36 months, taking the provider total from about 69,000 to about 106,000—from just under half to roughly 70% of the required pace. The two benchmarks bracket the plausible gains from expanding walkable payment access: not enough to meet the target on their own, but enough to raise the subsidy's contribution substantially.

The threshold pattern in Figure 4 sharpens the design lesson. What seems to matter most is whether a village has an agent within walking distance. A first agent in an unserved village is therefore likely to yield larger returns than further densifying the network where coverage already exists.

8.2 Payment Infrastructure as Complementary Public Capital

Why might private deployment fall short of the social optimum? Agents locate where transaction volume covers their operating costs, so low population density and limited mobile money use weaken the private incentive to enter remote rural areas. These are also the places where sparse payment access most weakens digitally delivered programs.

Our evidence shows that this wedge is not specific to solar. In the PAYGO market, better agent access amplifies a price subsidy by lowering the cost of the repeated payments the contract requires. In Novissi, the same mechanism runs in reverse: digital transfers have larger effects on the outcomes that require converting balances into spending power. Because the two programs sit on opposite sides of the conversion—one collects digital payments, the other disburses them—they suggest a simple rule for which programs the friction burdens most: exposure rises with how often a program requires converting between cash and digital money and with the distance to the nearest agent, and falls as the size of each conversion grows relative to the cost of a trip.

The social return to rural payment infrastructure therefore exceeds the commissions that make up the operator's private return. An agent in an underserved village raises the take-up of subsidized electricity and the realized value of digital transfers, and would do the same for other programs delivered through digital channels. Payment infrastructure can be viewed as a form of complementary public capital, akin to roads or telecommunications: much of its value comes from the downstream activities it enables.

Payment infrastructure and price subsidies should therefore be designed jointly. Where households remain far from an agent, bringing one within walking distance raises the marginal return of the subsidy program. Togo's *Réseau Granulaire* illustrates this effect, and the second benchmark above quantifies what generalizing it could add. Whether further agent deployment is optimal ultimately depends on operating costs, which we do not observe, and on the broader set of programs that run through digital channels.

9 Conclusion

Governments in low-income countries increasingly deliver benefits and collect payments through digital systems, yet households still transact mainly in cash. This paper shows that the cost of bridging this gap—reaching the physical network of mobile money agents that converts cash into digital balances—is a quantitatively important bottleneck for policy effectiveness. In rural Togo, transport costs to reach mobile money agents are equivalent to 19% of the monthly price of electricity access for the median customer, rising to 39% at the 75th percentile. This friction depresses take-up, which can thus distort measured price elasticities and weaken the effectiveness of policies delivered through digital payment channels.

Our main solar analysis exploits two sequential government interventions in Togo—a staggered solar home system subsidy and a government-led expansion of mobile money agents—to establish three main results. First, the subsidy more than doubles solar adoption, but its effect is nearly three times larger in villages with a nearby mobile money agent: 0.96 additional applications per village-month where an agent is present within 3 km, compared with just 0.35 where one is not. This gap of 0.61 applications is significant at the 5% level and survives flexible controls for village wealth, population density, and market access. The subsidy's effect drops once the nearest agent lies beyond walking distance and flattens thereafter. Second, the arrival of a mobile money agent raises adoption by 0.64 applications per village-month—about as large as that gap, as our second prediction indicates, so variation in the timing of payment access accounts for the cross-sectional difference. Third, the subsidy enables existing customers to consolidate payments and buy electricity in bulk, but this adjustment is concentrated in villages with nearby agents, where payment schedules are most flexible. Together, these results show that the subsidy's effect is larger where agents are present and that agents raise adoption under the subsidy—patterns consistent with the two policies acting as complements, one relaxing the household budget constraint and the other the payment-infrastructure constraint. Better payment access raises the adoption gains from a given subsidy.

The mechanism is not specific to electricity. Combining the randomised evaluation data from Togo's Novissi emergency cash transfer with our mobile money agent data, we find that a one-standard-deviation (SD) increase in canton-level mobile money access raises the treatment effect on financial health by 0.05 SD. The interaction is small and insignificant for food security, the outcome with the largest average effect (0.07 SD). Households appear to put the transfer toward food first whatever the cost of reaching an agent, so what varies with access is what is left over after the trip and accumulates as savings. The heterogeneity is confined to the outcomes that require converting the balance into cash, and in a horse race against distance to the nearest financial institution, distance to the nearest market, and population density, only agent access carries it. Novissi thus runs the same friction in reverse: the subsidy required households to pay into the digital system and the transfer required them to cash out of it, and both were worth less where the agent network was far.

These findings have a direct implication for the role of the state. Mobile money agent networks are built by private operators whose expansion decisions are governed by profitability. Remote, sparsely populated rural areas, where complementary public programs are most needed, are where private incentives to build agent coverage are weakest. But those are also places where the public return to agent coverage is large, because agents amplify the effectiveness of policies that rely on digital payment channels. Public investment in last-mile payment infrastructure is therefore justified not by its direct returns alone, but by the social return it raises across the programs delivered through it. As governments across Sub-Saharan Africa accelerate the shift toward digital service delivery, that return is likely to grow. Where and how to invest in this infrastructure, and how to coordinate it with the programs it supports, is a direction for future research.

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Tables and Figures

Table 1: Summary Statistics

	Mean (1)	Std. Dev. (2)	Observations (3)
Panel A. Household Characteristics			
Income source:			
Agriculture	0.708	0.455	52,878
Own business	0.164	0.370	52,878
Employee	0.089	0.284	52,878
Other	0.040	0.196	52,878
Previous energy source:			
Modern lighting	0.018	0.135	52,878
Traditional lighting	0.921	0.270	52,878
None	0.035	0.185	52,878
Kit type:			
Basic	0.490	0.500	52,878
Plus	0.196	0.397	52,878
Premium	0.309	0.462	52,878
Age at adoption	38.8	11.7	52,847
Female Applicant	0.031	0.173	52,878
Panel B. SHS Payment Behaviour (Monthly)			
Any payment	0.607	0.488	820,875
Number of payments	1.136	1.617	820,875
Multiple payments	0.236	0.425	820,875
Any usage	0.811	0.391	820,875
Usage share	0.714	0.399	820,851
Panel C. Village-Level Adoption (Monthly)			
Any application	0.181	0.385	92,904
Number of applications	0.521	2.050	92,904
Panel D. Village Characteristics			
MoMo agent distance, 2019 (km)	6.5	5.0	2,212
Nearby MoMo agent, 2019 ($\leq$3km)	0.272	0.445	2,212
Relative Wealth Index	-0.258	0.341	2,212
Population density (per km^2)	216.9	533.0	2,212
Distance to market (km)	3.2	2.9	2,212

Notes: This table reports summary statistics for pay-as-you-go solar home system (SHS) customers and villages. *Panel A* reports household characteristics measured at registration, with one observation per customer. Modern lighting comprises the grid, generator and solar panels; traditional lighting comprises battery lanterns, flashlights, and kerosene lamps or candles. *Panel B* reports SHS payment and usage behaviour at the customer-month level. *Panel C* reports village-level adoption at the village-month level: *Any application* is an indicator for at least one new customer application in the village-month, and *Number of applications* is the number of new applications. *Panel D* reports village characteristics: distance to the nearest MoMo agent (km) and an indicator for a MoMo agent within 3 km, both measured in 2019; the Relative Wealth Index; population density; and distance to the nearest market (km).

Table 2: Subsidy Effects on Adoption

	Monthly Applications for Solar Home Systems			
	Main Specification	Robustness Specifications		
	(1)	(2)	(3)	(4)
Panel A. All Villages				
Subsidy	0.473***	0.500***	0.497***	0.512***
	(0.067)	(0.071)	(0.070)	(0.086)
Pre-policy control mean	0.397	0.397	0.397	0.397
p-value: joint test of pre-trends	[0.481]	[0.609]	[0.607]	[0.590]
Villages	2,212	2,212	2,212	2,212
Observations	15,484	15,484	15,484	15,484
Panel B. Villages with Access to Mobile Money Agents				
Subsidy	0.958***	1.014***	1.005***	1.012***
	(0.222)	(0.237)	(0.238)	(0.234)
Pre-policy control mean	0.509	0.509	0.509	0.509
p-value: joint test of pre-trends	[0.801]	[0.677]	[0.688]	[0.698]
Villages	602	602	602	602
Observations	4,214	4,214	4,214	4,214
Panel C. Villages without Access to Mobile Money Agents				
Subsidy	0.349***	0.348***	0.334***	0.345***
	(0.091)	(0.092)	(0.086)	(0.096)
Pre-policy control mean	0.313	0.313	0.313	0.313
p-value: joint test of pre-trends	[0.346]	[0.355]	[0.377]	[0.351]
Villages	1,610	1,610	1,610	1,610
Observations	11,270	11,270	11,270	11,270
p-value: test of equality (B = C)	[0.018]	[0.014]	[0.014]	[0.011]
Village fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Wealth controls	No	Yes	Yes	Yes
Population density controls	No	No	Yes	Yes
Distance to market controls	No	No	No	Yes

Notes: This table reports the average effect of subsidy eligibility on monthly village-level applications for solar home systems, estimated using the interaction-weighted event-study estimator. The sample covers 2,212 villages observed monthly from December 2018 to June 2019. Treatment is defined by the staggered rollout of subsidy eligibility across three groups of districts (March, May, and July 2019), with the last group serving as the control. Panel A includes all villages. Panel B restricts to villages within 3 km of a mobile money agent at baseline; Panel C restricts to villages without such access. The p-value for the test of equality (B = C) tests whether the subsidy effect differs across the two subsamples. The *Main Specification* includes village and time fixed effects; the *Robustness Specifications* sequentially add controls for village-level wealth, population density, and distance to the nearest market. Standard errors are in parentheses, clustered at the district level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Mobile Money Agent Effects on Adoption

	Monthly Applications for Solar Home Systems			
	Main Specification	Robustness Specifications		
	(1)	(2)	(3)	(4)
Panel A. All Mobile Money Agent Expansion Villages				
Mobile Money Agent Arrival	0.642***	0.647***	0.647***	0.663**
	(0.239)	(0.242)	(0.244)	(0.260)
Pre-policy control mean	0.500	0.500	0.500	0.500
p-value: joint test of pre-trends	[0.151]	[0.165]	[0.215]	[0.228]
Villages	100	100	100	100
Observations	1,499	1,499	1,499	1,499
Panel B. All Mobile Money Agent Expansion Villages Excluding the May 2021 Cohort				
Mobile Money Agent Arrival	0.671**	0.667**	0.679**	0.693**
	(0.311)	(0.313)	(0.310)	(0.322)
Pre-policy control mean	0.667	0.667	0.667	0.667
p-value: joint test of pre-trends	[0.393]	[0.457]	[0.589]	[0.500]
Villages	94	94	94	94
Observations	1,409	1,409	1,409	1,409
Village fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Wealth controls	No	Yes	Yes	Yes
Population density controls	No	No	Yes	Yes
Distance to market controls	No	No	No	Yes

Notes: This table reports the average effect of mobile money agent arrival on monthly village-level applications for solar home systems, estimated using the interaction-weighted event-study estimator. The sample consists of 100 villages that received their first mobile money agent within 3 km through the government-led expansion program between September 2019 and May 2021. The analysis is conducted at the bimonthly frequency; outcomes are averaged across both months within each bimonth to maintain comparability with the monthly subsidy analysis. Villages receiving an agent in January–May 2021 serve as the control group. Panel A includes all 100 expansion villages. Panel B excludes the 6 villages in the May 2021 cohort, restricting the control group to the 6 villages treated in January 2021. The *Main Specification* includes village and time fixed effects; the *Robustness Specifications* sequentially add controls for village-level wealth, population density, and distance to the nearest market. Standard errors are in parentheses, clustered at the village level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Subsidy and Mobile Money Agent Effects on Payment Behaviour

	Any Payment (1)	Number of Payments (2)	Multiple Payments (3)	Conditional: Number of Payments (4)	Conditional: Multiple Payments (5)
Panel A. All Villages					
Subsidy	-0.025	-0.255***	-0.030**	-0.313***	-0.032**
	(0.020)	(0.052)	(0.014)	(0.057)	(0.015)
Pre-policy control mean	0.713	1.318	0.266	1.857	0.376
p-value: joint test of pre-trends	[0.915]	[0.446]	[0.692]	[0.082]	[0.978]
Customers	7,493	7,493	7,493	6,719	6,719
Observations	46,543	46,543	46,543	32,740	32,740
Panel B. Villages with Access to Mobile Money Agents					
Subsidy	-0.017	-0.309***	-0.046***	-0.374***	-0.045**
	(0.021)	(0.058)	(0.017)	(0.066)	(0.018)
Pre-policy control mean	0.734	1.402	0.277	1.921	0.382
p-value: joint test of pre-trends	[0.467]	[0.235]	[0.674]	[0.074]	[0.527]
Customers	2,994	2,994	2,994	2,689	2,689
Observations	18,807	18,807	18,807	13,410	13,410
Panel C. Villages without Access to Mobile Money Agents					
Subsidy	-0.027	-0.189***	-0.015	-0.237***	-0.021
	(0.026)	(0.049)	(0.015)	(0.060)	(0.018)
Pre-policy control mean	0.691	1.229	0.253	1.785	0.370
p-value: joint test of pre-trends	[0.955]	[0.793]	[0.527]	[0.651]	[0.664]
Customers	4,499	4,499	4,499	4,030	4,030
Observations	27,736	27,736	27,736	19,330	19,330
Panel D. All Mobile Money Agent Expansion Villages					
Mobile Money Agent Arrival	-0.044	0.128	-0.002	0.234	0.018
	(0.044)	(0.242)	(0.043)	(0.258)	(0.036)
Pre-policy control mean	0.908	2.063	0.510	2.273	0.561
p-value: joint test of pre-trends	[0.110]	[0.329]	[0.053]	[0.789]	[0.264]
Customers	535	535	535	529	529
Observations	5,669	5,669	5,669	5,069	5,069
p-value: test of equality (B = C)	[0.748]	[0.027]	[0.057]	[0.057]	[0.246]

Notes: This table reports the average effects of subsidy eligibility and mobile money agent arrival on the payment behaviour of existing customers. *Any Payment* is a binary indicator for whether the customer made any payment that period; *Number of Payments* is the number of payments made; *Multiple Payments* is a binary indicator for whether the customer made more than one payment. The *Conditional* columns report the latter two outcomes conditional on the customer making at least one payment. Panels A–C estimate the subsidy effect at the customer-month level: Panel A includes all villages; Panel B restricts to villages within 3 km of a mobile money agent at baseline; Panel C restricts to villages without such access. Panel D estimates the effect of mobile money agent arrival at the customer-bimonth level in the expansion villages. Samples and treatment definitions follow Tables 2 and 3; all effects are estimated using the interaction-weighted event-study estimator. The p-value for the test of equality (B = C) tests whether the subsidy effect differs across the two subsamples. All specifications include customer and time fixed effects. Standard errors are in parentheses, clustered at the district level in Panels A–C and at the village level in Panel D. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Heterogeneous Treatment Effects of Novissi by Mobile Money Access

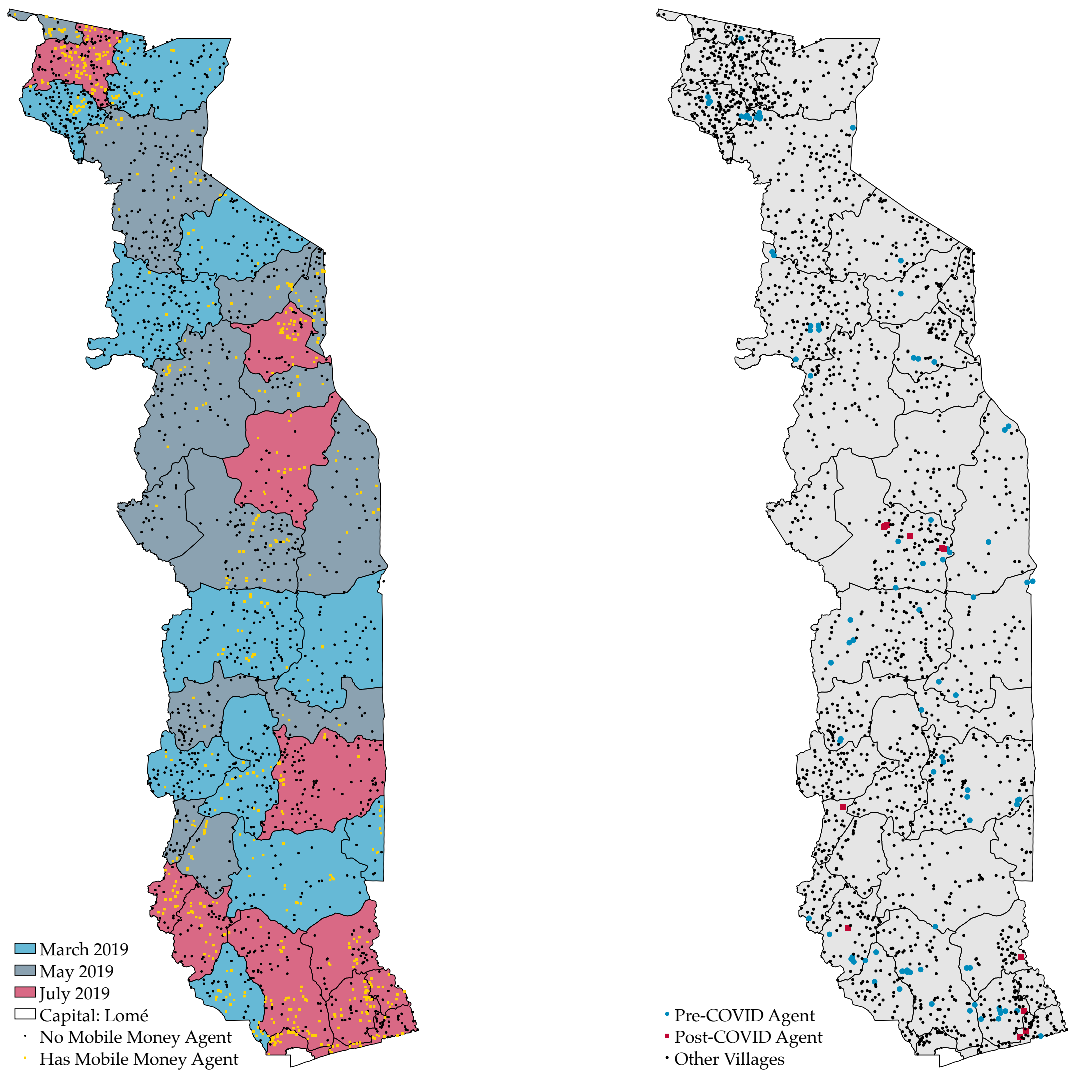
	Food Security (1)	Financial Health (2)	Health Care Access (3)	Other Outcomes (4)
Panel A. Main Treatment Effects (Matched Sample)				
Treatment	0.074***	0.035	0.007	0.056**
	(0.022)	(0.024)	(0.023)	(0.022)
Panel B. Heterogeneous Effects by Mobile Money Access (Baseline, 3 km)				
Treatment	0.073***	0.031	0.004	0.056**
	(0.022)	(0.024)	(0.023)	(0.022)
Treatment × MoMo Access (3 km)	0.013	0.052**	0.036	0.003
	(0.022)	(0.025)	(0.024)	(0.023)
MoMo Access (3 km)	−0.025*	−0.050***	−0.015	0.003
	(0.015)	(0.019)	(0.017)	(0.017)
Panel C. Robustness: Alternative Distance Thresholds				
Treatment × MoMo Access (2 km)	0.005	0.044*	0.030	−0.010
	(0.023)	(0.025)	(0.024)	(0.023)
Treatment × MoMo Access (4 km)	0.010	0.041	0.046**	0.001
	(0.022)	(0.025)	(0.023)	(0.022)
Panel D. Horse Race: MoMo Access vs Alternative Infrastructure Measures				
Treatment × MoMo Access (3 km)	0.026	0.078**	0.029	−0.002
	(0.028)	(0.033)	(0.032)	(0.029)
Treatment × Prox. Financial Inst.	−0.009	−0.031	0.001	0.020
	(0.030)	(0.033)	(0.030)	(0.031)
Treatment × Prox. Market	−0.027	−0.018	0.009	−0.013
	(0.024)	(0.030)	(0.026)	(0.025)
Treatment × Pop. Density	0.007	0.001	0.009	−0.006
	(0.026)	(0.025)	(0.026)	(0.025)
Control mean	−0.004	−0.002	0.013	0.005
Observations	9,284	9,284	9,284	9,284
Strata fixed effects	Yes	Yes	Yes	Yes
Enumerator fixed effects	Yes	Yes	Yes	Yes
Survey week fixed effects	Yes	Yes	Yes	Yes

Notes: Each outcome is a standardized index (mean zero, unit standard deviation in the control group). Columns (1)–(3) are the outcomes whose realisation requires converting the transfer into cash: food security, financial health, and health care access. Column (4) combines the four remaining outcome indices (financial inclusion, mental health, perceived status, and labor supply) following the index construction of the original study Aiken et al. (2025): each component index is standardized to the weighted control-group mean and standard deviation, the standardized components are summed, and the sum is standardized again to the weighted control group. Panel A reproduces the main treatment-effect specification of the original Novissi evaluation on the matched sample. Panel B adds the interaction with *MoMo Access*, the share of villages in the respondent's canton with at least one mobile money agent within 3 km, standardized to mean zero and unit standard deviation across cantons. Panel C re-estimates the interaction under alternative definitions of agent access (each row a separate regression per outcome): thresholds of 2 km and 4 km. Panel D reports a horse race in which the treatment interaction with mobile money access competes, within a single regression per outcome, with treatment interactions with proximity to the nearest financial institution, proximity to the nearest market, and population density (all standardized; all main effects included). All regressions include strata, enumerator, and survey week fixed effects, and are weighted by the sampling and response weights from the original study. Heteroskedasticity-robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 1: Districts of Togo by Policy Rollout

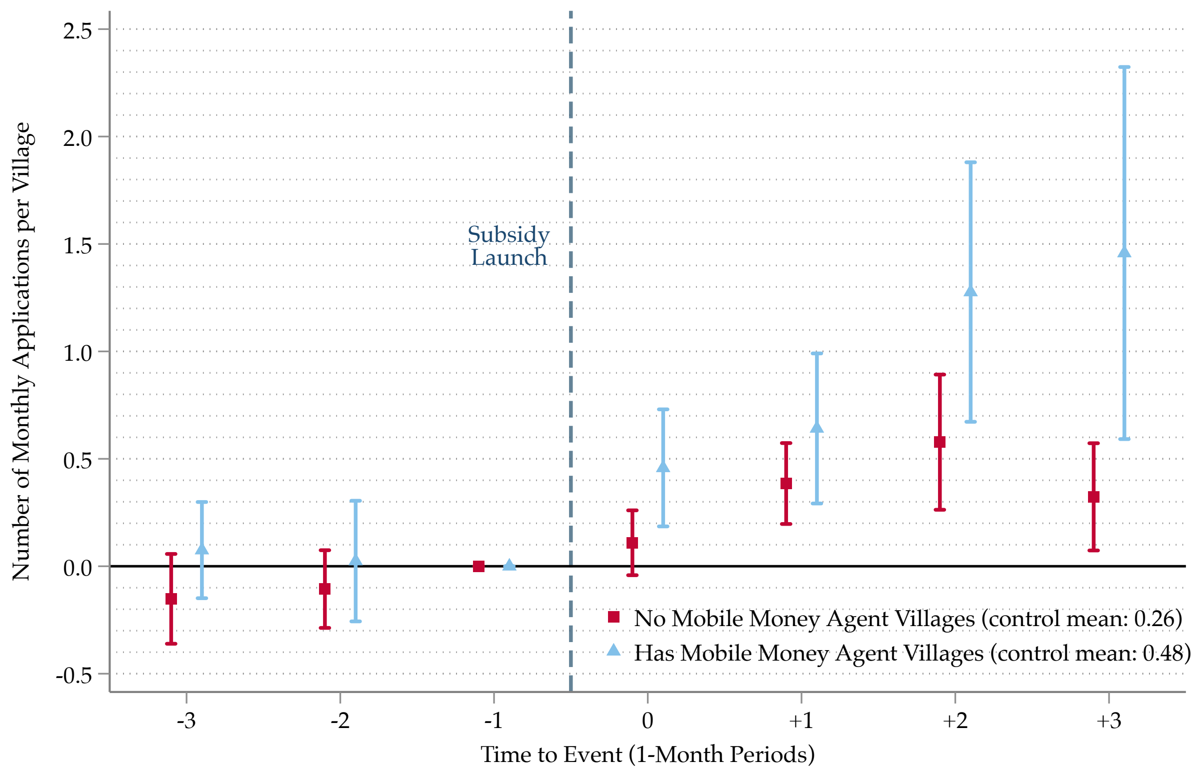
Panel A. Subsidy Rollout by District

Panel B. Mobile Money Agent Rollout Villages



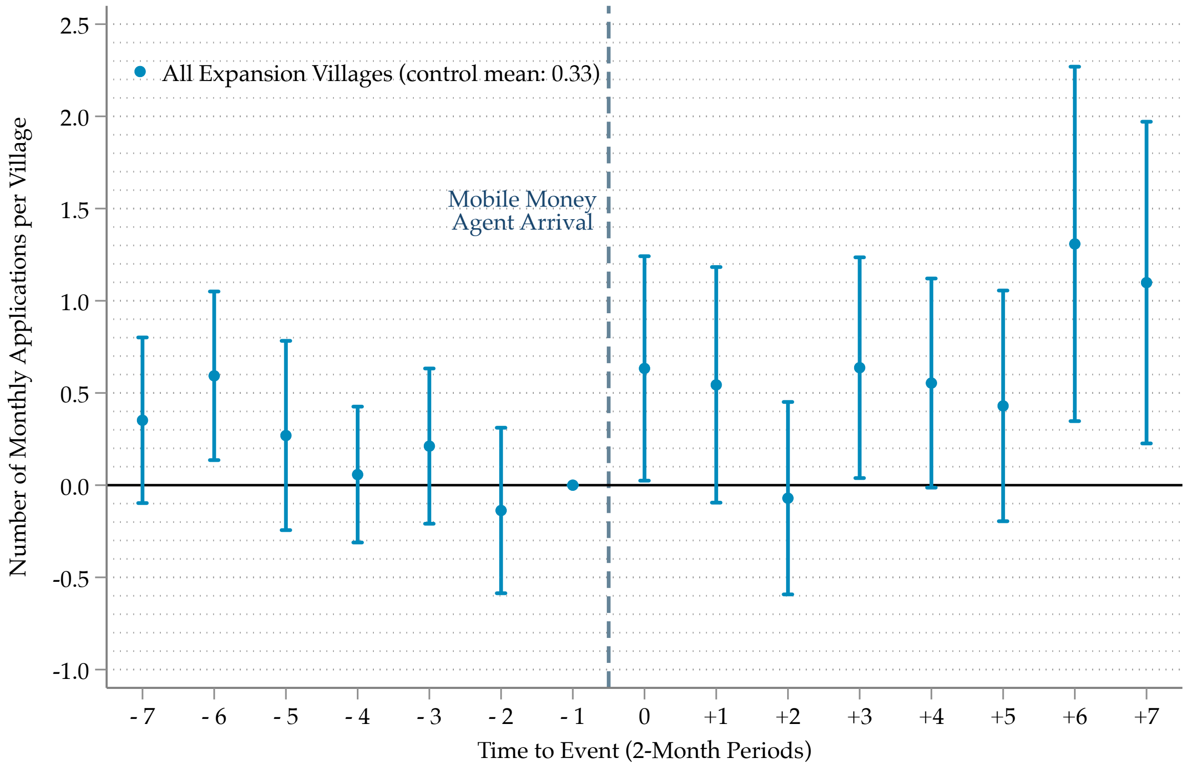
Notes: Panel A illustrates the spatial distribution of the subsidy program rollout across Togo's districts. Districts shaded in the light blue represent the first phase, where the program launched in March 2019. Districts in dark blue indicate the second phase, beginning in May 2019. The red districts correspond to the final phase, initiated in July 2019. Panel B depicts the locations of villages that were part of the mobile money agent expansion program.

Figure 2: Heterogeneous Subsidy Effect on Adoption by Mobile Money Agent Access



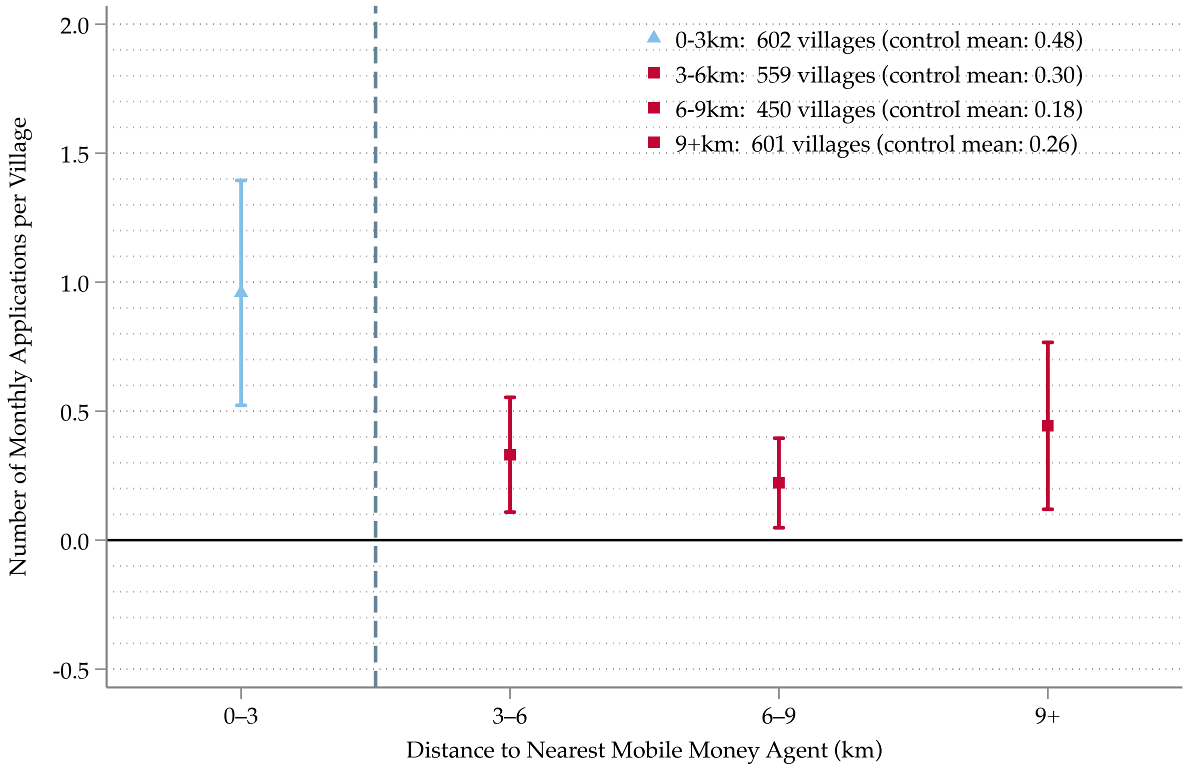
Notes: This figure displays village-level heterogeneous ITT effects of subsidy eligibility on solar home system (SHS) adoption, estimated using the interaction-weighted event-study estimator with all event-study terms and fixed effects fully interacted with baseline mobile money agent access. The effect for villages with a mobile money agent within 3 km prior to subsidy launch is shown in blue, and the effect for villages without an agent in red. The outcome is the number of SHS applications per month in each village. The sample, treatment, and control groups follow Table 2; the mean of the outcome in each control group prior to subsidy launch is reported in the legend. Vertical bars are 95% confidence intervals based on standard errors clustered at the district level; the dashed vertical line marks the subsidy launch.

Figure 3: Mobile Money Agent Effect on Adoption



Notes: This figure displays village-level dynamic ITT effects of mobile money agent arrival on solar home system (SHS) adoption, estimated using the interaction-weighted event-study estimator. The analysis is conducted at the bimonthly frequency; the outcome is the average number of SHS applications per month in each village. The sample, treatment, and control groups follow Table 3; the mean of the outcome in the control group prior to agent arrival is reported in the legend. Vertical bars are 95% confidence intervals based on standard errors clustered at the village level; the dashed vertical line marks the arrival of the mobile money agent.

Figure 4: Subsidy Effect on Adoption by Distance to Nearest Mobile Money Agent



Notes: This figure displays village-level heterogeneous ITT effects of subsidy eligibility on solar home system (SHS) adoption, estimated separately for four groups of villages defined by their pre-subsidy distance to the nearest mobile money agent: 0–3 km (with a mobile money agent), 3–6 km, 6–9 km, and 9+ km. Each group is estimated using the interaction-weighted event-study estimator; the outcome is the number of SHS applications per month in each village, and the sample, treatment, and control groups follow Table 2. Vertical bars are 95% confidence intervals based on standard errors clustered at the district level; the dashed vertical line marks the subsidy launch.

Appendices For Online Publication Only

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A Appendix Tables and Figures

Table A1: Summary Statistics: Transport Costs Between Villages

	Mean (1)	SD (2)	P25 (3)	Median (4)	P75 (5)	Minimum (6)	Observations
Distance (km)	8.7	5.9	3.9	7.4	11.6	1.5	123
One-way moto fare (2019 CFA)	672	420	331	580	829	83	124
Cost per km (2019 CFA/km)	93	58	56	83	113	20	123
Travel time, moto (min.)	31	25	12	20	45	5	124
Walking time (min.)	95	79	35	80	120	10	124

Notes: Data from a transport cost survey of origin villages across five regions of Togo. For each origin, enumerators identified the nearest village with a mobile money agent and up to three additional nearby villages. Distance is computed from Google Maps. Cost per km is the ratio of the one-way moto-taxi fare to the Google Maps distance. Monetary values are deflated from 2023 to 2019 CFA using World Bank CPI for Togo.

Table A2: Baseline Electrification by Subsidy Rollout Wave

	March 2019 (1)	May 2019 (2)	July 2019 (3)	F-statistic (4)
Electrified households (%)	5.00	16.54	23.18	26.37***
	(0.00)	(6.58)	(7.83)	
Prefectures	11	13	11	35

Notes: The table summarizes the electrification rates of the districts within the three groups. This is based on data collected in 2016 from the Government of Togo (data is unavailable for one out of thirty-six districts). The first three columns show the mean values of the district averages for each of the three groups of districts treated at different times, with standard deviations in parentheses. The electrification rates are provided by the Government in ranges, and so the midpoint of the range is used for each district average (all districts in Group 1 have electrification rates between 0 and 10%, hence the 5% mean with no standard deviation). The fourth column shows the F-statistic of the one-way analysis-of-variance (ANOVA) model to test for differences in means across the three groups. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Baseline Characteristics by Subsidy Rollout Wave

	Subsidy Rollout Wave			Test of Equality
	March 2019 (1)	May 2019 (2)	July 2019 (3)	p-value (4)
Panel A. Household Characteristics				
Income source:				
Agriculture	0.843	0.812	0.724	[0.227]
Own business	0.064	0.080	0.128	[0.102]
Employee	0.072	0.086	0.120	[0.387]
Other	0.020	0.021	0.028	[0.715]
Previous energy source:				
Modern lighting	0.026	0.014	0.015	[0.593]
Traditional lighting	0.962	0.956	0.971	[0.707]
None	0.001	0.023	0.005	[0.390]
Kit type:				
Basic	0.355	0.323	0.237	[0.117]
Plus	0.287	0.294	0.207	[0.251]
Premium	0.358	0.383	0.556	[0.038]
Age at adoption	40.5	39.6	42.5	[$<$0.001]
Female Applicant	0.008	0.014	0.035	[0.014]
Observations	2,815	2,697	1,981	
Panel B. SHS Payment Behaviour (averaged, through February 2019)				
Any payment	0.711	0.670	0.671	[0.548]
Number of payments	1.078	0.954	1.130	[0.216]
Multiple payments	0.193	0.163	0.219	[0.170]
Any usage	0.987	0.984	0.990	[0.448]
Usage share	0.933	0.929	0.953	[0.049]
Observations	2,486	2,410	1,741	
Panel C. Village-Level Adoption (averaged, through February 2019)				
Any application	0.081	0.083	0.093	[0.457]
Number of applications	0.279	0.325	0.220	[0.115]
Observations	826	677	709	
Panel D. Village Characteristics				
MoMo agent distance, 2019 (km)	8.1	6.8	4.5	[0.003]
Nearby MoMo agent, 2019 ($\leq$3km)	0.172	0.230	0.429	[$<$0.001]
Relative Wealth Index	-0.313	-0.310	-0.145	[0.055]
Population density (per km^2)	132.8	93.8	432.3	[0.026]
Distance to market (km)	3.4	3.6	2.6	[0.072]
Observations	826	677	709	
Districts (clusters)	11	13	12	

Notes: This table reports baseline characteristics by subsidy rollout wave (March, May, and July 2019); waves were assigned at the district level. Columns (1)–(3) report means by wave; column (4) reports the p-value (in brackets) of a joint test of equality of the three means, from a regression of each characteristic on rollout-wave indicators with standard errors clustered at the district (assignment) level. Panels A–B are computed on the estimation sample of the customer-level analyses: customers who registered before the first rollout wave and were active at baseline. *Panel A* reports characteristics measured at registration. Modern lighting comprises the grid, generator and solar panels; traditional lighting comprises battery lanterns, flashlights, and kerosene lamps or candles. Kit type is chosen at adoption and may itself respond to the subsidy; it is reported for completeness. *Panel B* reports payment and usage behaviour averaged to one observation per customer over the pre-rollout months, excluding the installation (first) month. Usage share is the fraction of days the system is active. *Panel C* reports village-level adoption averaged to one observation per village over the pre-rollout months: the share of months with at least one new application and the mean number of new applications per month. *Panel D* reports village characteristics measured at baseline: distance to the nearest MoMo agent (km) and an indicator for an agent within 3 km, the Relative Wealth Index, population density, and distance to the nearest market (km).

Table A4: Baseline Characteristics of Mobile Money Expansion Villages

	MoMo Agent Expansion		Test of Equality
	September 2019 to March 2020 (1)	January to May 2021 (2)	p-value (3)
Panel A. Household Characteristics			
Income source:			
Agriculture	0.701	0.662	[0.641]
Own business	0.132	0.150	[0.814]
Employee	0.121	0.175	[0.258]
Other	0.046	0.013	[0.136]
Previous energy source:			
Modern lighting	0.013	0.000	[0.040]
Traditional lighting	0.949	1.000	[0.024]
None	0.029	0.000	[0.153]
Kit type:			
Basic	0.343	0.375	[0.751]
Plus	0.154	0.175	[0.762]
Premium	0.503	0.450	[0.547]
Age at adoption	41.0	41.3	[0.868]
Female Applicant	0.018	0.025	[0.772]
Observations	455	80	
Panel B. SHS Payment Behaviour (averaged, through August 2019)			
Any payment	0.796	0.818	[0.697]
Number of payments	1.278	1.764	[0.113]
Multiple payments	0.263	0.403	[0.009]
Any usage	0.988	0.982	[0.431]
Usage share	0.909	0.902	[0.782]
Observations	395	65	
Panel C. Village-Level Adoption (averaged, through August 2019)			
Any application	0.116	0.155	[0.351]
Number of applications	0.405	0.439	[0.851]
Observations	88	12	
Panel D. Village Characteristics (January 2019)			
MoMo agent distance, 2019 (km)	7.7	6.4	[0.081]
Nearby MoMo agent, 2019 ($\leq$3km)	0.000	0.000	–
Relative Wealth Index	-0.328	-0.293	[0.703]
Population density (per km^2)	128.7	191.5	[0.458]
Distance to market (km)	3.4	1.6	[0.001]
Observations	88	12	

Notes: This table reports baseline characteristics for villages in the MoMo agent expansion sample: villages that eventually receive an agent within 3 km and have no other agent within 3 km. Treated villages received their first agent before the COVID-19 pandemic; control villages received their first agent in 2021. The two groups correspond exactly to the treated and control cohorts of the event-study analysis (Table 3). Column (3) reports the p-value (in brackets) of a test of equality of the two means, from a regression of each characteristic on the treatment indicator with standard errors clustered at the village (assignment) level. Panels A–B are computed on the estimation sample of the customer-level analyses: customers in sample villages who registered before the expansion began and were active at baseline. *Panel A* reports characteristics measured at registration. Modern lighting comprises the grid, generator and solar panels; traditional lighting comprises battery lanterns, flashlights, and kerosene lamps or candles. Kit type is chosen at adoption and may itself respond to the policies; it is reported for completeness. *Panel B* reports payment and usage behaviour averaged to one observation per customer over the pre-expansion months, excluding the installation (first) month. Usage share is the fraction of days the system is active. *Panel C* reports village-level adoption averaged to one observation per village over the pre-expansion months: the share of months with at least one new application and the mean number of new applications per month. *Panel D* reports village characteristics measured at baseline: distance to the nearest MoMo agent (km); an indicator for an agent within 3 km, which is zero for all sample villages by construction; the Relative Wealth Index; population density; and distance to the nearest market (km).

Table A5: Baseline Characteristics by Access to Mobile Money Agents

	Mobile Money Agent Access, 2019		Test of Equality
	Villages with Access to Mobile Money Agents (1)	Villages without Access to Mobile Money Agents (2)	p-value (3)
Panel A. Household Characteristics			
Income source:			
Agriculture	0.777	0.817	[0.045]
Own business	0.099	0.079	[0.072]
Employee	0.102	0.082	[0.084]
Other	0.022	0.022	[0.970]
Previous energy source:			
Modern lighting	0.022	0.016	[0.332]
Traditional lighting	0.956	0.966	[0.234]
None	0.014	0.007	[0.284]
Kit type:			
Basic	0.262	0.345	[0.007]
Plus	0.267	0.269	[0.910]
Premium	0.471	0.385	[0.008]
Age at adoption	41.0	40.5	[0.241]
Female Applicant	0.020	0.015	[0.238]
Observations	2,994	4,499	
Panel B. SHS Payment Behaviour (averaged, through February 2019)			
Any payment	0.679	0.691	[0.490]
Number of payments	1.087	1.020	[0.129]
Multiple payments	0.200	0.181	[0.180]
Any usage	0.988	0.986	[0.263]
Usage share	0.944	0.932	[0.028]
Observations	2,678	3,959	
Panel C. Village-Level Adoption (averaged, through February 2019)			
Any application	0.118	0.073	[<0.001]
Number of applications	0.396	0.229	[<0.001]
Observations	602	1,610	
Panel D. Village Characteristics (January 2019)			
MoMo agent distance, 2019 (km)	1.3	8.5	[<0.001]
Nearby MoMo agent, 2019 ($\leq$3km)	1.000	0.000	–
Relative Wealth Index	-0.043	-0.339	[<0.001]
Population density (per km^2)	494.7	113.0	[<0.001]
Distance to market (km)	1.7	3.8	[<0.001]
Observations	602	1,610	

Notes: This table reports baseline characteristics for villages with and without access to a mobile money agent, where access is defined as having an agent within 3 km in 2019, before the subsidy rollout. The two groups correspond exactly to the subsamples of Panels B and C of Table 2. Column (3) reports the p-value (in brackets) of a test of equality of the two means, from a regression of each characteristic on the access indicator with standard errors clustered at the village, the level at which access varies. Agent placement is not random: this table documents the composition of the two groups underlying the heterogeneity results, and is not a randomisation check. Panels A–B are computed on the estimation sample of the customer-level subsidy analyses, as in Table A3: customers who registered before the subsidy rollout and were active at baseline. *Panel A* reports characteristics measured at registration. Modern lighting comprises the grid, generator and solar panels; traditional lighting comprises battery lanterns, flashlights, and kerosene lamps or candles. Kit type is chosen at adoption and may itself respond to the policies; it is reported for completeness. *Panel B* reports payment and usage behaviour averaged to one observation per customer over the pre-rollout months, through February 2019, excluding the installation (first) month. Usage share is the fraction of days the system is active. *Panel C* reports village-level adoption averaged to one observation per village over the same pre-rollout months: the share of months with at least one new application and the mean number of new applications per month. *Panel D* reports village characteristics measured in January 2019: distance to the nearest MoMo agent (km); an indicator for an agent within 3 km, which equals one in column (1) and zero in column (2) by construction because it defines the split, so no test is reported; the Relative Wealth Index; population density; and distance to the nearest market (km).

Table A6: Subsidy Effects on Adoption by Timing of Mobile Money Access

	Monthly Applications for Solar Home Systems			
	Agent before Subsidy		No Agent during Subsidy	
	Before 2018	2018	2020	After 2020
	(1)	(2)	(3)	(4)
Subsidy	0.959***	0.677***	0.318	0.320***
	(0.234)	(0.228)	(0.250)	(0.095)
Pre-policy control mean	0.506	0.542	0.354	0.190
p-value: joint test of pre-trends	[0.880]	[0.824]	[0.650]	[0.235]
Villages	540	62	102	1,372
Observations	3,780	434	714	9,604
Village fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes

Notes: This table reports the average effect of subsidy eligibility on monthly village-level applications for solar home systems, ordering villages by when a mobile money agent arrived relative to the subsidy. The sample, treatment definition, and estimation follow Table 2, using the interaction-weighted event-study estimator, with access defined at the main 3 km cutoff. Agent locations are observed as annual start-of-year snapshots. Columns (1) and (2) restrict to villages with an agent within 3 km at baseline and together comprise Panel B of Table 2: column (1) covers villages already served at the start of 2018, and column (2) villages whose agent arrived during 2018, the year preceding the subsidy. Columns (3) and (4) cover villages with no agent within 3 km during the subsidy rollout: column (3) villages whose agent arrived during 2020, after the rollout had ended, and column (4) villages with no agent as of the start of 2021, comprising 89 whose agent arrived during 2021 and 1,283 with no agent within 3 km as of the start of 2022, the last snapshot in our data. Because columns (3) and (4) both lacked an agent throughout the rollout and differ only in whether operators subsequently selected the village, the contrast between them reflects selection into agent placement rather than access to payment infrastructure. The 136 villages whose agent arrived during 2019 are omitted, since the annual snapshots cannot establish whether they had an agent for part of the rollout. All specifications include village and time fixed effects. Standard errors are in parentheses, clustered at the district level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A7: Subsidy Effects on Adoption with Different Distance Cutoffs

	Monthly Applications for Solar Home Systems		
	Cutoff Distance (d) for Access to Mobile Money Agent		
	$d \leq 2km$ (1)	$d \leq 3km$ (2)	$d \leq 4km$ (3)
Panel A. Villages with Access to Mobile Money Agents			
Subsidy	1.160***	0.958***	0.777***
	(0.357)	(0.222)	(0.181)
Pre-policy control mean	0.615	0.509	0.485
p-value: joint test of pre-trends	[0.788]	[0.801]	[0.853]
Villages	426	602	808
Observations	2,982	4,214	5,656
Panel B. Villages without Access to Mobile Money Agents			
Subsidy	0.363***	0.349***	0.357***
	(0.085)	(0.091)	(0.095)
Pre-policy control mean	0.302	0.313	0.281
p-value: joint test of pre-trends	[0.406]	[0.346]	[0.367]
Villages	1,786	1,610	1,404
Observations	12,502	11,270	9,828
p-value: test of equality (A = B)	[0.040]	[0.018]	[0.058]
Village fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes

Notes: This table reports the average effect of subsidy eligibility on monthly village-level applications for solar home systems, varying the distance cutoff used to define access to a mobile money agent. The sample, treatment definition, and estimation follow Table 2, using the interaction-weighted event-study estimator. Each column redefines a village as having access if a mobile money agent is located within the indicated cutoff distance ($d \leq 2$, 3, or 4 km): Panel A restricts to villages with such access, and Panel B to villages without. The p-value for the test of equality (A = B) tests whether the subsidy effect differs across the two subsamples. All specifications include village and time fixed effects. Standard errors are in parentheses, clustered at the district level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8: Mobile Money Agent Effects on Adoption with Different Distance Cutoffs

	Monthly Applications for Solar Home Systems		
	Cutoff Distance (d) for Access to Mobile Money Agent		
	$d \leq 2km$ (1)	$d \leq 3km$ (2)	$d \leq 4km$ (3)
Mobile Money Agent Arrival	0.569**	0.642***	0.310
	(0.264)	(0.239)	(0.273)
Pre-policy control mean	0.363	0.500	0.386
p-value: joint test of pre-trends	[0.209]	[0.151]	[0.252]
Treated Villages	69	88	101
Control Villages	13	12	15
Villages	82	100	116
Observations	1,229	1,499	1,740
Village fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes

Notes: This table reports the average effect of mobile money agent arrival on monthly village-level applications for solar home systems, varying the distance cutoff used to define receipt of a mobile money agent. The sample, treatment definition, and estimation follow Table 3, using the interaction-weighted event-study estimator at the bimonthly frequency. Each column redefines a village as having received an agent through the expansion program if one is located within the indicated cutoff distance ($d \leq 2$, 3, or 4 km). The sample comprises all mobile money agent expansion villages. All specifications include village and time fixed effects. Standard errors are in parentheses, clustered at the village level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Subsidy Effects on Adoption for Mobile Money Agent Expansion Villages

	Monthly Applications for Solar Home Systems			
	Main Specification	Robustness Specifications		
	(1)	(2)	(3)	(4)
Panel A. Mobile Money Agent Expansion Villages				
Subsidy	0.326	0.319	0.314	0.342
	(0.328)	(0.339)	(0.315)	(0.314)
Pre-policy control mean	0.198	0.198	0.198	0.198
p-value: joint test of pre-trends	[0.314]	[0.383]	[0.394]	[0.317]
Villages	100	100	100	100
Observations	700	700	700	700
Village fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Wealth controls	No	Yes	Yes	Yes
Population density controls	No	No	Yes	Yes
Distance to market controls	No	No	No	Yes

Notes: This table estimates the subsidy effect on monthly village-level applications for the villages that later received a mobile money agent through the government-led expansion program. This specification tests whether the subsidy effect in the expansion villages prior to agent arrival resembles that of other villages without mobile money access, helping to verify that the expansion villages are an appropriate comparison group. The treatment definition and estimation follow Table 2, using the interaction-weighted event-study estimator at the village-month level. The *Main Specification* includes village and time fixed effects; the *Robustness Specifications* sequentially add controls for village-level wealth, population density, and distance to the nearest market. Standard errors are in parentheses, clustered at the district level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A10: Mobile Money Agent Effects on Adoption Pre-COVID

	Monthly Applications for Solar Home Systems			
	Main Specification	Robustness Specifications		
	(1)	(2)	(3)	(4)
Panel A. Mobile Money Agent Expansion Villages (up to May 2020)				
Mobile Money Agent Arrival	0.625**	0.594**	0.612**	0.680**
	(0.245)	(0.254)	(0.259)	(0.294)
Pre-policy control mean	0.500	0.500	0.500	0.500
p-value: joint test of pre-trends	[0.153]	[0.166]	[0.217]	[0.229]
Villages	100	100	100	100
Observations	1,199	1,199	1,199	1,199
Panel B. Mobile Money Agent Expansion Villages (up to March 2020)				
Mobile Money Agent Arrival	0.520**	0.491*	0.488*	0.523*
	(0.245)	(0.267)	(0.271)	(0.289)
Pre-policy control mean	0.500	0.500	0.500	0.500
p-value: joint test of pre-trends	[0.153]	[0.166]	[0.217]	[0.229]
Villages	100	100	100	100
Observations	1,099	1,099	1,099	1,099
Village fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Wealth controls	No	Yes	Yes	Yes
Population density controls	No	No	Yes	Yes
Distance to market controls	No	No	No	Yes

Notes: This table tests the robustness of the mobile money agent expansion results (Table 3) to restricting the post-treatment window to the pre-COVID period. Panel A uses data through May 2020 (two months after the onset of COVID-19 in Togo); Panel B uses data through March 2020 (the month COVID-19 restrictions began). Both panels estimate the average effect of mobile money agent arrival at the bimonthly village level, following Table 3 and using the interaction-weighted event-study estimator. The *Main Specification* includes village and time fixed effects; the *Robustness Specifications* sequentially add controls for village-level wealth, population density, and distance to the nearest market. Standard errors are in parentheses, clustered at the village level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A11: Subsidy and Mobile Money Agent Effects on Adoption under Alternative Estimators

	Monthly Applications for Solar Home Systems			
	Main Estimator	Alternative Estimators		
	Sun and Abraham (1)	Callaway and Sant'Anna (2)	Borusyak, Jaravel and Spiess (3)	de Chaisemartin and D'Haultfoeuille (4)
Panel A. Villages with Access to Mobile Money Agents				
Subsidy	0.958***	0.956***	0.806***	0.810***
	(0.222)	(0.217)	(0.188)	(0.191)
Pre-policy control mean	0.509	0.509	0.509	0.509
p-value: joint test of pre-trends	[0.801]	[0.225]	[0.475]	[0.942]
Villages	602	602	602	602
Observations	4,214	4,214	4,214	4,214
Panel B. Villages without Access to Mobile Money Agents				
Subsidy	0.349***	0.353***	0.407***	0.326***
	(0.091)	(0.093)	(0.100)	(0.091)
Pre-policy control mean	0.313	0.313	0.313	0.313
p-value: joint test of pre-trends	[0.346]	[0.257]	[0.540]	[0.443]
Villages	1,610	1,610	1,610	1,610
Observations	11,270	11,270	11,270	11,270
Panel C. All Mobile Money Agent Expansion Villages				
Mobile Money Agent Arrival	0.642***	0.557**	0.487**	0.477*
	(0.239)	(0.269)	(0.198)	(0.270)
Pre-policy control mean	0.500	0.500	0.500	0.500
p-value: joint test of pre-trends	[0.151]	[0.171]	[0.102]	[0.153]
Villages	100	100	100	100
Observations	1,499	1,499	1,499	1,499
Comparison group	Last treated	Last treated	Last treated	Last treated
Village fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes

Notes: This table reports the adoption results of Tables 2 and 3 under alternative estimators for staggered treatment timing. The outcome, sample, fixed effects, and clustering are held fixed across columns, so that the columns differ only in the estimator. Column (1) is the interaction-weighted estimator of Sun and Abraham (2021) used throughout the paper; column (2) the doubly robust estimator of Callaway and Sant'Anna (2021); column (3) the imputation estimator of Borusyak, Jaravel and Spiess (2021); and column (4) the estimator of De Chaisemartin and d'Haultfoeuille (2020). Panels A and B report the subsidy effect at the village-month level for villages with and without a mobile money agent within 3 km at baseline, corresponding to Panels B and C of Table 2; Panel C reports the mobile money agent arrival effect at the village-bimonth level for all expansion villages, corresponding to Panel A of Table 3. All four columns use the last treated cohort as the comparison group. For column (3) this is imposed on the estimation sample, since the imputation estimator otherwise also draws on not-yet-treated observations; the resulting sample is marginally smaller than the one reported. Each column reports the equally weighted average of the dynamic treatment effects over the four months following subsidy eligibility (Panels A and B) and the eight two-month periods following agent arrival (Panel C), matching the corresponding rows of the main tables. The exception is column (4), which reports that estimator's average total effect per treated unit, weighting horizons by length of exposure rather than equally. Villages and Observations refer to the common estimation sample of column (1). The test of equality across Panels A and B reported in the main tables has no counterpart in the alternative estimators and is therefore omitted. Standard errors are in parentheses, clustered at the district level in Panels A and B and at the village level in Panel C. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A12: Subsidy and Mobile Money Agent Effects on Installations

	Monthly Installations for Solar Home Systems			
	Main Specification	Robustness Specifications		
	(1)	(2)	(3)	(4)
Panel A. Villages with Access to Mobile Money Agents				
Subsidy	0.884***	0.941***	0.938***	0.946***
	(0.234)	(0.247)	(0.248)	(0.243)
Pre-policy control mean	0.513	0.513	0.513	0.513
p-value: joint test of pre-trends	[0.711]	[0.670]	[0.666]	[0.675]
Villages	602	602	602	602
Observations	4,214	4,214	4,214	4,214
Panel B. Villages without Access to Mobile Money Agents				
Subsidy	0.253***	0.254***	0.235**	0.245**
	(0.095)	(0.095)	(0.094)	(0.101)
Pre-policy control mean	0.302	0.302	0.302	0.302
p-value: joint test of pre-trends	[0.185]	[0.189]	[0.175]	[0.150]
Villages	1,610	1,610	1,610	1,610
Observations	11,270	11,270	11,270	11,270
Panel C. All Mobile Money Agent Expansion Villages				
Mobile Money Agent Arrival	0.627***	0.631***	0.634***	0.656***
	(0.225)	(0.228)	(0.230)	(0.243)
Pre-policy control mean	0.500	0.500	0.500	0.500
p-value: joint test of pre-trends	[0.122]	[0.118]	[0.151]	[0.171]
Villages	100	100	100	100
Observations	1,499	1,499	1,499	1,499
p-value: test of equality (A = B)	[0.013]	[0.010]	[0.008]	[0.006]
Village fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Wealth controls	No	Yes	Yes	Yes
Population density controls	No	No	Yes	Yes
Distance to market controls	No	No	No	Yes

Notes: This table replicates the adoption analysis of Tables 2 and 3 using installations (SHS units physically installed) rather than applications as the outcome. Panels A and B estimate the subsidy effect at the village-month level: Panel A restricts to villages within 3 km of a mobile money agent at baseline; Panel B restricts to villages without such access. Panel C estimates the effect of mobile money agent arrival at the village-bimonth level in the expansion villages. Samples and treatment definitions follow those tables; all effects are estimated using the interaction-weighted event-study estimator. The p-value for the test of equality (A = B) tests whether the subsidy effect differs across the two subsamples. The *Main Specification* includes village and time fixed effects; the *Robustness Specifications* sequentially add controls for village-level wealth, population density, and distance to the nearest market. Standard errors are in parentheses, clustered at the district level in Panels A and B and at the village level in Panel C. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A13: Subsidy and Mobile Money Agent Effects on Adoption by Kit

	Monthly Applications for Solar Home Systems		
	Basic	Plus	Premium
	(1)	(2)	(3)
Panel A. Villages with Access to Mobile Money Agents			
Subsidy	0.432***	0.186**	0.343***
	(0.150)	(0.076)	(0.072)
Pre-policy control mean	0.100	0.084	0.327
p-value: joint test of pre-trends	[0.987]	[0.687]	[0.742]
Villages	602	602	602
Observations	4,214	4,214	4,214
Panel B. Villages without Access to Mobile Money Agents			
Subsidy	0.172***	0.064***	0.115***
	(0.064)	(0.018)	(0.037)
Pre-policy control mean	0.077	0.044	0.193
p-value: joint test of pre-trends	[0.044]	[0.539]	[0.362]
Villages	1,610	1,610	1,610
Observations	11,270	11,270	11,270
Panel C. All Mobile Money Agent Expansion Villages			
Mobile Money Agent Arrival	0.339*	0.098	0.205*
	(0.186)	(0.067)	(0.122)
Pre-policy control mean	0.202	0.060	0.238
p-value: joint test of pre-trends	[0.415]	[0.170]	[0.300]
Villages	100	100	100
Observations	1,499	1,499	1,499
p-value: test of equality (A = B)	[0.148]	[0.107]	[0.001]

Notes: This table reports the effects of subsidy eligibility (Panels A and B) and mobile money agent arrival (Panel C) on monthly village-level applications for solar home systems, disaggregated by kit type: the outcome in each column is the number of applications for *Basic*, *Intermediary*, and *Premium* kits, respectively. Panels A and B report the subsidy effect for villages with and without a mobile money agent within 3 km at baseline, corresponding to Panels B and C of Table 2; Panel C reports the mobile money agent arrival effect for all expansion villages, corresponding to Panel A of Table 3. Samples, treatment definitions, and estimation follow those tables, using the interaction-weighted event-study estimator. The p-value for the test of equality (A = B) tests whether the subsidy effect differs between villages with and without mobile money access. All specifications include village and time fixed effects. Standard errors are in parentheses, clustered at the district level in Panels A and B and at the village level in Panel C. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A14: Subsidy and Mobile Money Agent Effects on Adoption by Customer Types

	Monthly Applications for Solar Home Systems					
	Primary Income		Age		Gender	
	Agriculture (1)	Other (2)	18-34 (3)	35+ (4)	Men (5)	Women (6)
Panel A. Villages with Access to Mobile Money Agents						
Subsidy	0.639***	0.320***	0.349***	0.608***	0.887***	0.071***
	(0.139)	(0.111)	(0.094)	(0.142)	(0.209)	(0.022)
Pre-policy control mean	0.356	0.152	0.174	0.334	0.462	0.047
p-value: joint test of pre-trends	[0.608]	[0.230]	[0.665]	[0.767]	[0.784]	[0.867]
Villages	602	602	602	602	602	602
Observations	4,214	4,214	4,214	4,214	4,214	4,214
Panel B. Villages without Access to Mobile Money Agents						
Subsidy	0.301***	0.047	0.191***	0.159***	0.330***	0.019*
	(0.092)	(0.029)	(0.057)	(0.043)	(0.086)	(0.011)
Pre-policy control mean	0.259	0.053	0.102	0.211	0.281	0.032
p-value: joint test of pre-trends	[0.342]	[0.784]	[0.228]	[0.470]	[0.351]	[0.460]
Villages	1,610	1,610	1,610	1,610	1,610	1,610
Observations	11,270	11,270	11,270	11,270	11,270	11,270
Panel C. All Mobile Money Agent Expansion Villages						
Mobile Money Agent Arrival	0.389*	0.253***	0.340**	0.301**	0.480**	0.161**
	(0.213)	(0.088)	(0.136)	(0.137)	(0.198)	(0.080)
Pre-policy control mean	0.357	0.143	0.167	0.333	0.435	0.065
p-value: joint test of pre-trends	[0.280]	[0.423]	[0.473]	[0.145]	[0.130]	[0.260]
Villages	100	100	100	100	100	100
Observations	1,499	1,499	1,499	1,499	1,499	1,499
p-value: test of equality (A = B)	[0.061]	[0.010]	[0.147]	[0.005]	[0.020]	[0.053]

Notes: This table reports the effects of subsidy eligibility and mobile money agent arrival on monthly village-level applications, disaggregated by customer type. The outcome in each column is the number of applications from customers in the indicated group: primary income source (*Agriculture* vs. *Other*), age group (*18–34* vs. *35+*), and gender (*Men* vs. *Women*). Panels A and B estimate the subsidy effect for villages with and without baseline mobile money agent access (≤3 km), respectively; Panel C estimates the effect of mobile money agent arrival at the bimonthly level in the expansion villages. Samples and treatment definitions follow Tables 2 and 3; all effects are estimated using the interaction-weighted event-study estimator. The p-value for the test of equality (A = B) tests whether the subsidy effect differs across the two subsamples. All specifications include village and time fixed effects. Standard errors are in parentheses, clustered at the district level in Panels A and B and at the village level in Panel C. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A15: Subsidy and Mobile Money Agent Effects on Energy Usage

	Any Usage (1)	Majority Usage (2)	Usage Share (3)
Panel A. Villages with Access to Mobile Money Agents			
Subsidy	-0.019	-0.037*	-0.028
	(0.013)	(0.021)	(0.018)
Pre-policy control mean	0.990	0.974	0.955
p-value: joint test of pre-trends	[0.185]	[0.020]	[0.082]
Customers	2,994	2,994	2,994
Observations	18,807	18,807	18,807
Panel B. Villages without Access to Mobile Money Agents			
Subsidy	-0.002	-0.018	-0.012
	(0.026)	(0.035)	(0.032)
Pre-policy control mean	0.987	0.968	0.941
p-value: joint test of pre-trends	[0.763]	[0.185]	[0.085]
Customers	4,499	4,499	4,499
Observations	27,736	27,736	27,736
Panel C. All Mobile Money Agent Expansion Villages			
Mobile Money Agent Arrival	-0.027	-0.036	-0.012
	(0.028)	(0.041)	(0.042)
Pre-policy control mean	0.981	0.917	0.886
p-value: joint test of pre-trends	[0.006]	[0.008]	[0.017]
Customers	535	535	535
Observations	5,669	5,669	5,669
p-value: test of equality (A = B)	[0.412]	[0.471]	[0.543]

Notes: This table reports the effects of subsidy eligibility and mobile money agent arrival on customer-level energy usage. *Any Usage* is a binary indicator for whether the customer used any electricity that period; *Majority Usage* is a binary indicator for whether the customer used electricity on the majority of days; *Usage Share* is the share of days with electricity usage. Panels A and B estimate the subsidy effect at the customer-month level for villages with and without baseline mobile money agent access (≤3 km), respectively; Panel C estimates the effect of mobile money agent arrival at the customer-bimonth level for existing customers in the expansion villages. Samples and treatment definitions follow Tables 2 and 3; all effects are estimated using the interaction-weighted event-study estimator. The p-value for the test of equality (A = B) tests whether the subsidy effect differs across the two subsamples. All specifications include customer and time fixed effects. Standard errors are in parentheses, clustered at the district level in Panels A and B and at the village level in Panel C. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A16: Subsidy and Mobile Money Agent Effects on Payment Coordination

	Share of Paying Customers Using Third Party Payments (1)	Share of Payments by Third Parties (2)	Share of New Customers Using Third Party Payments (3)
Panel A. Villages with Access to Mobile Money Agents			
Subsidy	-0.123***	-0.110***	-0.087
	(0.044)	(0.040)	(0.102)
Pre-policy control mean	0.403	0.371	0.490
p-value: joint test of pre-trends	[0.911]	[0.849]	[0.124]
Villages	432	432	216
Observations	2,543	2,543	788
Panel B. Villages without Access to Mobile Money Agents			
Subsidy	-0.094*	-0.084*	-0.157*
	(0.048)	(0.048)	(0.095)
Pre-policy control mean	0.450	0.416	0.613
p-value: joint test of pre-trends	[0.042]	[0.016]	[0.308]
Villages	1,048	1,048	452
Observations	5,725	5,725	1,352
Panel C. All Mobile Money Agent Expansion Villages			
Mobile Money Agent Arrival	0.029	-0.018	0.043
	(0.064)	(0.045)	(0.115)
Pre-policy control mean	0.389	0.259	0.395
p-value: joint test of pre-trends	[0.384]	[0.318]	[0.028]
Villages	97	97	80
Observations	1,121	1,121	464
p-value: test of equality (A = B)	[0.602]	[0.649]	[0.633]

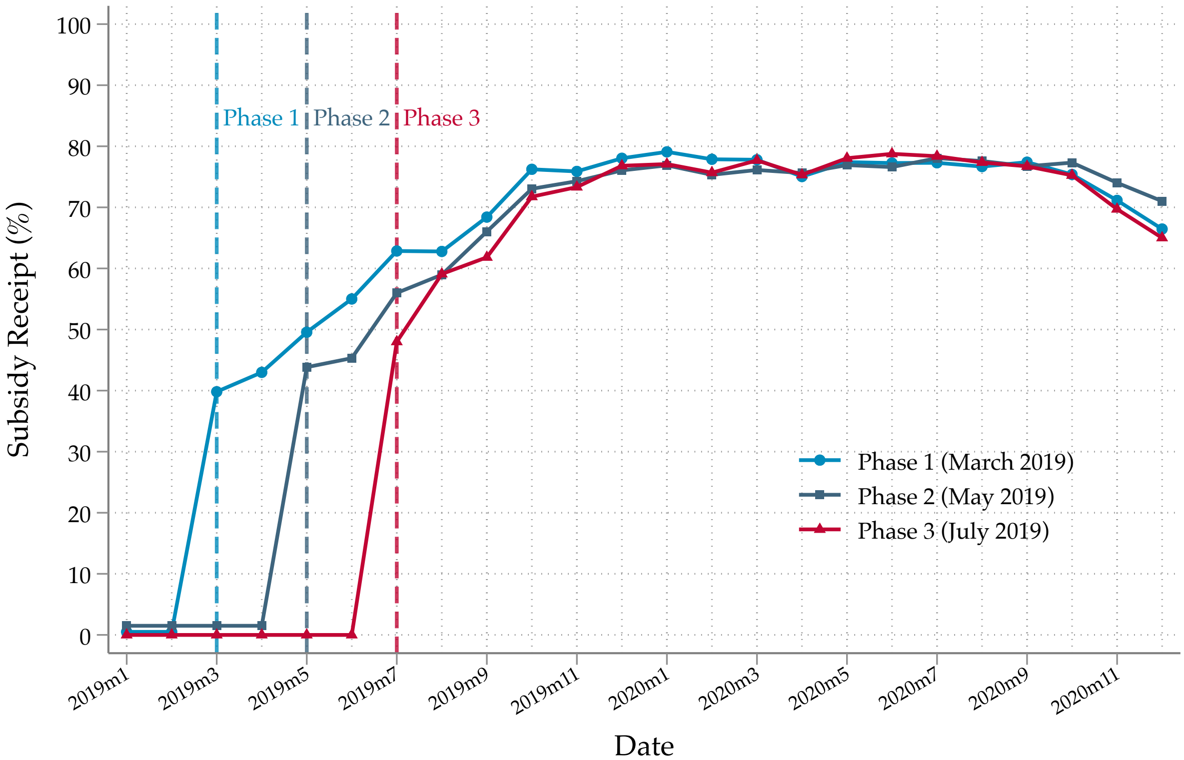
Notes: This table reports the effects of subsidy eligibility and mobile money agent arrival on third-party payment coordination at the village level. *Share of Paying Customers Using Third Party Payments* is the share of paying customers who use a third party's phone to make at least one payment; *Share of Payments by Third Parties* is the share of total payments made via a third party's phone; *Share of New Customers Using Third Party Payments* is the share of new customers whose first payment is made via a third party's phone. The sample is restricted to village-periods with at least one paying customer (first two outcomes) or at least one new customer (third outcome). Panels A and B estimate the subsidy effect at the village-month level for villages with and without baseline mobile money agent access ($\leq$3 km), respectively; Panel C estimates the effect of mobile money agent arrival at the bimonthly village level in the expansion villages, with analogous sample restrictions. Samples and treatment definitions otherwise follow Tables 2 and 3; all effects are estimated using the interaction-weighted event-study estimator. The p-value for the test of equality (A = B) tests whether the subsidy effect differs across the two subsamples. All specifications include village and time fixed effects. Standard errors are in parentheses, clustered at the district level in Panels A and B and at the village level in Panel C. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A17: Mobile Money Agent Effects on Subsidy Disbursement

	Share of Payments Getting the Subsidy (1)	Share of New Customers Getting the Subsidy (2)
Panel A. All Mobile Money Agent Expansion Villages		
Mobile Money Agent Arrival	-0.079	0.033
	(0.064)	(0.129)
Pre-policy post-subsidy control mean	0.559	0.526
p-value: joint test of pre-trends	[0.379]	[0.708]
Villages	97	80
Observations	1,121	464

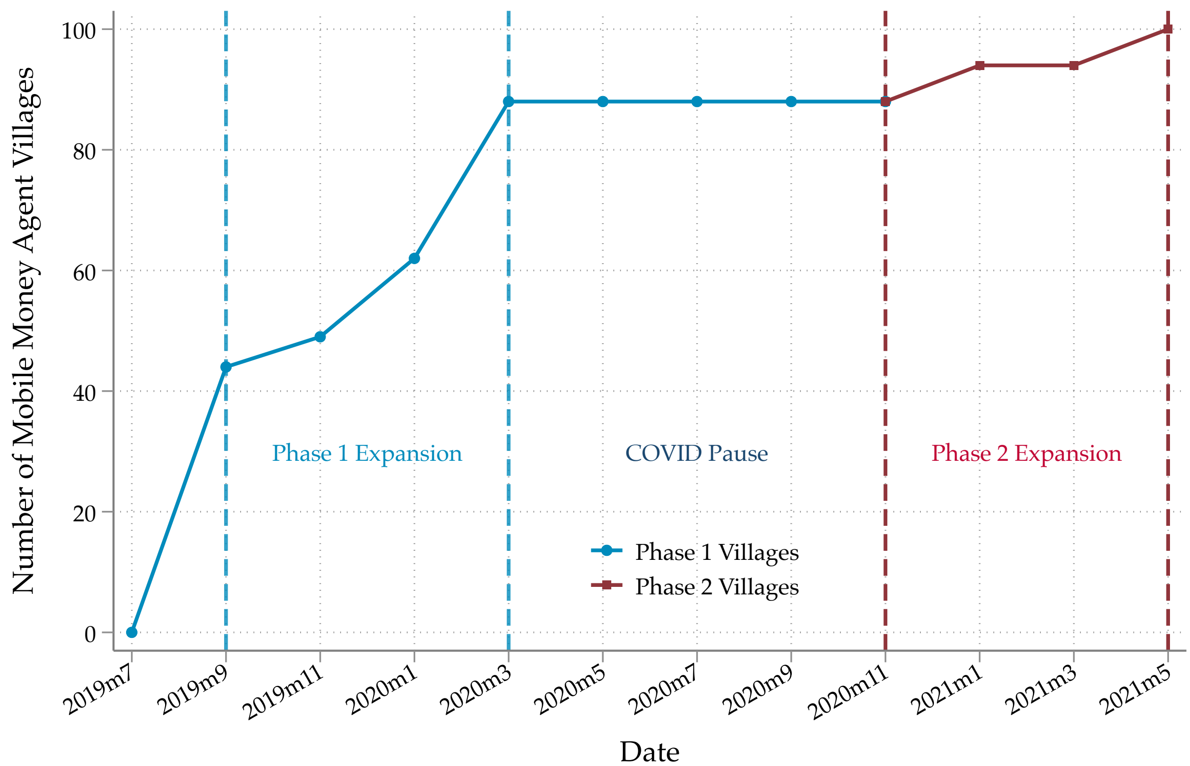
Notes: This table tests whether mobile money agent arrival affects subsidy disbursement rates at the village level, using only the post-subsidy period within the mobile money agent expansion sample. *Share of Payments Getting the Subsidy* is the share of payments that received the subsidy; *Share of New Customers Getting the Subsidy* is the share of new customers who received the subsidy on their first payment; the sample for this second outcome is smaller because it conditions on village-bimonths with at least one new customer. The effect of agent arrival is estimated at the bimonthly village level, following Table 3 and using the interaction-weighted event-study estimator. All specifications include village and time fixed effects. Standard errors are in parentheses, clustered at the village level. The p-value for the joint test of pre-trends tests the null that all pre-treatment coefficients are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure A1: Subsidy Policy Rollout



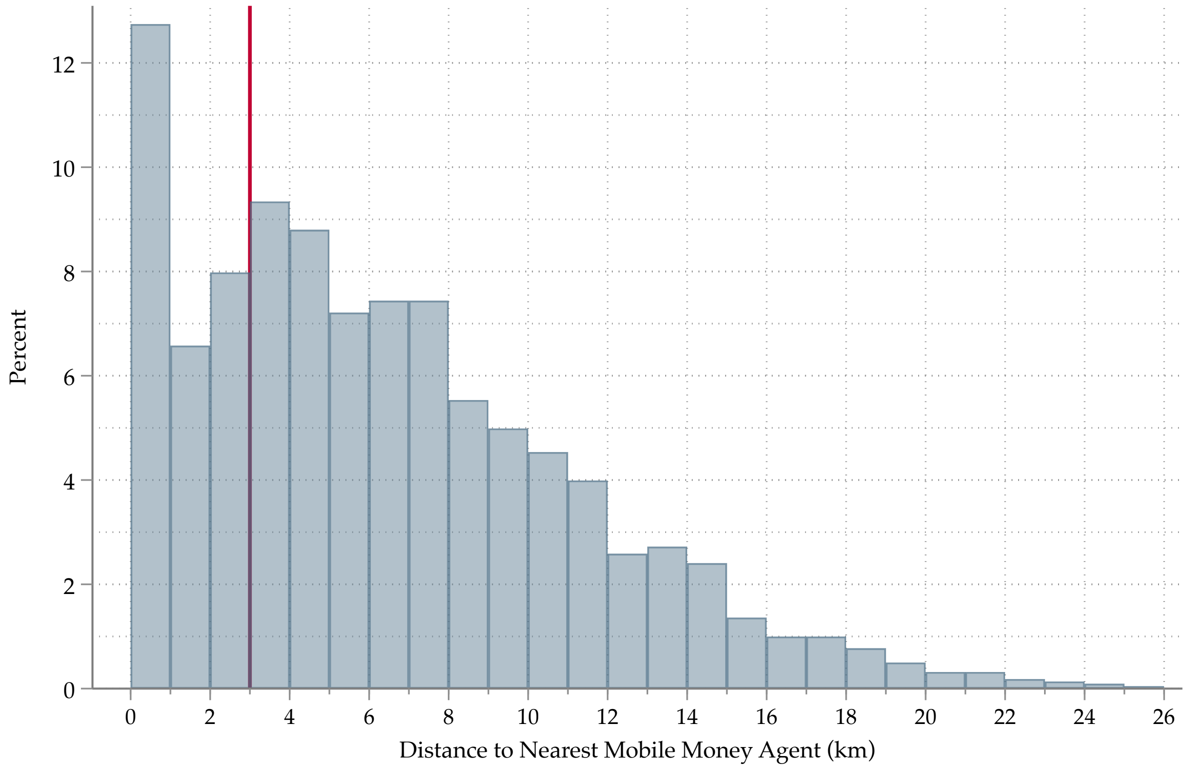
Notes: This figure plots the share of solar home system (SHS) payments that received the subsidy top-up in each month, separately for the three subsidy rollout phases: Phase 1 (districts treated in March 2019), Phase 2 (May 2019), and Phase 3 (July 2019). Subsidy receipt is measured by matching the telco subsidy disbursement records to the solar company's payment data; a payment is subsidised only when it is made from the customer's own mobile money account linked to their solar account, so subsidy receipt is incomplete, particularly in the early months of the rollout.

Figure A2: Mobile Money Agent Policy Rollout



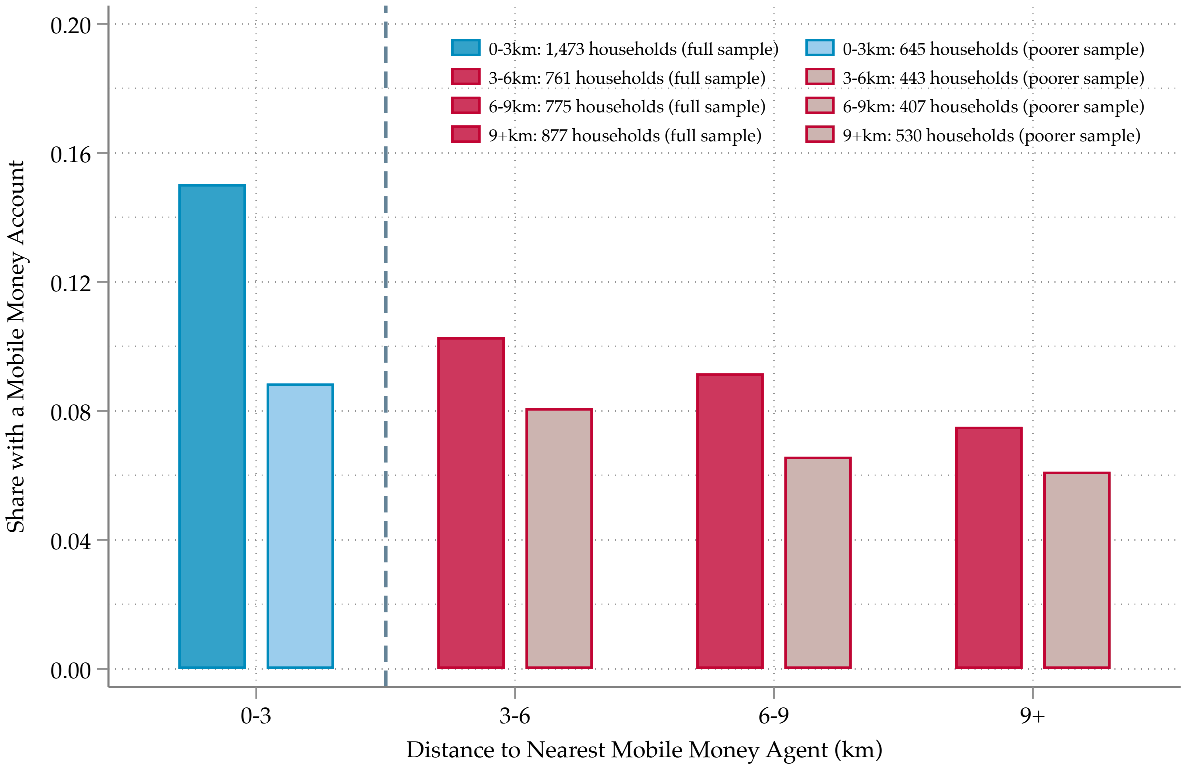
Notes: This figure plots the evolution of the number of villages with a mobile money agent within 3 km, due to the expansion campaign described in Section 1.5. The vertical lines demarcate the two waves of the expansion: the first between September 2019 and March 2020, and the second between November 2020 and May 2021.

Figure A3: Distance to Nearest Mobile Money Agent Across Study Villages, 2019



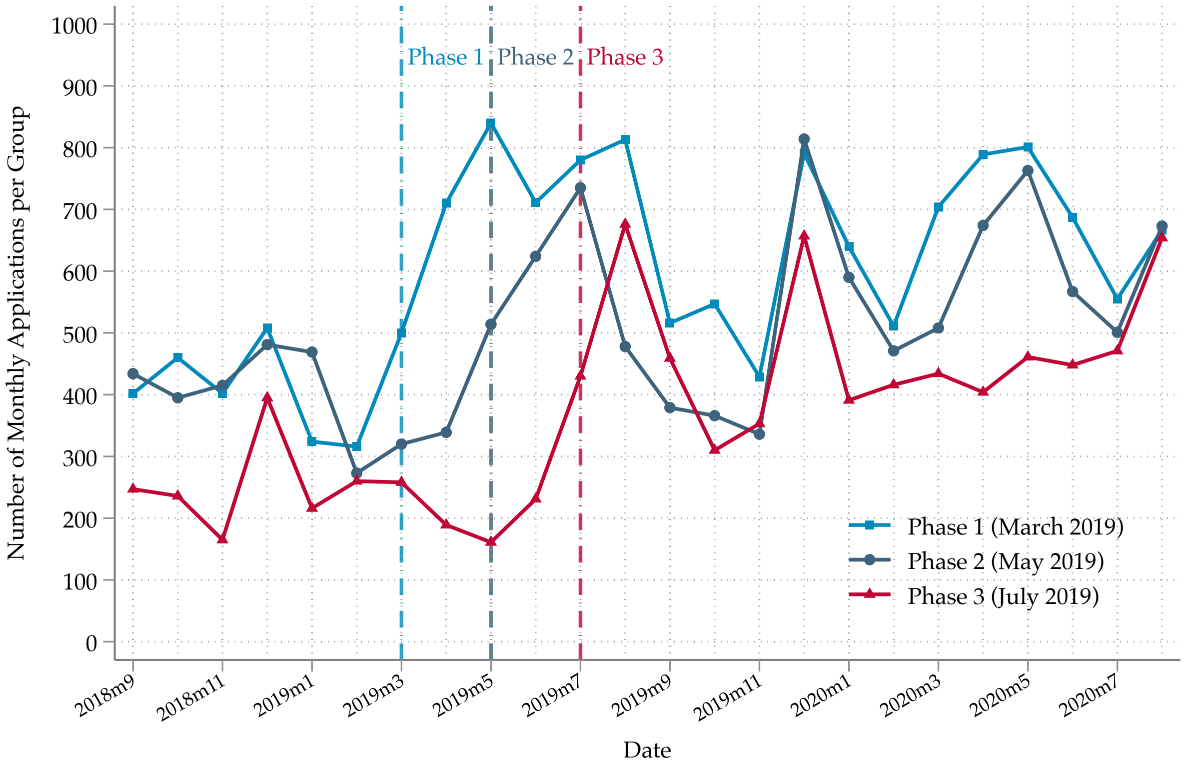
Notes: This figure plots the distribution of village distance to the nearest mobile money agent in 2019, before the subsidy was rolled out, across all villages in our data. Distances are computed by combining data from the Togolese government on the locations of all mobile money agents in the country with the GPS coordinates of the villages in our sample.

Figure A4: Rural Mobile Money Account Ownership by Distance to Nearest Agent, 2018-2019



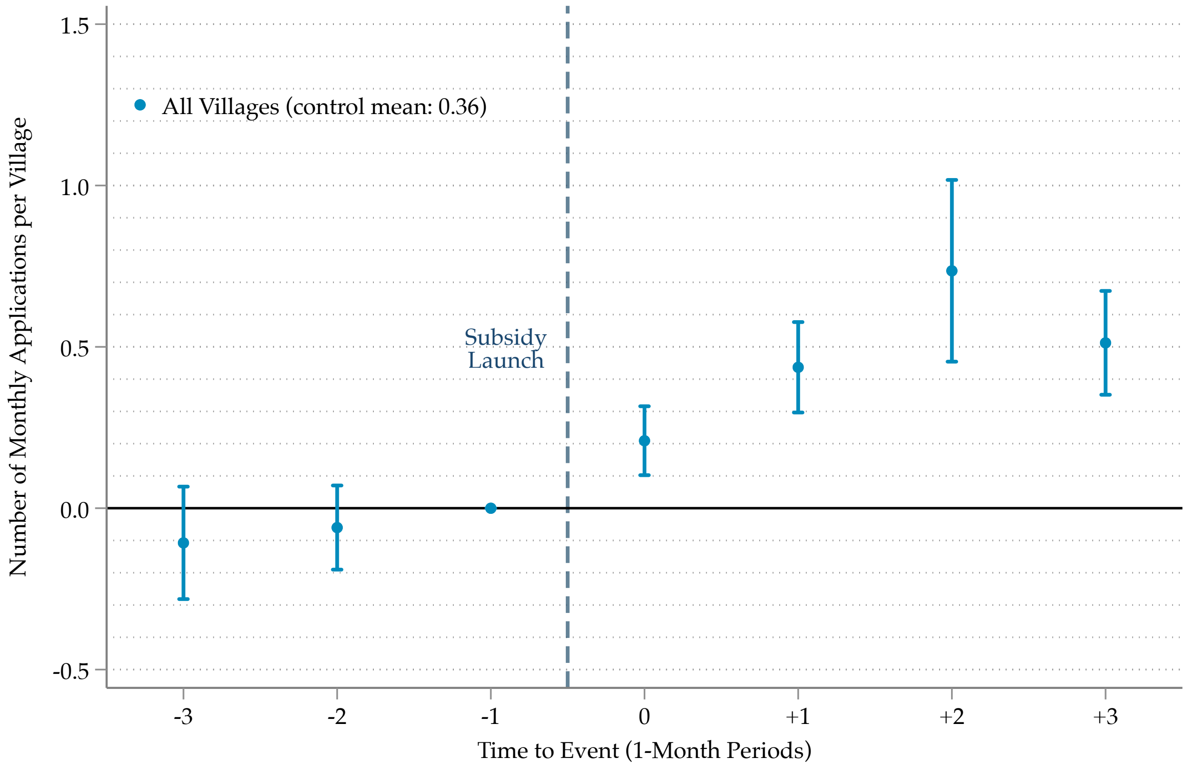
Notes: This figure plots the share of respondents reporting ownership of a mobile money account, by the distance from their survey cluster to the nearest mobile money agent. Data are from the 2018–2019 *Enquête Harmonisée sur les Conditions de Vie des Ménages* (EHCVM), a nationally representative household survey conducted by INSEED under the WAEMU household survey harmonisation programme (Institut National de la Statistique et des Études Économiques et Démographiques, 2019). The sample is restricted to the 333 rural survey clusters with GPS coordinates, covering 9,737 respondents in 3,886 households. Distances combine the GPS coordinates of the survey cluster with the data on mobile money agent locations in 2018. Solid bars report the full sample; shaded bars report respondents in households with per capita consumption below the national poverty line (273,619 CFA per person per year), 2,025 of the 3,886 households. Shares are weighted by household sampling weights.

Figure A5: Aggregate Adoption of Solar Home Systems



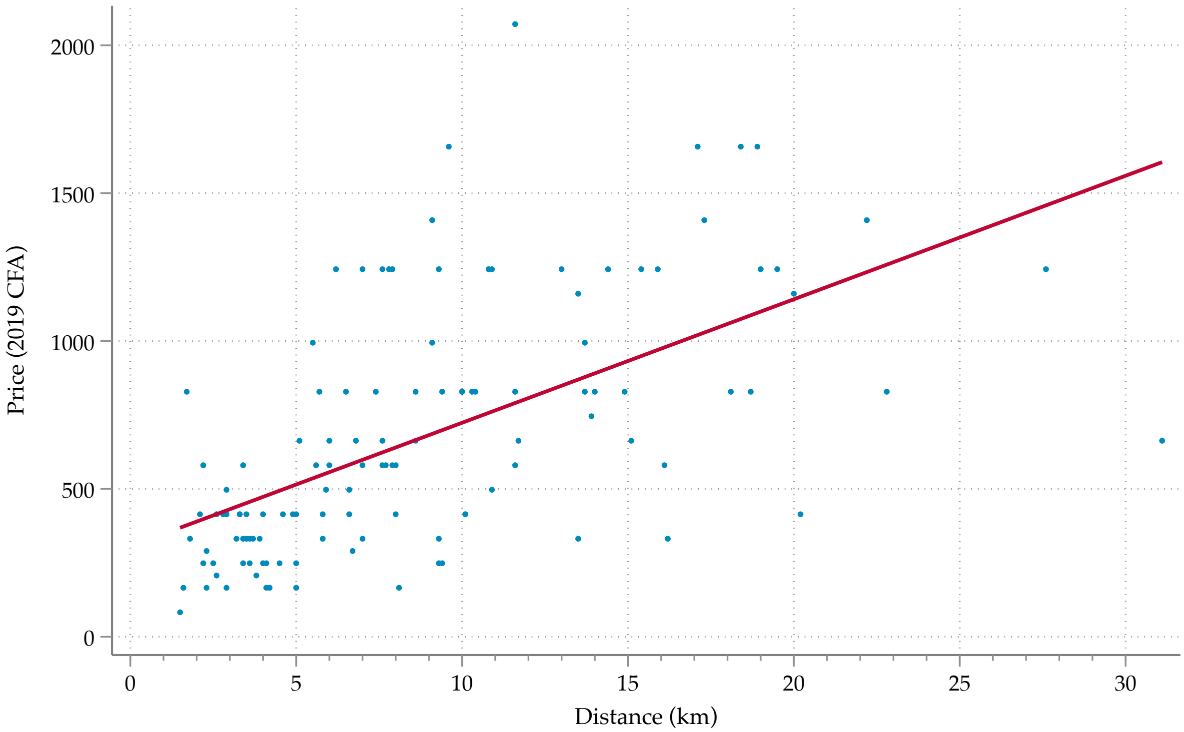
Notes: This figure plots the total number of new SHS applications in each month by subsidy rollout phase (see Section 4.1).

Figure A6: Subsidy Effects on Adoption



Notes: This figure displays village-level dynamic ITT effects of subsidy eligibility on solar home system (SHS) adoption, estimated using the interaction-weighted event-study estimator. The outcome is the number of SHS applications per month in each village. The sample, treatment, and control groups follow Table 2; the mean of the outcome in the control group prior to subsidy launch is reported in the legend. Vertical bars are 95% confidence intervals based on standard errors clustered at the district level; the dashed vertical line marks the subsidy launch.

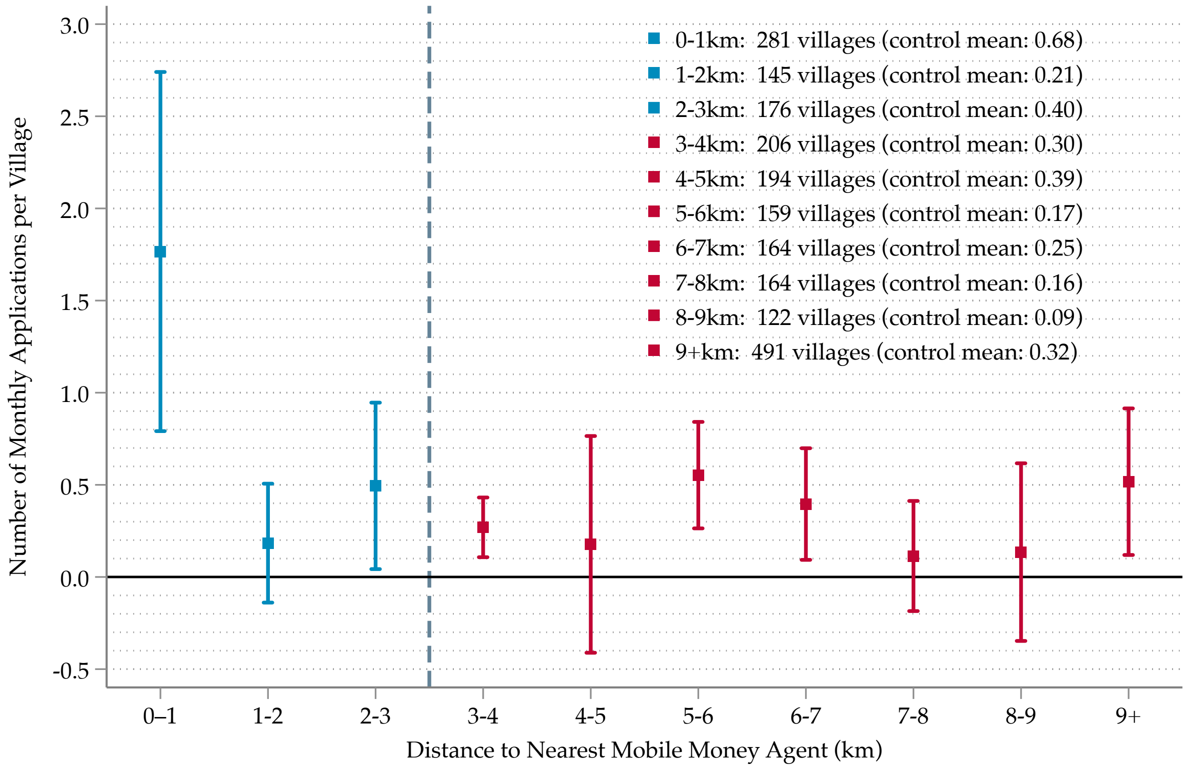
Figure A7: Travel Distance and Travel Cost



Linear fit: price = 306 + 42 × distance

Notes: This figure shows the relationship between travel distance and monetary travel cost based on the transport cost data collected in rural Togo (see Section 3.5). Each observation corresponds to one hypothetical trip elicited from respondents during the transport cost survey. Distance is measured in kilometers and travel cost in 2019 CFA francs, deflated from the 2023 survey using the World Bank consumer price index for Togo.

Figure A8: Subsidy Effect on Adoption by 1km Increment Distances to Nearest Agents



Notes: This figure displays village-level heterogeneous ITT effects of subsidy eligibility on solar home system (SHS) adoption, estimated separately for ten groups of villages defined by their pre-subsidy distance to the nearest mobile money agent, in 1 km bands from 0–1 km to 9+ km. Each group is estimated using the interaction-weighted event-study estimator; the outcome is the number of SHS applications per month in each village, and the sample, treatment, and control groups follow Table 2. Vertical bars are 95% confidence intervals based on standard errors clustered at the district level; the dashed vertical line marks the subsidy launch. Table 2 reports the corresponding average effects.

B Information Campaigns

A potential concern is that the estimated effect of the subsidy may partly reflect changes in information rather than changes in price. This concern is especially relevant in rural settings, where awareness of new technologies may be limited. To assess this possibility, we examine the timing of marketing campaigns conducted by the solar provider around the subsidy rollout. Because the available data cover only the provider's campaigns, this exercise should be interpreted as a partial check on information-based confounding rather than a full test of all information channels.

The campaign data are aggregated at the store level. The provider operated 22 stores, each of which was assigned responsibility for one or more districts, yielding campaign information for 29 districts in total. These data do not allow us to observe which villages within a district were targeted in a given period. Our analysis therefore captures broad district-level marketing activity rather than village-level exposure.

We group campaigns into three categories: (i) *market* campaigns, which involve proactive outreach in major market areas; (ii) *radio* campaigns, including advertisements and radio shows; and (iii) *other* campaigns, which combine festivals, children-focused events, and door-to-door outreach. Appendix Figure B1 plots the number and type of campaigns conducted in each bi-month period, while Appendix Table B1 reports the share of District $\times$ Bi-month observations in which each type of campaign occurred. Market visits and radio campaigns are the most common forms of outreach.

The key question is whether provider marketing intensified at the time the subsidy was introduced. Columns (2) and (3) of Appendix Table B1 compare the two bi-month periods immediately before and after the subsidy launch in each district. We find no evidence of an increase in market campaigns or in the residual "other" category around rollout. If anything, the share of districts receiving radio campaigns declines modestly in the post-launch period, with the difference significant at the 10% level. Overall, the descriptive evidence suggests that the subsidy was not systematically accompanied by a contemporaneous increase in the provider's recorded marketing activity.

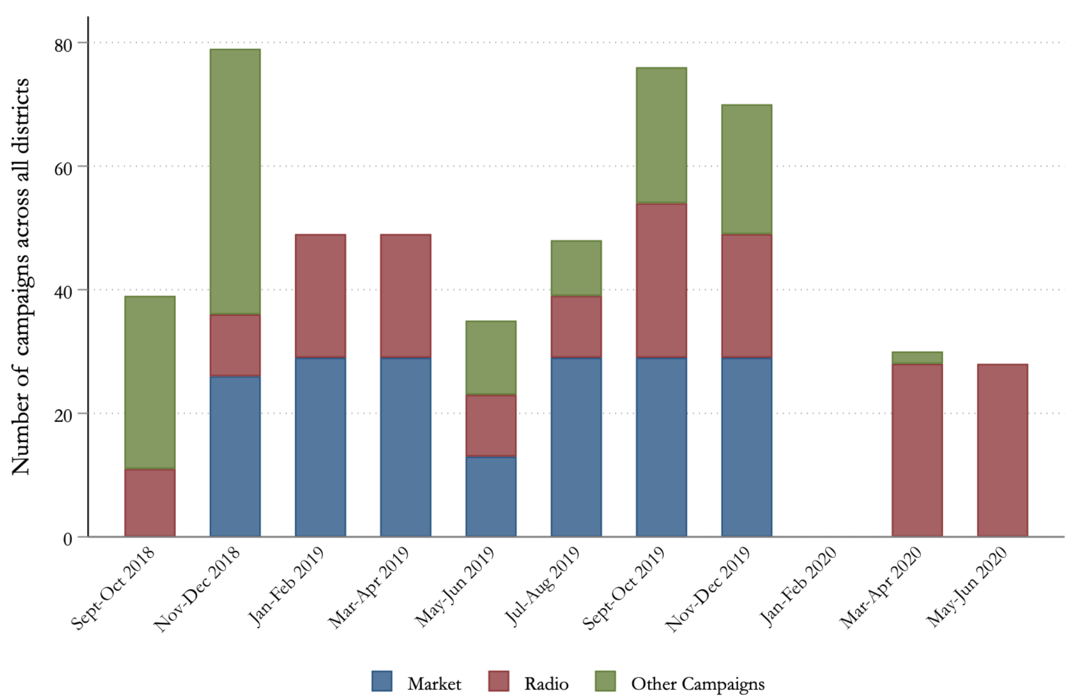
Taken together, these patterns reduce the concern that our subsidy estimates are driven by a coincident increase in observed provider-led campaigns. At the same time, this evidence has important limitations. First, the data are measured at the store/district level and cannot capture within-district targeting. Second, the data cover campaigns conducted by the solar company, not broader information efforts by the government or other actors. We therefore view this section as a descriptive check showing that the provider's recorded district-level marketing activity does not appear to rise at subsidy launch, rather than as a definitive test that information played no role.

Table B1: Percentage of District x Bi-months targeted by marketing campaigns

	(1) September 2018 - May 2020	(2) 2 months pre-subsidy	(3) 2 months post-subsidy	(4) T-stat difference
Market	55.76	73.33	83.33	(1.14)
	(49.74)	(44.98)	(37.90)	
Radio	55.15	56.67	50.00	(-1.86*)
	(49.81)	(50.40)	(50.85)	
Other campaigns	33.86	17.24	20.69	(-0.33)
	(47.40)	(38.44)	(41.23)	
N	319	29	29	

Notes: The table summarizes the proportion of districts targeted by various marketing campaigns across each bi-month period. The first column shows this proportion over the full sample period (September 2018 - May 2020). The second and third columns show the proportion in the two months pre- and post- subsidy launch (these specific months vary by prefecture). The fourth column shows the t-stat of the difference between the two months pre- and post-subsidy, where the following OLS regression equation is performed $Y_i = \beta Post - Subsidy + \epsilon_i$, with standard errors clustered at the district-level (with the sample restricted only to two months pre- and post- subsidy launch). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B1: Number of District-level Information Campaigns



Notes: This figure plots the number and type of information campaigns conducted by the solar company around the time of the subsidy rollout.

C Extension: Payment Infrastructure and Cash Transfers

C.1 Background on the Novissi Program

In April 2020, the Togolese government launched *Novissi* ("solidarity" in Ewe), a nationwide digital cash transfer program designed to support informal workers affected by the COVID-19 pandemic (Aiken et al., 2022). The program targeted the 100 poorest cantons in Togo, identified through a machine learning algorithm that combined satellite imagery and mobile phone metadata with survey data to produce a poverty map. Within these cantons, eligible individuals were identified using phone-based proxy means testing and received monthly transfers. Transfers were distributed between November 2020 and August 2021.

Critically, Novissi transfers were delivered exclusively through mobile money. To receive a payment, beneficiaries needed a registered mobile money account, and to convert the digital transfer into cash for daily purchases, they needed to visit a mobile money agent. This design feature creates a direct link between the density of the mobile money agent network and the effective value of the transfer: in cantons with limited agent access, beneficiaries face higher costs—in time, transport, and forgone earnings—to cash out their transfers.

Aiken et al. (2025) evaluate the impacts of Novissi using a randomised controlled trial embedded within the program, in which approximately 49,000 individuals were randomly assigned to receive or not receive the transfer. Using follow-up phone surveys, the authors find positive and significant effects on food security, mental health, and a combined index of seven well-being outcomes, though they document geographic heterogeneity, with stronger effects in the Savanes region.

C.2 Data and Empirical Strategy

We combine two data sources. First, the individual-level experimental data from Aiken et al. (2025), which contains treatment assignment, sampling weights, enumerator and strata identifiers, and seven standardised outcome indices for 9,511 individuals across 260 cantons: food security, financial health, financial inclusion, mental health, perceived social status, health care access, and labour supply, as well as a combined index of all seven measures.[36] Second, we use government administrative data on the GPS location of every mobile money agent in Togo, matched to the universe of 2,686 villages across 385 cantons. For each village, we compute the distance to the

[36]Following Aiken et al. (2025), outcome indices are constructed using the methodology of Bryan, Chowdhury and Mobarak (2021). *Food security* captures the number of meals, frequency of food restrictions, and consumption of specific food items over the past week. *Financial health* measures savings, the duration a household could sustain itself without income, and the amount of money obtainable in an emergency. *Financial inclusion* is the fraction of household adults with a bank account or mobile money account. *Mental health* is the Kessler K6 nonspecific distress scale, measuring the frequency of feeling nervous, hopeless, restless, that everything was an effort, extremely sad, or worthless over the past seven days. *Perceived status* captures self-rated and community-perceived socioeconomic standing on a 1–5 scale. *Health care access* measures whether the household obtained health care when needed, whether it was received at a hospital, and the ability to bring a child to the hospital. *Labour supply* measures hours worked and income received in the past week. Each index is standardised to have zero mean and unit variance in the control group.

Table C1: Mobile Money Agent Access Across Cantons in the Novissi Sample

	Mean	SD	P25	Median	P75	N
Panel A. Share of Villages with Agent within Threshold						
Within 1 km	0.283	0.296	0.000	0.211	0.442	236
Within 2 km	0.371	0.322	0.062	0.333	0.586	236
Within 3 km	0.478	0.355	0.155	0.485	0.750	236
Within 5 km	0.652	0.343	0.429	0.667	1.000	236
Within 10 km	0.884	0.269	1.000	1.000	1.000	236
Panel B. Canton-Level Average Distance to Nearest Agent (km)						
Mean across villages	4.57	4.65	2.00	3.35	5.45	236
Panel C. Canton Characteristics						
Number of villages	8.11	5.63	4.00	7.00	11.00	236
Share with no agent $\leq$ 3 km	19.5% of cantons (46 of 236)					236
Share with any agent $\leq$ 3 km	80.5% of cantons (190 of 236)					236
Share with all villages $\leq$ 3 km	17.8% of cantons (42 of 236)					236

Notes: Each observation is a canton in the matched Novissi sample—the cantons successfully linked between the Novissi individual data and the mobile money agent data (see Section C.2). Panel A reports the share of villages in each canton with at least one mobile money agent (any operator) within the indicated distance in 2020. Panel B reports the canton-level average distance (in km) to the nearest agent, computed as the mean across the canton's villages. Panel C reports canton characteristics: the number of villages, and the share of cantons in which no village, at least one village, or all villages lie within 3 km of an agent. Columns report the cross-canton mean, standard deviation, 25th/50th/75th percentiles, and the number of cantons (N). All measures refer to 2020, the year of the Novissi program.

nearest mobile money agent of any operator in 2020, the year of the Novissi program.

We construct a canton-level measure of mobile money access: the share of villages within each canton that have at least one mobile money agent within 3 kilometres. This distance threshold is consistent with the definition used in our main analysis of the agent expansion policy (Section 1.5), where we define a village as having direct access when an agent operates within 3 km. We match canton names across the two datasets, successfully linking 236 of 260 cantons (covering 9,284 of 9,511 individuals, or 97.6%).

Appendix Table C1 reports summary statistics for mobile money access across the cantons in the Novissi sample. There is substantial variation: the average canton has 47.8% of its villages within 3 km of a mobile money agent, but the distribution ranges from 0% to 100%. Roughly one in five cantons (19.5%) has no village within 3 km of an agent, while only 17.8% have full coverage. The median canton-level average distance to the nearest agent is 3.4 km, but the 75th percentile is 5.5 km, indicating that a sizeable fraction of the population in the poorest cantons faces non-trivial travel distances to access mobile money services.[37] These distances are comparable to those documented in Section 1.1 for the broader population of Togolese villages.

We estimate the following specification, which interacts the randomly assigned treatment in-

[37]Since we observe only each respondent's canton, not their village, we assign each canton the mean distance to the nearest agent across its constituent villages. At the individual level, the sampling-weighted median distance is 3.1 km.

dicator with canton-level mobile money access:

$$Y_i = \beta_1 \,\text{Treatment}_i + \beta_2 \,\text{MoMoAccess}_{c(i)} + \beta_3 \,(\text{Treatment}_i \times \text{MoMoAccess}_{c(i)}) + \alpha_s + \gamma_t + \delta_{e(i)} + \varepsilon_i, \tag{C1}$$

where Y_i is a standardised outcome index for individual i; Treatment$_i$ is the randomly assigned indicator; MoMoAccess$_{c(i)}$ is the share of villages with an agent within 3 km in canton c, standardised to have mean zero and unit standard deviation; α_s are strata fixed effects; δ_e are enumerator fixed effects; γ_t are survey week fixed effects. Observations are weighted using the sampling and response weights from Aiken et al. (2025). The coefficient of interest is β_3: if payment infrastructure is complementary to cash transfer effectiveness, we expect $\beta_3 > 0$ for outcomes where the transfer operates through monetary channels.

Table 5 in the main text reports the estimates. Appendix Table C2 reproduces its Panels A and B with the seven outcome indices reported separately, so that the components of the combined index in column (4) of Table 5 can be inspected individually.

Table C2: Heterogeneous Treatment Effects of Novissi by Mobile Money Access: All Seven Outcomes Separately

	Food Security (1)	Financial Health (2)	Health Care Access (3)	Financial Inclusion (4)	Mental Health (5)	Perceived Status (6)	Labour Supply (7)
Panel A. Main Treatment Effects (Matched Sample)							
Treatment	0.074***	0.035	0.007	0.006	0.071***	0.039*	0.014
	(0.022)	(0.024)	(0.023)	(0.022)	(0.018)	(0.022)	(0.025)
Control mean	−0.004	−0.002	0.013	0.004	0.007	−0.004	0.005
Observations	9,284	9,284	9,284	9,284	9,284	9,284	9,284
Panel B. Heterogeneous Effects by Mobile Money Access							
Treatment	0.073***	0.031	0.004	0.006	0.073***	0.038*	0.013
	(0.022)	(0.024)	(0.023)	(0.022)	(0.018)	(0.022)	(0.025)
Treatment × MoMo Access	0.013	0.052**	0.036	0.001	−0.020	0.015	0.011
	(0.022)	(0.025)	(0.024)	(0.022)	(0.019)	(0.023)	(0.028)
MoMo Access	−0.025*	−0.050***	−0.015	0.008	0.008	0.008	−0.016
	(0.015)	(0.019)	(0.017)	(0.016)	(0.014)	(0.017)	(0.023)
Control mean	−0.004	−0.002	0.013	0.004	0.007	−0.004	0.005
Observations	9,284	9,284	9,284	9,284	9,284	9,284	9,284
Strata fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Enumerator fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Survey week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports Panels A and B of Table 5 for each of the seven outcome indices separately; Table 5 combines columns (4)–(7) into a single index. Panel A reproduces the main treatment-effect specification of the original Novissi evaluation, estimated on the matched sample—the individuals in cantons successfully linked to the mobile money data—so that its estimation sample coincides with Panel B. Panel B adds the interaction with mobile money access, estimated by OLS on the same matched sample. Each outcome is a standardised index (mean zero, unit standard deviation in the control group), grouped into consumption and assets (food security and financial health) and other outcomes (health care access, financial inclusion, mental health, perceived status, and labour supply), as indicated in the column headers. *Treatment* is the randomly assigned indicator for receiving the Novissi cash transfer. *MoMo Access* is the share of villages in the respondent's canton with at least one mobile money agent within 3 km, standardised to mean zero and unit standard deviation across cantons. All regressions include strata, enumerator, and survey week fixed effects, and are weighted by the sampling and response weights from the original study. Heteroskedasticity-robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.3 Transaction Costs and the Effective Value of Cash Transfers

The complementarity between mobile money access and cash transfer effectiveness documented above is consistent with the transaction cost mechanism analysed in the main body of this paper. To make this link concrete, we compare the magnitude of the Novissi transfers to the transport costs that beneficiaries face when cashing out.

Novissi transfers were 8,620 CFA ($15.50) for women and 7,450 CFA ($13) for men per month. To convert these digital transfers into cash, beneficiaries had to travel to the nearest mobile money agent. Using the transport cost data described above, where a round trip costs approximately 650 CFA plus 90 CFA per kilometre (a 325 CFA fixed fare plus 44 CFA per kilometre, each way), we can estimate the share of each transfer consumed by transport costs at different distances. Monetary values here are deflated from the 2023 survey to 2021 CFA using the World Bank consumer price index for Togo, so that they are expressed in the price year of the transfers.

At the median canton-level distance to the nearest agent (3.4 km), the estimated round-trip moto-taxi fare is approximately 950 CFA, representing about 12% of the average transfer.[38] At the 75th percentile (5.5 km), the round-trip cost rises to approximately 1,140 CFA, or 14% of the transfer. For the 46 cantons (19.5%) where no village has an agent within 3 km—where the median distance is 9.4 km—a single round trip costs approximately 1,490 CFA, consuming 19% of the transfer value.

Moreover, the cost is not purely monetary: at the median distance, a round trip by moto-taxi takes approximately 45 minutes, while walking would require around two and a half hours. For the 19.5% of cantons with no agent within 3 km, travel times are even longer. This means that the *effective* value of the transfer—the amount actually available for consumption after accounting for cash-out costs—is systematically lower in areas with poor mobile money coverage, and beneficiaries face a non-trivial time cost on top of the monetary one.

This mechanism provides a natural explanation for the heterogeneous effects documented in Table 5. In cantons with better mobile money access, beneficiaries can cash out more cheaply and more frequently, retaining a larger share of the transfer for consumption, savings, and health care expenditures. The positive and significant interaction on financial health—an outcome that depends directly on the ability to convert digital money into purchasing power—is consistent with this channel; the interaction on health care access is positive but not significant at the 3 km threshold (0.04, s.e. 0.02) and significant at 4 km (Panel C of Table 5).

[38]The average transfer is 8,035 CFA, the midpoint of the men's (7,450 CFA) and women's (8,620 CFA) amounts.